

**AECOM**

Contract No.	DD-2023-118		
Project	Diriyah Gate II Infrastructure Package 1		
Doc. Title	Method Statement & Risk Assessment for		
	Cast In-Situ Concretes works for potable water Pump Station02		
Doc. No.	IP-INF-054-0000-AIC-MST-CE-000032	Rev:02	Date: 14-01-2025

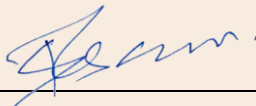
Jacobs

Document Title: Method Statement for
Cast In-Situ Concretes works for potable water Pump Station02
Rev: 01
Date 14-01-2025

		Contract No.	DD-2023-118				
		Project	Diriyah Gate II Infrastructure Package 1				
		Doc. Title	Method Statement & Risk Assessment for Cast In-Situ Concretes works for potable water Pump Station 02				
		Doc. No.	IP-INF-054-0000-AIC-MST-CE-000032	Rev: 02	Date:14-01-2025		

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02	14-01-2025	Issued for Approval

Document Review				
Approval	Name	Position	Signature	Date
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Reviewed by	V.Srinivasa Reddy	QA/QC Manager		14-01-2025
Reviewed by	Abdullah Sabri	Construction Director		14-01-2025
Approved by	Sherif Abdulsattar Ahmed	Project Director		14-01-2025

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1 Purpose

The purpose of this Method Statement is to outline the procedures, responsibilities, and safety measures to be followed during the construction of a pump station. This document aims to ensure

that the construction is carried out systematically, efficiently, and safely, complying with relevant standards, specifications, and regulations. The Method Statement covers all stages of the pump station construction, concrete work foundation works, structural works.

2 Scope of work

The scope of this method statement is for Pump Station Cast In-Situ Concretes works include but not limited to Blinding, Reinforcement, Formwork installations, Scaffolding, Concrete placement, Curing, finishing, application of Geotextile, Water stop, Screed, etc.

2.1 Reference

2.1.1 Contract Specification

Ref. DD-2023-118_Diriyah Gate II Infrastructure Package 1_Vol 2 – Specification (4 of 9)

2.1.2 IFC drawing

2.1.3 Related Drawing

- IP-INF-054-0000-JCB-DWG-ST-100-2000101-01
- IP-INF-053-0000-JCB-DWG-ST-100-2000103-03
- IP-INF-054-0000-JCB-DWG-ST-200-2002220-04
- IP-INF-054-0000-JCB-DWG-ST-200-2002221-02
- IP-INF-054-0000-JCB-DWG-ST-200-2002222-02
- IP-INF-054-0000-JCB-DWG-ST-200-2002223-04
- IP-INF-054-0000-JCB-DWG-ST-200-2002224-02
- IP-INF-054-0000-JCB-DWG-ST-400-2002225-03
- IP-INF-054-0000-JCB-DWG-ST-400-2002226-04
- IP-INF-054-0000-JCB-DWG-ST-600-2002227-02
- IP-INF-054-0000-JCB-DWG-ST-600-2002228-02
- IP-INF-054-0000-JCB-DWG-ST-700-2002229-02
- These Drawing are all reference for below mentioned MTS

a. Applicable Codes and Standards:

- DGCL HSE Specifications
- DGCL Environment Requirements for Contract & DGCL Sustainability Requirements
- Ministry of Transport (MOT)

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- National Water Company (NWC)
- Riyadh Municipality (AMANA)
- Ministry of Municipal and Rural Affairs (MOMRA)

b. Abbreviations used in method statement.

Definitions	Abbreviations
Client	DGCL
PMC	AECOM
Engineer	Jacobs
Contractor	AICC (Al-Ayuni Investment and Contracting Company)
CM	Construction Manager
QA	Quality Assurance
QC	Quality Control
HSE	Health, Safety & Environment
RAMS	Risk Assessment and Method Statement
LS	Land Surveyor
SE	Site Engineer
ITP	Inspection and Test Plan
NCR	Non-Conformance Report
QCP	Quality Control Plan
MSS	Method Statement Submittal
TS	Technical Submittal Request
SD	Shop Drawing
MS	Material Submittal
IR	Inspection Request
RFI	Request for Information
GIS	Geographic Information System
FDT	Field density test
PPE	Personal protective equipment

2.2 Responsibility

2.2.1 Project Director

- The Project Director has the overall responsibility for assuring that the work is continuously performed safely, meets the customer 's Quality requirements, and is completed within schedule.

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- The Project Director is responsible for assuring that adequate tools, equipment, and qualified manpower, as well as foremen and supervisors, are provided to implement the work without delays and perform the work “right the first time.
- Ensure project procedures are identified, developed, and implemented and are adequate and effective. Be continually alerted to means and methods of improving the project team's operations.
- Coordinate with other construction activities to minimize conflicts.
- Communicate with relevant stakeholders to coordinate hoarding installation activities, including contractors, subcontractors, and site personnel.

2.2.2 Director Construction

AICC Construction Director shall be fully responsible for all procedures applicable to the project and shall visibly demonstrate the priority of safety in the execution of works, including setting a good example as follows:

- Responsible for all employees both AICC and rental companies to stop their work in the presence of unsafe acts or unsafe conditions.
- Review work method statements and precautions and attain approval prior to commencing work.
Inspect the work at the site and provide suggestions and instructions to correct deficiencies and ensure that the appropriate supervisor or foreman responds immediately to take corrective actions and perform follow-up to ensure corrective actions have been taken.
- The overall responsibility for the implementation of the Execution and HSE program during the construction of this project.
- The Construction Director has a role to play in delivering the strongest health and safety. Cultures across all project sites The Construction Director have the responsibility for ensuring that the work is performed safely in line with the program and shall report directly to the client and his representative.
- He will attend all weekly site performance information meetings.
- Attending management meetings, department-wide H&S meetings, toolbox talks, participating in behavioral observations, inspections, and audits.
- Assuring adequate resources (people, equipment, budgets) for the effect management of health and safety within business lines.

2.2.3 Project Health & safety Manager

- AICC Safety Manager is responsible for ensuring that AICC staff are complying with the requirements of the approved safety plan and procedures, including conducting routine safety audits and toolbox meetings and obtaining approval before using and participating in the regular safety meetings.
- In all work, the safety manager must demonstrate the importance of safety, including setting a good example and taking the following steps:
- Lead safety meetings and contribute information to improve the working environment.

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- Oversee the proper use of personal protective equipment, safety devices, and safeguards by all employees under his supervision.
- Make sure that the safest method is being used for each job activity on their work site.
- Educate employees under his supervision about proper work methods, sequence of operations, risks, and precautions to take by holding toolbox meetings, in-house and life critical training for them.
- To perform site inspections to identify potential hazards in his area of responsibility and take corrective action.
- Assure that only trained and qualified personnel are assigned to heavy equipment. operations and other specific jobs requiring qualified workers.
- Be familiar with the Client/consultant project Safety Manual and other legal requirements.as per Saudi labor law.
- Ensure that good housekeeping in his respective work area.
- Recognize and reward personnel who eliminate hazards by their initiatives.
- In an emergency or when immediate action is required by safety personnel, immediately assist them.

2.2.4 Environmental Manager

The Environment Manager must clearly demonstrate the importance of environmental requirements. in all earthworks, including setting a good example as follows:

- Ensuring compliance with legislation and the requirements of this Plan.
- Liaising with all discipline Managers/Supervisors on the day-to-day management of earthworks to comply with environmental risks and issues to close out.
- Assure that environmental requirements are complied with, qualified personnel are assigned to heavy. Equipment operation, and they don't harm to flora and fauna in the work area.
- Monitoring implementation and compliance with the AICC CESMP.
- Ensuring personnel are made aware of their requirements under this Plan.
- Coordinating any environmental monitoring required by this Plan.
- Ensuring environmental records relevant to the works are generated and maintained.
- Leading environmental site inspections and systems audits and ensured corrective actions were closed out.
- Leading environmental incident investigations and ensuring corrective actions are closed out.
- Responsible for ensuring the preparation of site-specific environmental and regulatory documentation and for regular review and improvement of such documentation, as appropriate.
- Responsible for ensuring that appropriate levels of environmental expertise and resources are available to conduct environmental management functions.
- Responsible for delivering appropriate training to the workforce in line with AL-Diriyah Framework & CESMP for the activity.
- Provides management support for identification, correction, and resolution of instances. of environmental non-compliance and to prevent future recurrences.

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- Verifies Non-Compliance Record (NCR) issued by Environmental Field Monitor and issues it concurrently to the Environmental Manager, Construction Manager, the AL-Diriyah Project Manager and HSE Manager.

2.2.5 Construction Manager

- The Construction Manager will oversee the allocation and management of project resources including labor, equipment, and materials.
- The Construction Manager will lead and manage the project team, including site supervisors, engineers, and other personnel.
- The Construction Manager will be responsible for ensuring that the construction activities meet the required quality standards and specifications.
- The Construction Manager will be responsible for promoting a safe working environment on the construction site.
- The Construction Manager will ensure that all project documentation is accurate, complete, and up to date.
- The Construction Manager must clearly demonstrate the importance of safety in all their work, including setting a good example as follows:
 - Report on the safety and quality of work performed on the job and contribute information to improving the condition of the site.
 - Before starting work on the site, evaluate the work area.
 - Conduct daily site inspections to ensure all client requirements have been met.
 - He must ensure that all employees are aware of the activities they are going to perform, and They are qualified to participate in those activities.
 - Ensure compliance with Quality and Safety requirements of the client/consultant.
 - Attend management meetings, department-wide H&S meetings, toolbox talks, and participate in behavioral observations, inspections, and audits.
 - Responsible for direct coordination with front-line Engineers and subcontractors.
 - Responsible for the execution of the quality work as per project specifications.
 - Responsible for implementing the agreed risk assessment/ method statements.
 - Responsible for monitoring and reviewing the total direct person-hours, and the production.
 - Responsible for making daily progress reports and monitoring.
 - Responsible for preparing three weeks/months look ahead schedule.

2.2.6 Site Engineer

- The Project Engineer will familiarize himself with this program and ensure its implementation:

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- Will be responsible for executing the construction project according to the approved plans, specifications, and schedules.
- Will be ensuring that the construction work meets the project requirements, codes, and the required quality standards.
- Will provide technical support and guidance to the project team to interpret and clarify construction drawings, specifications, and other technical documents.
- Will organize their work activities so that work is carried out to be required standard with minimum risk to men, equipment, and materials.
- Coordinate with the QA/QC Department during the daily work process where the activities are involved.
- Will be familiar with work permit procedures.
- Will give precise instructions on responsibilities for correct work methods.
- Plan for and ensure good housekeeping practices are maintained.
- Check that power equipment and hand tools are maintained in good condition.
- Position equipment effectively and ensure that electricity supply is installed, used, maintained correctly.
- Coordinate with other contractors on-site to ensure that the correct procedures are followed.
- Confirm that all his crew is implementing all HSE requirements as instructed by the HSE manager.

2.2.7 QA/QC Engineer / Inspector (Civil)

- The QA/QC Engineer / Inspector (Civil), will be responsible for the following activities:
- Ensure that the project complies with relevant codes, regulations, and industry standards.
- Maintain accurate records and documentation related to quality control activities, including inspection reports, test results, and non-conformance reports.
- Monitor and verify compliance with applicable codes, regulations, and industry standards.
- Assess the quality of workmanship and construction processes to identify potential issues or areas for improvement.
- Provide training and guidance to workers and subcontractors on quality control requirements and procedures.
- Conduct daily site surveillance, identify, record, and communicate deficiencies to the team to avoid reoccurrence.
- Conduct work inspections in coordination with the construction team to verify compliance with drawings and specifications, ensure records are maintained, and raise WIRs for completed and verified works.
- Ensure that only approved materials, tools, and equipment are used.

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2.2.8 QA/QC Manager

- Ensure that all works are executed in compliance with the relevant specifications and Contract Conditions.
- Prepare ITP (Inspection & test Plan) in line with project specifications and attain approval prior to commencing work.
- Periodically inspect the work site and provide suggestions and instructions to correct deficiencies.
- Coordinate with Surveyor to ensure that all levels are as per the plan.
- Coordinate with the Site engineer and the Construction Manager to ensure that all works are implemented in compliance with the approved PQP and ITP.

2.2.9 Health & Safety Officer

- Follow the HSE Manager's instructions in implementing the requirements detailed in the Client/Consultant project Safety Manual and other legal requirements (e.g., Saudi labor law).
- Provide site induction and safety orientation to new employees and visitors arriving on site.
- Communicate safety rules and safe work procedures to employees through toolbox meetings, safety talks, information boards, and reports.
- Review work method statements, job safety analysis, recommend safety precautions and safety measures to minimize potential hazards as low as reasonably practicable.
- Perform periodic site inspections, reporting unsafe conditions to responsible personnel, providing recommendations to correct deficiencies, and conducting follow-up audits to verify the implementation of agreed corrective actions.
- Maintain personnel certificates and ensure their validity.
- Ensure that a permanent record of all injuries, near misses, fires, motor vehicle accidents, property damage, and crane lift incidents are recorded and archived.
- Participate in incident investigation, safety meetings, emergency drills, and safety audits, and assist in providing training sessions.
- Ensure that all employees in the project are familiar with the potential hazards, emergencies procedures, and incident reporting procedures.

2.2.10 Supervisor

- Provide support to the Project and HSE team in all aspects of safety, occupational health, safety, and environmental issues.
- Organizing the equipment and manpower necessary to complete the project.
- Support the Health and Safety Manager in ensuring compliance with applicable stature and client's requirements and perform regular HSE audits in the areas of business and operations.
- Attend pre-job, planning, and client meetings as required and HSE-related matters.

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- Ensure training, toolbox meetings, and drills are implemented as part of the companies.
- HSE education and training program, i.e. emergency response systems, etc.
- Inspecting construction sites regularly to identify and eliminate potential safety hazards.
- Ensure all worksites are HSE compliant with Company policies and safety standards.
- Support the Project's HSE personnel in the development of work site HSE campaigns and good practices.

2.2.11 Surveyor

- The surveyor will be responsible for the following activities:
- The Surveyors will ensure that their team is correctly briefed for all aspects of the day's activity onsite, including safety aspects by means of a toolbox talk.
- Use GIS to create detailed site maps, including topographic data and property boundaries.
- Conduct property boundary surveys using GIS to determine the exact location of property lines.
- Collaborate with utility companies and use GIS to map the locations of underground utilities. (water, gas, electricity) to prevent damage during construction.
- Collect and update GIS data related to the project site, including land use, zoning, and environmental features.
- Maintain accurate records of all survey data, GIS data, and project-related documentation.

2.2.12 Foreman

- Will be responsible for supervising the construction crew, including laborers, equipment operators, and subcontractors.
- Will ensure that the work performed by the crew meets the required quality standards and specifications.
- Be familiar with the company's general safety requirements and safety program applicable to work in which subordinate workers are engaged.
- Incorporate safety instructions in routine orders and see they are obeyed.
- Correct unsafe activities in routine orders and see they are obeyed.
- Ensure that new employees are properly instructed in precautions to be taken before they allow the start of work.
- Commend men who by action or initiative eliminate hazards.
- Report unsafe conditions and any defects in equipment and ensure these are quickly remedied.
- Conduct toolbox meetings with the crew before the start of a new activity.
- Implement good housekeeping plans daily.

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2.2.13 Individual worker

- Use the correct tools and equipment for the jobs and use the protective clothing and equipment provided.
- Do nothing to endanger him or co-workers.
- Keep tools in good condition.
- Keep the area in which they are working in clean condition.
- Refrain from horseplay and abuse of safety devices, equipment, and welfare facilities.
- Report all unsafe conditions to the foremen or the site project engineer.

2.2.14 Operators

- Operate equipment safely and effectively for work activities.
- The certification of all Equipment Operators will be checked to ensure that they have the correct training for the equipment.
- 3rd Party TUV & Registration of Equipment shall be checked.
- Ensure that equipment is maintained in good and safe working condition.
- Perform regular equipment maintenance to ensure work continuity with high quality.

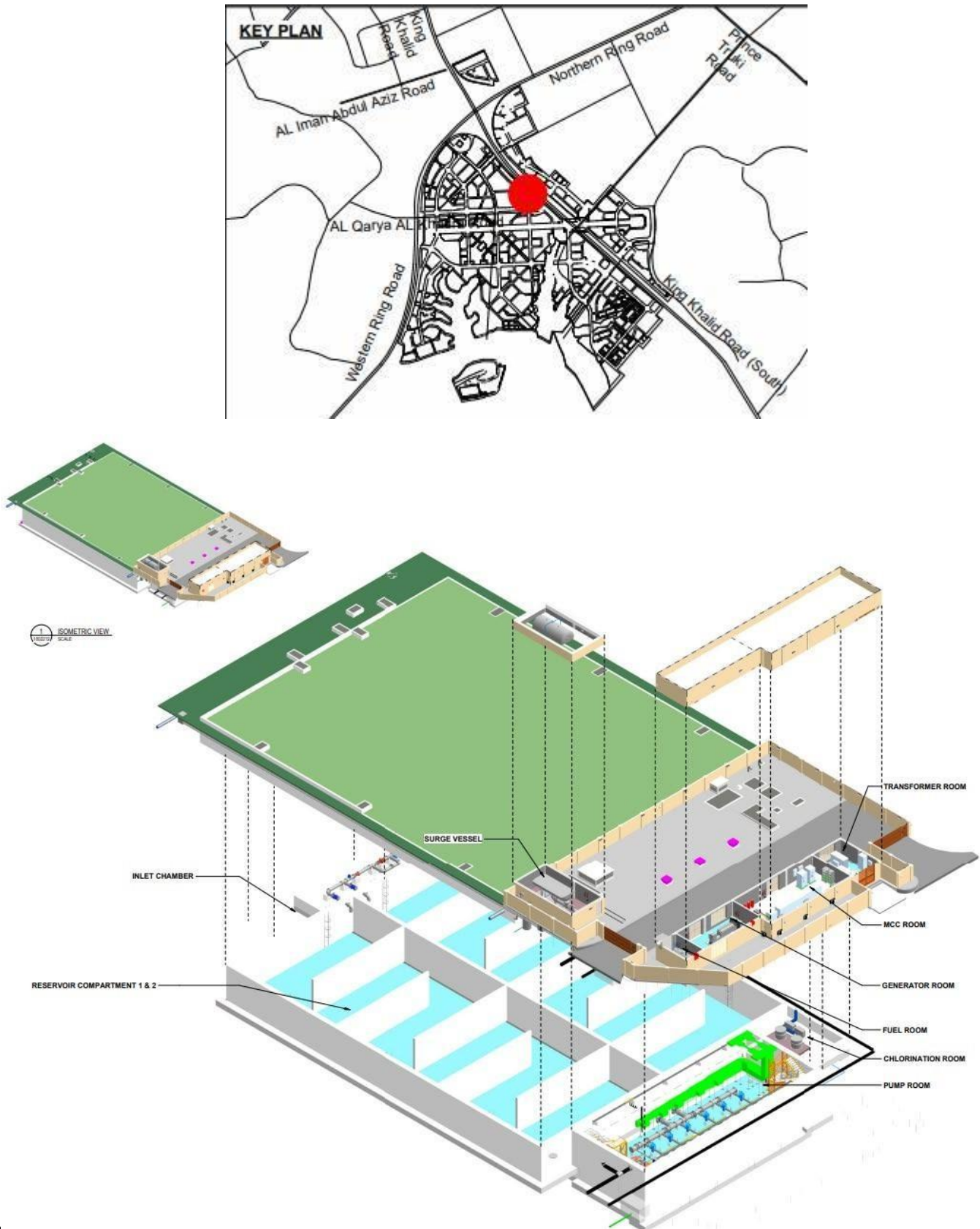
2.2.15 Supervision Arrangements

S. N	Description	Names	Contact details
01	Construction Manager	Mohamed Rehan	0508921768
02	HSE Engineer	Adnan Hakami	053388274
03	QA /QC Engineer Civil	Babar Ali	0549274764
04	QA /QC Engineer Earthwork	Fahim Ullah	0559062258
05	QC Inspector Civil	Sabir Ullah	0566230607
06	Safety Officer	Khuram Shahzad	0559062258
07	Senior Engineer	Mohamed biome	0502827322
08	Surveyor	Mohamed squab	0572666095
09	Scaffolding Inspector/Supervisor	Engr Rajan	0573017003
10	Site Foreman	Prem Kumar	0596902762
Note: Above listed manpower shall be assigned as work per requirements			

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3. Work Procedure (sequence of works)

3.1 Location and Work Limits (Area/Site Specific Layout)



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3.2 Site Planning & Preparation

- All necessary work permits and approvals such as UMA, NTS, MST, etc. for works will be obtained and copies will be kept at the site at work location.
- Before the start of the work, a pre-construction meeting shall be held to discuss and plan the safe work procedures. in compliance with Safety, Quality, and progress. A Site Supervisor will be assigned for the activity covered by this method statement, Risk assessment and emergency procedures & Plan will be provided to all operators and a Pre-Task Briefing will be signed.
- Temporary welfare shall be arranged on site near the working area and shall be shifted based on any change of working area.
- Vehicle parking areas and workers drop zones will be arranged by construction engineer on site based on daily activity areas and as per the traffic management plan.
- Exclusion zone for the equipment area and pedestrians' walkways shall be maintained.
- Assign of adequate number of Manpower, Equipment, tools, and Materials required for the activity and must be always ready.
- Quality personnel are to determine all quality requirements and initiate proactive to control quality of work.
- Quality personnel to coordinate with CSC Engineers for the schedule /Inspection of work.
- Housekeeping is always to be maintained at a high standard throughout the site.
- In case of any unexpected obstacles such as inclement weather or equipment malfunctions, the construction Team /safety team will follow the contingency plan.
- Identify potential hazards specific to the site, terrain, equipment, and activities.
- Be prepared for extreme temperatures, sudden storms, or visibility issues. .

3.3 Survey & Setting out

- Works will be done by an experienced land surveyor (LS) of (AICC) to determine the actual site coordinates based on Benchmarks provided by consultant, and to be able prepare a topographic survey that will describe the condition of the sites and shall be complied with the approved MSRA for survey works.
- All dimensions will be checked and verified on site to accommodate all specified tolerances and differences between actual site dimensions and dimensions shown on the working drawings provided by consultant.

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3.4 Blinding Concrete

- Survey setting out for Levels & Dimension of blinding concrete will be done in accordance with Approved Shop drawings and shall be inspected as per Approved ITP for Concrete Cast in Place Concrete.
- Permanent benchmarks/grids will be established based on survey control points provided by Surveyor using high precision survey instrument "Total Station".
- Shop Drawings with zones to match with blinding bays will be submitted to Engineer for approval.
- Preparation of formation level
- The finished surface of formation will be trimmed to the required profiles and levels within tolerance of +0, -25mm. additionally, deviation in surface flatness shall be limited to +/- 5mm measured along a 3m straight edge. No abrupt steps in surface level are permissible.
- Polyethylene sheet shall be laid above compacted soil as vapor retarder as per approved Material Submittal.

3.4.1 Formwork Installation for blinding concrete

- Set up formwork for plain concrete, ensuring accurate alignment and dimensions.
- Use appropriate formwork materials capable of withstanding concrete pressure.
- Proper bracing should be implemented for the formwork to prevent breakage during concrete pouring.

3.4.2 Preparation of FDT and Blinding Concrete

- A field density test is conducted before the concreting for the compacted soil by third party laboratory as per project specifications.
- Timbers shall be used to control the thickness of concrete. Steel pins will be driven to maintain the level of the finished blinding surface.
- WIR shall be raised for CSC consultant as per approved ITP Ref. Cast in Place Concrete.

3.4.3 Delivery of concrete

- Concrete for blinding shall be AS PER IP-INF-054-0000-JCB-DWG-ST-100-2000101-01 and as per approved Mix design.
- Fresh concrete tests (slump test, temperature checking) and obtain cylinder sample for compressive strength shall be done by third part at site.

3.4.4 Laying and Compacting of Blinding Concrete

- Thick blinding concrete will be laid in its appropriate location from concrete mixer directly using metal chute/manual or using concrete pumps where mixer is not accessible.

3.4.5 Finishing of Blinding Concrete

- Screeding: After pouring the concrete, a screed—a straight board or piece of metal—is used to level the surface by scraping off excess concrete. This ensures an even plane across the entire area.

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- Floating: Once screeding is done, a float—a flat tool with a handle—is used to smooth the surface and remove any imperfections. This process helps to consolidate the concrete and bring the cream (the paste-like layer of concrete) to the surface.
- Edging: Edging tools are used to create crisp edges along the perimeter of the concrete surface. This Levels will be adjusted and carefully checked related to the waterproofing system and its installation as per the IFC drawing.

3.4.6 Curing of Blinding Concrete

- After finishing, the blinding, concrete will be cured with water by covering with wet burlap cloth made from jute for 3-7 days as per ACI 308.1.

3.4.7 Inspection and acceptance of works

- Upon completion of blinding concrete, the finished area will be offered to Engineer for inspection and approval to proceed with subsequent activities.

4.0 CONCRETE

- 4.1 ALL CONCRETE WORKS SHALL CONFORM TO THE ACI 318-19. A GRADE OF C40/32 INDICATES THAT CONCRETE SHALL HAVE A CHARACTERISTIC COMPRESSIVE STRENGTH OF 40 N/mm² ESTABLISHED FROM TEST CUBES AT 28 DAYS. EQUIVALENT TO A COMPRESSIVE STRENGTH OF 32N/mm² ESTABLISHED FROM CYLINDER TESTS AT 28 DAYS. CONCRETE MIX DESIGN SHALL WITH ACI 301-05 AS FOLLOWS:

MIX NUMBER	1	2	3
GRADE	20/20	40/32	40/32
MIN. CEMENTITIOUS CONTENT kg/m ³	200	380	380
CEMENT TYPE	OPC	OPC	OPC
GGBS	-	65%	40%
MAX. FREE w/c RATIO	0.55	0.40	0.40
SLUMP (mm)	100 ±25	75 ±25	75 ±25
MAX. AGGREGATE SIZE (mm)	20	20	20

- *. 65% GGBS MAY BE REPLACED BY 50% GGBS + 5% SILICA FUME. THE CONTRACTOR HAS TO TEST EACH MIX AND SELECT THE ONE THAT SATISFIES THE SPECIFIED STRENGTH AND DURABILITY REQUIREMENT AS PER SPECIFICATION DESIGN OR OTHERWISE PROPOSE AN ALTERNATIVE MIX. FOR THE ENGINEERS REVIEW AND APPROVAL.

- ** AGGREGATE SHALL BE SELECTED TO MAINTAIN A COEFFICIENT OF THERMAL EXPANSION OF 10 ME/C° (MAXIMUM)

MIX 1 : USED FOR PLAIN CONCRETE BLINDING, SCREEDS AND MASS FILL.

MIX 2 : USED IN ALL REINFORCED CONCRETE SUBSTRUCTURE WORKS, (FOUNDATION AND BOUNDARY WALLS IN CONTACT WITH SOIL.

MIX 3 : USED IN ALL REINFORCED CONCRETE WORKS FOR SUPERSTRUCTURE.

3.4.8 Formwork installation and removal

The **removal of concrete formwork** also called as strike-off or stripping of formwork should be carried out only after the time when concrete has gained sufficient strength, at least twice the stress to which the concrete may be subjected to when the formworks are removed. It is also necessary to ensure the stability of the remaining formwork during formwork removal.

Make sure that the approval by the safety personnels is taken before removal of formwork.

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Concrete Formwork Removal Time

The rate of hardening of concrete or the concrete strength depends on temperature and affects the formwork removal time. For example, time required for removal of concrete in winter will be more than time required during summer.

Special attention is required for formwork removal of flexural members such as beams and slabs. As these members are subjected to self-load as well as live load even during construction, they may deflect if the strength gained is not sufficient to handle to loads.

To estimate the strength of concrete before formwork removal, the tests on concrete cubes or cylinders should be carried out. The concrete cubes or cylinders should be prepared from the same mix as that of the structural members and cured under same circumstances of temperature and moisture as that of structural member.

When it is ensured that the concrete in the structural members has gained sufficient strength to withstand the design load, only then formworks should be removed. If possible, the formworks should be left for longer time as it helps in curing.

Removal of formwork from the concrete section should not make the structural element to:

- Collapse under self-load or under design load
- deflect the structural member excessively in short or the long term
- physically damage the structural member when formwork is removed.
 - **The following points must be kept in mind during formwork removal whether the structure will be prone to:**
- freeze thaw damage
- cracks formation due to thermal contraction of concrete after formwork striking.

If there is a significant risk of any of the above damage, it is better to delay the removal time of formwork. If formwork must be removed for optimizing the concrete construction activities, then these structures must be insulated well to prevent such damage.

It shall be kept in mind that the area is barricade before removal of formwork.

It must be made sure that the dismantling of formwork is under scaffolding supervisor and by the permission of the scaffolding approved engineer in charge at site.

Make sure all safety precautions been taken prior to starting formwork removal.

The above precautions and measures shall be taken once the formwork installation shall be started.

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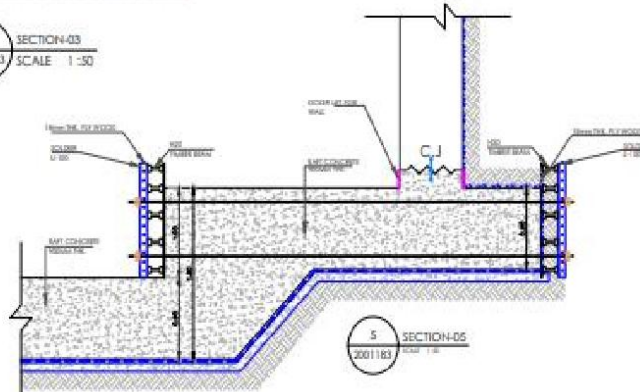
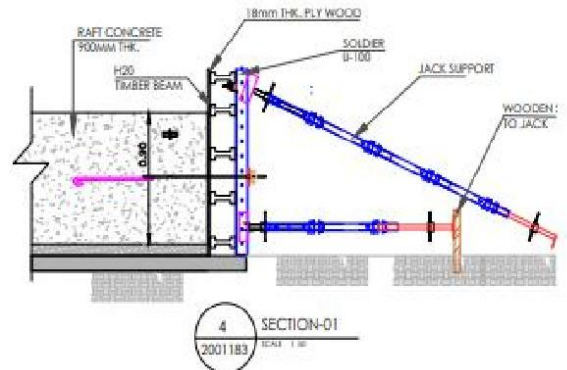
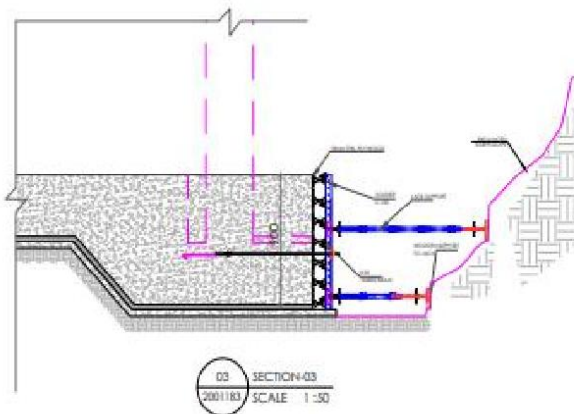
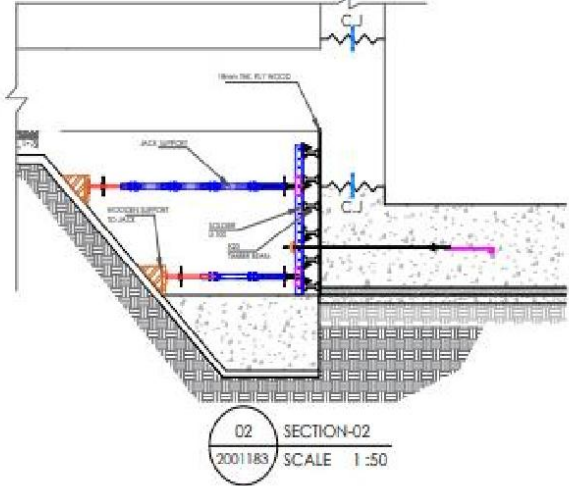
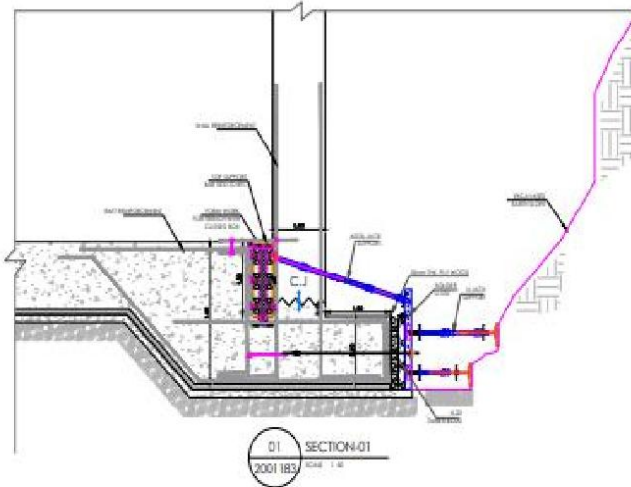
3.4.9 Scaffolding Design

Acrow scaffolding approved general design is as follows.

The same design will be followed for the pump station 02 and shall be shared in detail once we shall get by the third party (ACROW).

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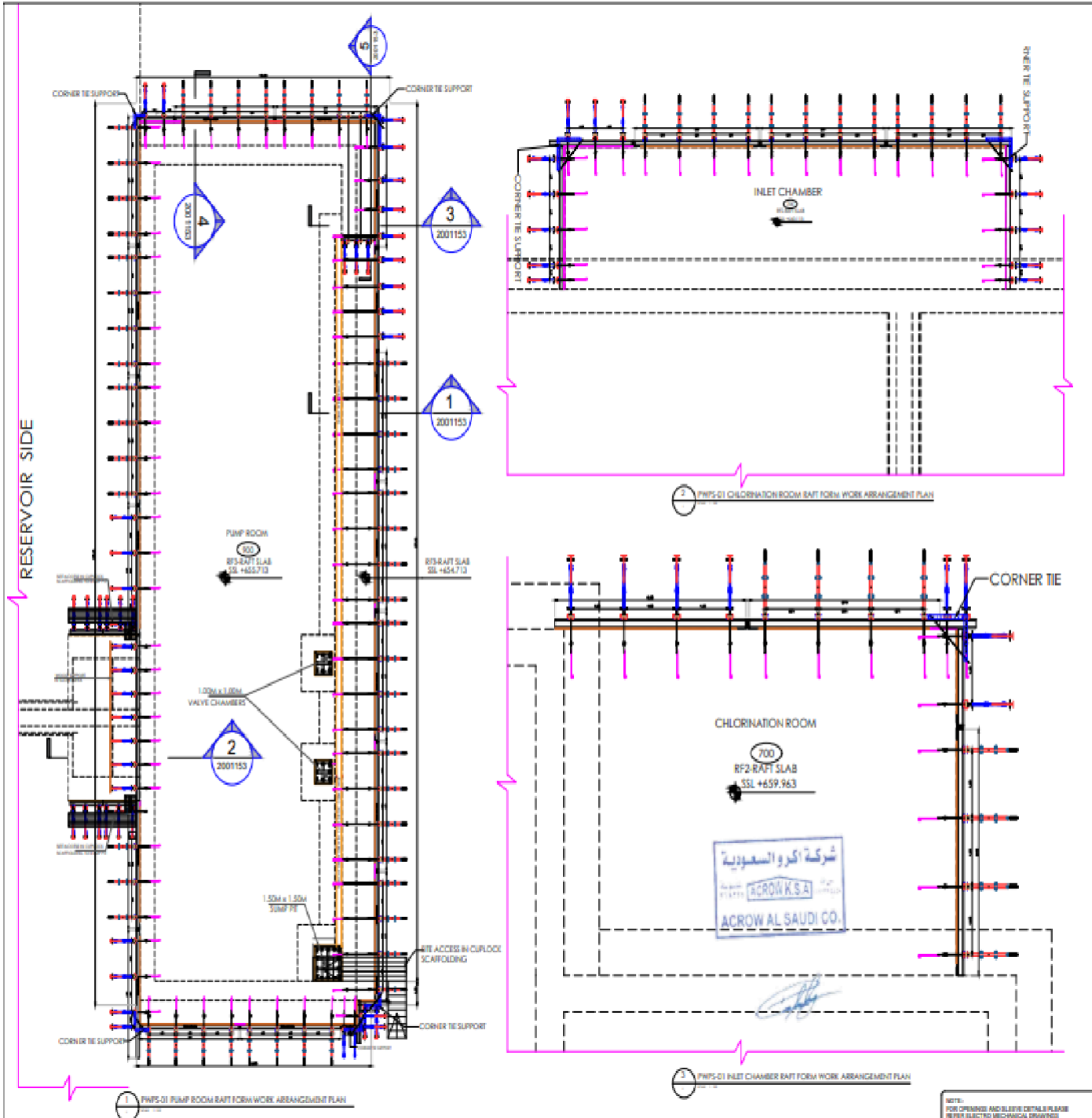
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NOTE:
FOR OFFICE AND SUBMIT DETAILS PLEASE
REFER ELECTRICAL/MECHANICAL DRAWINGS

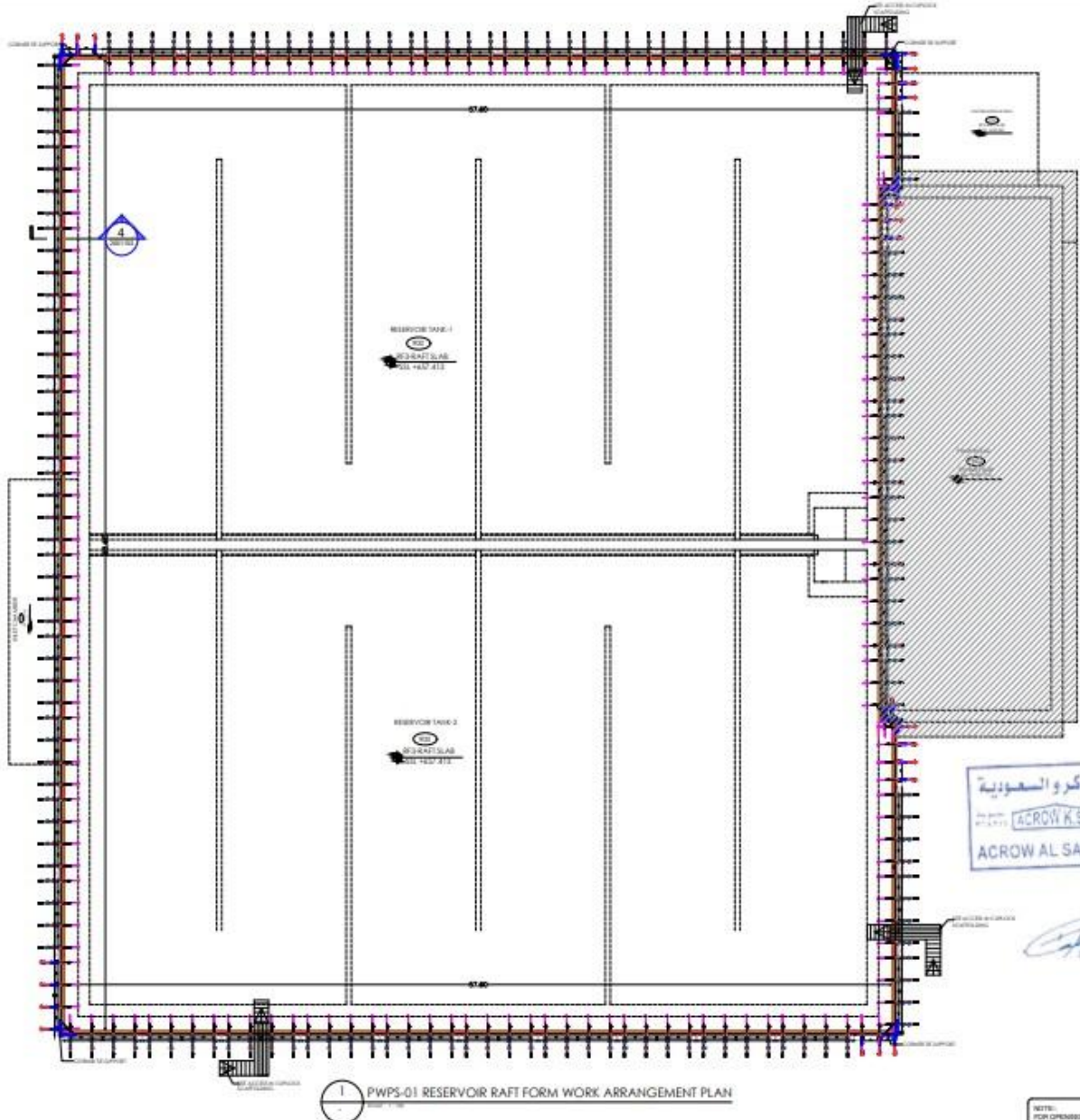
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3.5 Water Proofing

- **Apply Water proofing system above blinding concrete as per approval materials and approved method of statement.**
- **refer to approved MTS:**
IP-INF-054-0000-AIC-MTS-CE-000028

3.6 Screed Concrete:

- Survey setting out for Levels & Dimension of screed concrete will be done in accordance with Approved Shop drawings and shall be inspected as per Approved ITP for Concrete Cast in Place Concrete.
- Permanent benchmarks/grids will be established based on survey control points provided by Surveyor using high precision survey instrument "Total Station".
- Shop Drawings with zones to match with screed bays will be submitted to Engineer for approval.
- Preparation of formation level
- The finished surface of formation will be trimmed to the required profiles and levels within tolerance of +0, -25mm. additionally, deviation in surface flatness shall be limited to +/- 5mm measured along a 3m straight edge. No abrupt steps in surface level are permissible.

3.6.1 Formwork Installation for screed concrete

- Set up formwork for screed concrete, ensuring accurate alignment and dimensions.
- Use appropriate formwork materials capable of withstanding concrete pressure.
- Proper bracing should be implemented for the formwork to prevent breakage during concrete pouring.

3.6.2 Delivery of concrete

- Concrete for screed shall be as per approved Mix design.

3.6.3 Laying and compacting of Screed Concrete

- Thick blinding concrete will be laid in its appropriate location from concrete mixer directly using metal chute/manual or using concrete pumps where mixer is not accessible.

3.6.4 Finishing of Screed Concrete

- **Screeding:** After pouring the concrete, a screed—a straight board or piece of metal—is used to level the surface by scraping off excess concrete. This ensures an even plane across the entire area.
- **Floating:** Once screeding is done, a float—a flat tool with a handle—is used to smooth the surface and remove any imperfections. This process helps to consolidate the concrete and bring the cream (the paste-like layer of concrete) to the surface.
- **Edging:** Edging tools are used to create crisp edges along the perimeter of the concrete surface.

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3.6.5 Curing of Screed Concrete

- After finishing, the Screed, concrete will be cured with water by covering with wet burlap cloth made from jute.

3.7 Formwork for Reinforcement Concrete

- Reference Contract Specification Ref. DD-2023-118_Diriyah Gate II Infrastructure Package 1_Vol 2– Specification (4 of 9)

3.7.1 Pre-Installation Planning

- Design and Engineering: Ensure the formwork design meets the load requirements and is approved by third party engineer.
- Site Inspection: Inspect the site for uneven ground, obstacles, and other hazards that could affect formwork stability.

3.7.2 Personal Protective Equipment (PPE)

- Workers should wear appropriate PPE, including helmets, gloves, safety glasses, and steel-toed boots.
- Use fall protection equipment like harnesses and lifelines when working at heights.

3.7.3 Proper Training

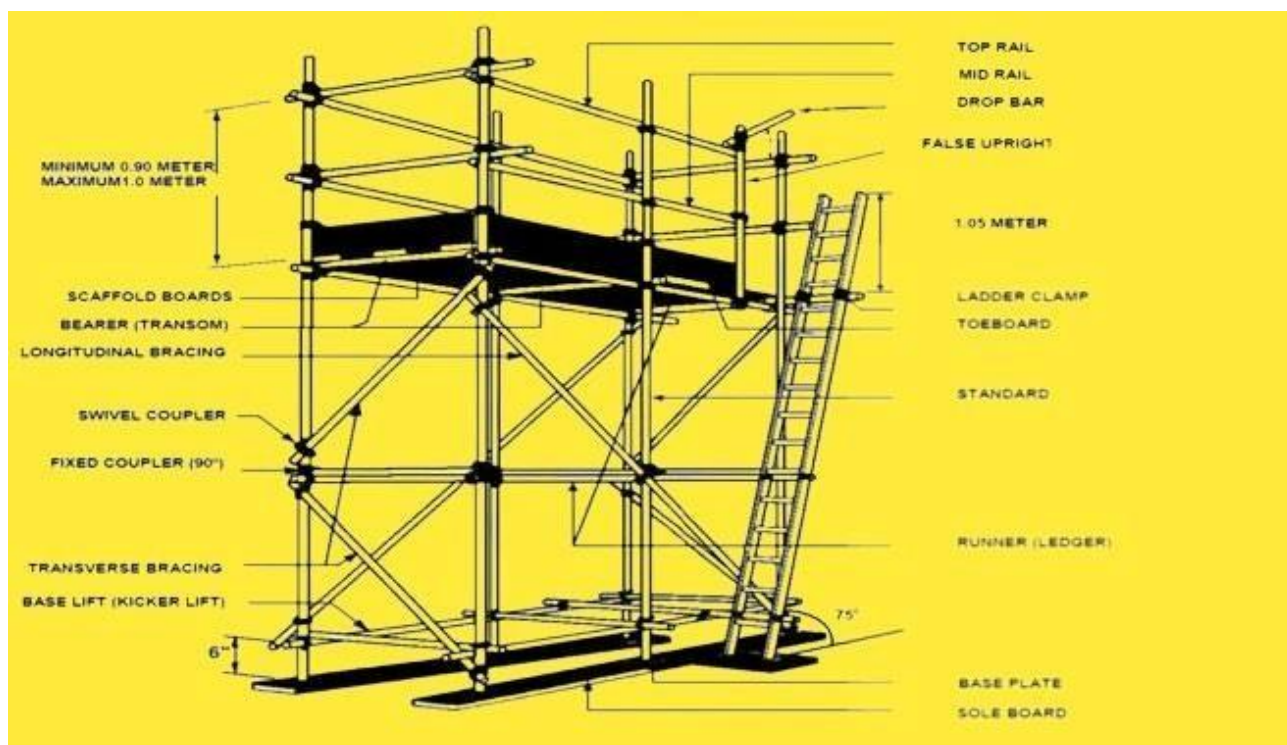
- Workers should be trained in formwork installation procedures, safety practices, and the use of tools and equipment by third party engineer.

3.7.4 Formwork Assembly

- Quality Materials: Use materials that are in good condition and meet the required specifications.
- Correct Assembly: Follow the manufacturer's instructions and engineering plans precisely. Ensure all components are securely fastened.
- Bracing: Properly brace formwork to prevent movement or collapse during concrete pouring.
- Ladders and Scaffolding: Ensure ladders and scaffolding used during installation are stable and secure.
- Safe access, edge protection, support & props will be provided.
- Regular safety briefings and drills shall be conducted regularly.
- Proper bracing should be implemented for the formwork to prevent breakage during concrete pouring.
- Formwork material shall be inspected to ensure they comply with specified requirement and are properly cleaned prior to use.
- Construct formwork so concrete members are of size, shape, alignment, elevation, and position indicated, within tolerance limits of ACI 117.

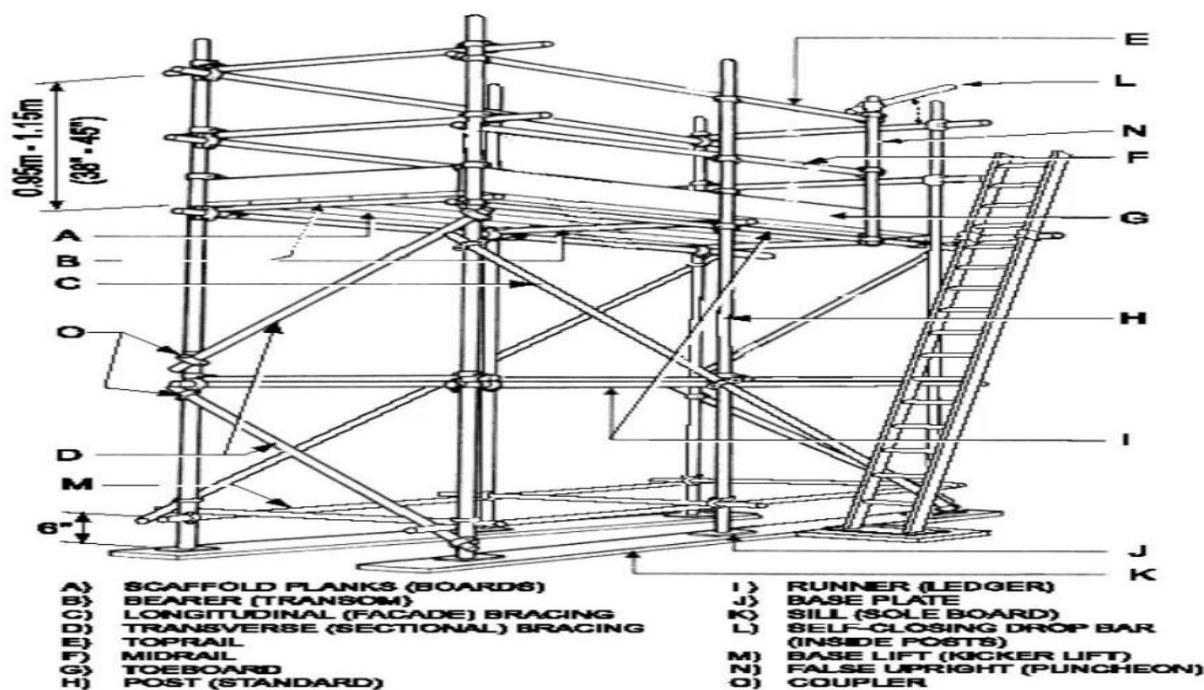
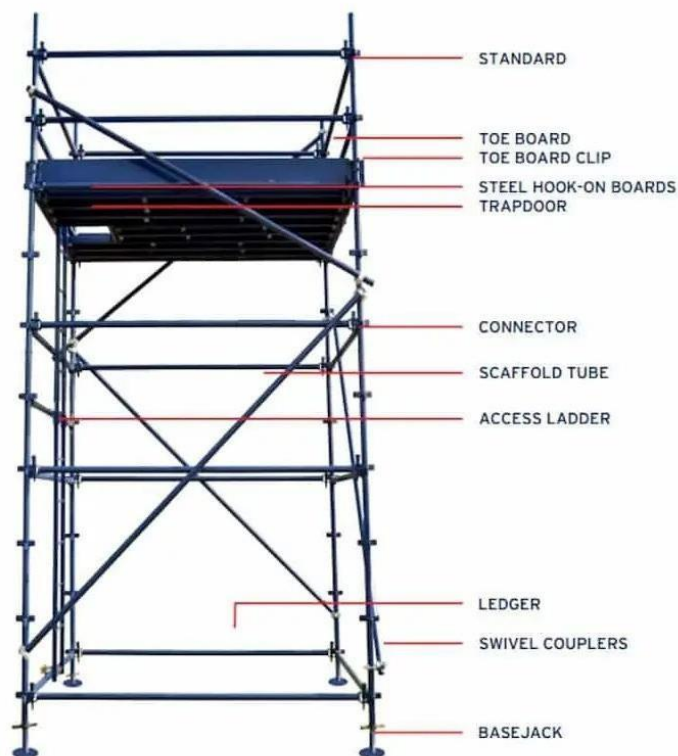
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- Prior to placement of forms, survey lines & grade control shall be checked for accuracy using levelling machines & total station with reference to the benchmark given by Surveyor & approved Drawings.
- Construct forms tight enough to prevent loss of concrete mortar.
- Complete form work shall be inspected for compliance with design drawings, specification tolerance shown on the approved shop drawings shall apply.
- Formwork shall be properly anchored to the shores below to prevent upward or horizontal movement of any part of the formwork system during concrete placement.
- Formworks for wall & top slab, we will use cup lock standard scaffolding system approved by Engineer and inspected by Scaffolding competent person & inspector. We will erect scaffolding under supervision of competent person and after reaching required height we must place formworks (first H beam over top jack and then we will place 18mm plywood on H beam manually) over scaffolding with survey level as per drawing. We will provide access & egresses for working crews at interval of 25 ft. We will provide proper working platform for the workers to works safely at height with safety harness.



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**Figure II.9.1
SCAFFOLD TERMINOLOGY**

- Chamfer exterior corners and edges of concrete receiving applied waterproofing membranes. Chamfer strips shall be inspected to verify proper placement.
- Forms shall be inspected before the reinforcing steel is placed to ensure that the dimensional & the location of the concrete member will conform to the approved drawings.

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- Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete.
- Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.
- AICC Quality team will do internal inspection for the installation of Steel reinforcement and formwork works to verify acceptance of activity and will submit WIR for CSC approval.
- After placing of walls formworks, it should be checked and ensure by surveyor, formworks alignment and its verticality as per approved drawings. Proper support and tie rod will be provided prior to release D-Shackles from the formworks to hold the loads.
- Form surface shall be coated with specified form release agent.
- Form shall be re used in a condition that it should be maintained, no deformity, clean & free from any chemicals etc. that might affect the surface & the quality of the concrete.

3.7.5 Monitoring During Installation

- Supervision: Experienced supervisors should monitor the installation process to ensure compliance with safety procedures.
- Weather Conditions: * Be cautious of weather conditions, especially wind, which can affect the stability of formwork.

3.7.6 Load Management

- Concrete Pouring: Monitor the rate and volume of concrete being poured to prevent overloading the formwork.
- Vibration: Ensure proper vibration techniques to avoid disturbing the formwork structure.

3.7.7 Inspection and Maintenance

- Regular Inspections: Inspect formwork regularly for signs of wear, damage, or instability. Repair or replace damaged components immediately.
- Post-Pouring Inspection: After pouring concrete, inspect the formwork for signs of dis

3.7.8 Lifting formworks (Walls & Top Slab)

- Typically involves using specialized equipment such as cranes, hoists inspected by third party with valid legal documents TUV to raise the formwork safely and efficiently. Prior to lifting, thorough inspection of the formwork and the lifting equipment is necessary to ensure everything is in good condition. Additionally, proper planning and coordination among the construction team are essential to ensure a smooth and safe lifting operation. Once the formwork is lifted, it can be moved to the next section of the construction site or dismantled, depending on the project requirements.
- All workers shall be trained by formwork supplier / third party on safe erection and working methodology. Well trained workers will be performed the activity of formworks installation.
- All formworks will be lifted/erected as per lifting calculation and safe lifting plan with approved lifting tackles as per formworks design. Location of lifting tackles and hooks on formworks will be used as per designer and technical consideration mentioned in the design drawings of formworks.

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Lifting Belt/Sling with D-Shackles will be used for formworks lifting with approved third party and as per SWL will be used for the safe working operation. All lifting tackles will be inspected and approved by Lifting competent person prior to start the works.

Erection of walls formworks will be commenced with the help of 25 tons crane followed by Rigger, Lifting Competent Person & presence of TWC. Location of hook hanging points inside H beam will be followed as per designer's consideration.

3.7.9 Permit to Load

- After complete installation of plywood on entire slab area, we will invite our TWC and Formworks designer to inspect our formworks is as per design or not? Whether it will take the load as designed? After successful inspection and satisfaction TWC should release the permit to load, then only we can start loading the rebar for top slab.
- Form shall be re used in a condition that it should be maintained, no deformity, clean & free from any chemicals etc. that might affect the surface & the quality of the concrete.

3.7.10 Formwork methodology

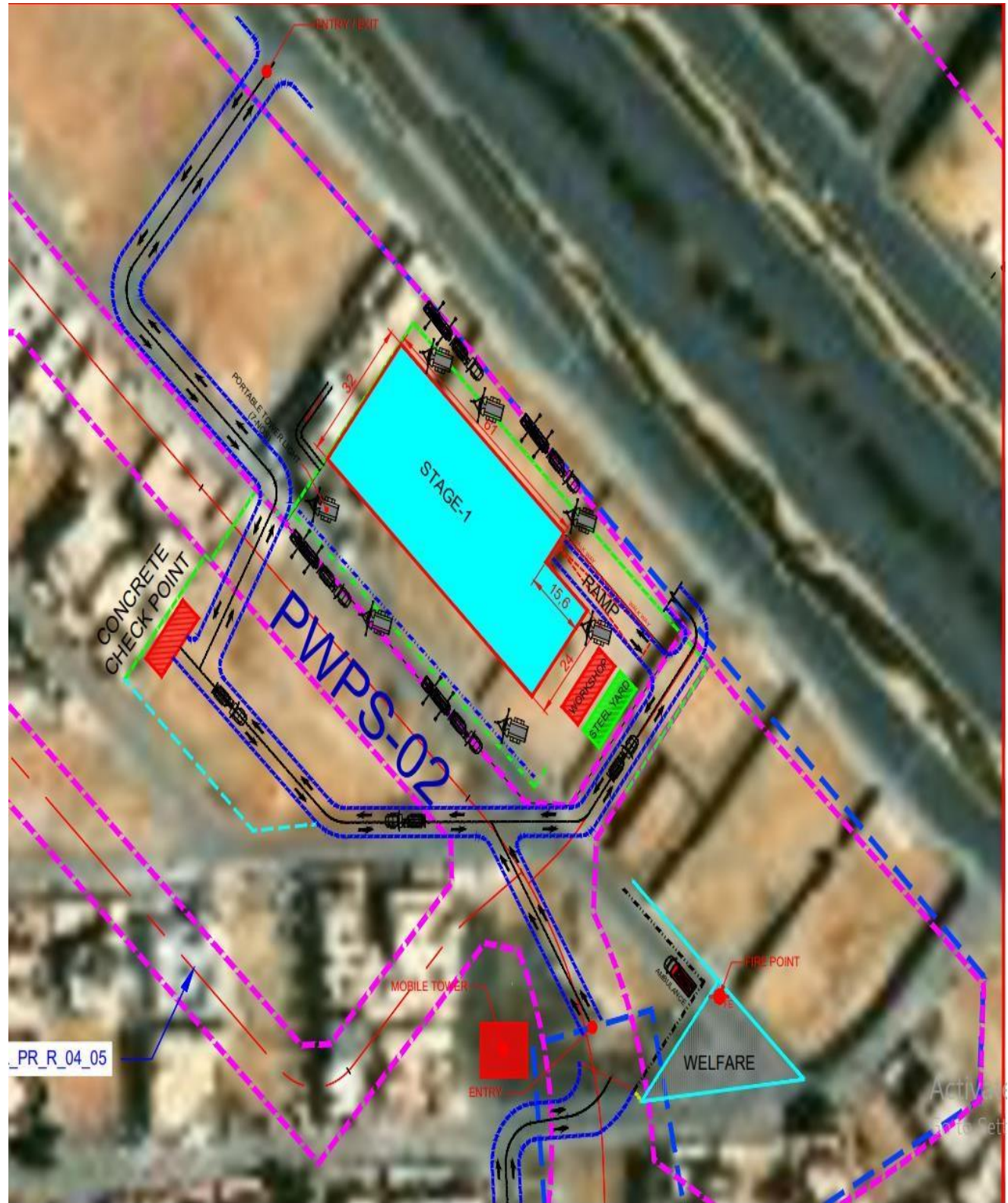
In Addition to the above formwork precautions and procedure we shall do the extra precaution as the access, props, protruding bars are concerned. Proper safe access will be provided for works at heights. Works on edges and openings shall be protected by proper side railings and supports.

Proper edge protection will be provided and shall be supervised by the scaffolding competent individuals.

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3.7.11 Site Layout

Updated site layout is as under



3.8 Steel reinforcement & installation (water stop)

Reference Contract Specification Ref. Ref. DD-2023-118_Diriyah Gate II Infrastructure Package

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- All reinforcement shall be Type 2 high yield deformed bars grade (420N/mm²) as per project specifications.
- Rebar's shall be cut and bend as per bar bending schedule and fabricated rebar's shall be stacked properly with tags for proper identification.
- The shifting of the rebar's with mobile crane, Boom truck or manually workers will be used for placing at working location.
- lap lengths and anchorage lengths of reinforcement shall be as per aci 318-14. additional
- lapping if required to be provided with engineer's approval. the minimum lap length of reinforcement shall be as per IFC Drawing

(IP-INF-054-0000-JCB-DWG-ST-100-2000103-03).

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3.8.1 Minimum Cover

- to ends and sides of reinforcement unless otherwise indicated on Drawings to be as per IFC Drawing (IP-INF-054-0000-JCB-DWG-ST-100-2000101-01)

5.5 CLEAR COVER TO REINFORCEMENT INCLUDING LINKS, STIRRUPS, AND TIES SHALL BE AS FOLLOWS:-

a) STRUCTURE IN CONTACT WITH SOIL

STRIP FOOTINGS/RAFT	=	75mm
WALL AND COLUMN	=	50mm
SLAB AT GRADE LEVEL	=	50mm
PRECAST FOOTING	=	75mm

b) SUPER STRUCTURE

COLUMNS	=	40mm
BEAMS	=	40mm
SLABS	=	30mm
WALLS	=	40mm

ALL CONCRETE ELEMENTS IN CONTACT WITH WATER 50mm

LAP AND ANCHORAGE LENGTHS FOR REINFORCEMENT

- LAP LENGTHS AND ANCHORAGE LENGTHS OF REINFORCEMENT SHALL BE AS PER ACI 318-19.
- THE TENSION DEVELOPMENT LENGTH AND TENSION SPLICES SHOWN ARE APPLICABLE WHEN AT LEAST ONE OF THE FOLLOWING CONDITIONS IS SATISFIED:
 - THAT CLEAR SPACING OF BARS BEING DEVELOPED OR SPLICED NOT LESS THAN (DB), CLEAR COVER AT LEAST (DB) AND STIRRUPS OR TIES THROUGHOUT LD NOT LESS THAN THE CODE MINIMUM ; OR,
 - CLEAR SPACING OF BARS BEING DEVELOPED OR SPLICED AT LEAST (2 * DB) AND CLEAR COVER AT LEAST (DB)

WHERE NONE OF THESE CONDITIONS IS SATISFIED, TENSION DEVELOPMENT LENGTH AND TENSION SPLICE SHALL BE CALCULATED AS PER CODE REQUIREMENTS.
- THE LENGTHS ARE BASED ON THE MIN COVER TO THE BAR LISTED ON NOTE 5.5
- VALUES FOR SLABS AND WALLS APPLY ONLY FOR BAR CLEAR SPACING OF 100mm OR GREATER. VALUES FOR BEAMS AND COLUMNS APPLY ONLY FOR BAR CLEAR SPACINGS OF 80mm OR GREATER. FOR SPACINGS LESS THAN THESE VALUES REFER TO THE DRAWINGS.
- TABULATED VALUES ARE FOR NORMAL WEIGHT CONCRETE. THE VALUES ON THIS SHEET DO NOT APPLY TO LIGHT-WEIGHT CONCRETE.
- WHERE ACTUAL CONDITIONS DIFFER FROM THE CLEAR COVER SHOWN ON THE TYPICAL DETAILS OR DIFFER FROM PROVIDED SCHEDULED BAR SIZE, MINIMUM SPACING AND/OR f_c , LENGTHS SHALL BE ADJUSTED ONLY WITH THE APPROVAL OF THE STRUCTURAL ENGINEER OF RECORD.
- WHERE BARS OF DIFFERENT SIZES ARE LAP SPLICED IN TENSION. THE LAP LENGTH SHALL BE THE LARGER OF THE TENSION LAP SPLICE LENGTH (L_{dt}) OF THE SMALLER BAR AND THE TENSION DEVELOPMENT LENGTH (L_{dt}) OF THE LARGER BAR.
- WHERE BARS OF DIFFERENT SIZES ARE LAP SPLICED IN COMPRESSION, THE COMPRESSION LAP LENGTH (L_{dc}) SHALL BE THE LARGER OF THE COMPRESSION DEVELOPMENT LENGTH (L_{dc}) OF THE LARGER BAR OR THE COMPRESSION LAP SPLICE LENGTH OF THE SMALLER BAR.
- FOR LAP SPLICE LENGTH THE TENSION DEVELOPMENT LENGTH (L_d) IS TO BE MULTIPLIED BY 1.3.

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3.8.2 Splices and Development Length and Standard Hook

- **should be staggered as per IFC Drawing Ref IP-INF-054-0000-JCB-DWG-ST-100-2000103-03**
- After the rebar placement in position, it will be tied at intersections with specified binding wire to form the reinforcement cage for the structure.
- Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.
- All bars and cast in anchors bolts will be accurately located and securely fixed in the correct position by using template & per the surveying settings out.
- For safe, the vertical layers of reinforcement bars in the wall shall be fixing by crossing and tie enough with steel bar to avoid collapsing and fall.
- Distributing the steel crossing between layer with equal distance to keep steel alignment.
- Location, spacing, size supports, and cover shall be checked during the installation process and QC will make final inspection prior to pouring.

TENSION DEVELOPMENT LENGTH, Ldt

TENSION DEVELOPMENT LENGTH, Ldt (mm)				
BAR SIZE	HORIZONTAL BAR WITH LESS THAN 300mm CONCRETE BELOW BAR OR VERTICAL BAR		HORIZONTAL BAR WITH MORE THAN 300mm CONCRETE BELOW BAR	
	CONCRETE CYLINDER STRENGTH (fc)		CONCRETE CYLINDER STRENGTH (fc)	
	32 MPa	≥40 MPa	32 MPa	≥40 MPa
T10	480	415	600	535
T12	555	495	720	645
T14	645	580	840	750
T16	740	660	960	860
T20	1140	1020	1480	1325
T22	1250	1120	1625	1455
T25	1420	1270	1850	1655
T28	1590	1425	2070	1850
T32	1820	1630	2365	2115

* VALUES FOR T40 BARS ARE DEVELOPMENT LENGTHS ONLY. THESE BAR SIZES ARE NOT TO BE LAP SPICED.

COMPRESSION SPLICE LENGTH, Lsc

COMPRESSION SPLICE LENGTH SCHEDULE, Lsc (mm)	
BAR SIZE	COMPRESSION SPLICE LENGTH (mm) (APPLICABLE TO ALL CONCRETE GRADES AND BAR TYPES)
T10	410
T12	495
T14	575
T16	660
T20	820
T22	905
T25	1025
T28	1150
T32	1315

* VALUES FOR T40 BARS ARE DEVELOPMENT LENGTHS ONLY. THESE BAR SIZES ARE NOT TO BE LAP SPICED.

TENSION SPLICE LENGTH, Lst

TENSION DEVELOPMENT LENGTH SCHEDULE, Ldt (mm)		
BAR SIZE	CONCRETE CYLINDER STRENGTH, fc	
	32 MPa	≥ 40 MPa
T10	600	535
T12	720	645
T14	840	750
T16	960	860
T20	1480	1325
T22	1625	1455
T25	1850	1655
T28	2070	1850
T32	2365	2115

COMPRESSION DEVELOPMENT LENGTH, Ldc

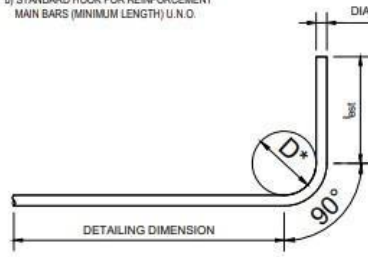
COMPRESSION DEVELOPMENT LENGTH SCHEDULE, Ldc (mm)	
BAR SIZE	COMPRESSION DEVELOPMENT LENGTH, Ldc (mm)
T10	215
T12	260
T14	305
T16	345
T20	430
T22	475
T25	540
T28	605
T32	690

**AECOM**

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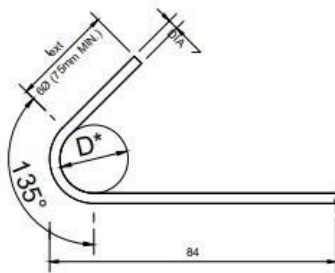
Jacobs**HOOKS FOR REINFORCEMENT MAIN BARS, STIRRUPS AND TIES****a) BARS SELECTION**

DESIGNATION	AREA	UNIT WEIGHT
mm.	mm ²	kg/m
Ø 10	79	0.617
Ø 12	113	0.888
Ø 16	201	1.578
Ø 20	314	2.466
Ø 25	491	3.853
Ø 32	805	6.319

**b) STANDARD HOOK FOR REINFORCEMENT
MAIN BARS (MINIMUM LENGTH) U.N.O.****90° HOOK**

BARS IZE	D*	l _{ext}	L _{dh}
	mm.	mm.	mm.
Ø 10	60	120	210
Ø 12	75	150	255
Ø 16	100	200	340
Ø 20	120	250	420
Ø 25	150	300	525
Ø 32	260	400	675

D* = MINIMUM BEND
DIAMETER
TO ACI 318M-19

**c) STANDARD HOOK FOR
STIRRUPS AND TIES**

BARS IZE	D*	l _{ext}	L _{dh}
	mm.	mm.	mm.
Ø 8	32	75	170
Ø 10	40	75	210
Ø 12	48	75	255
Ø 16	64	100	340

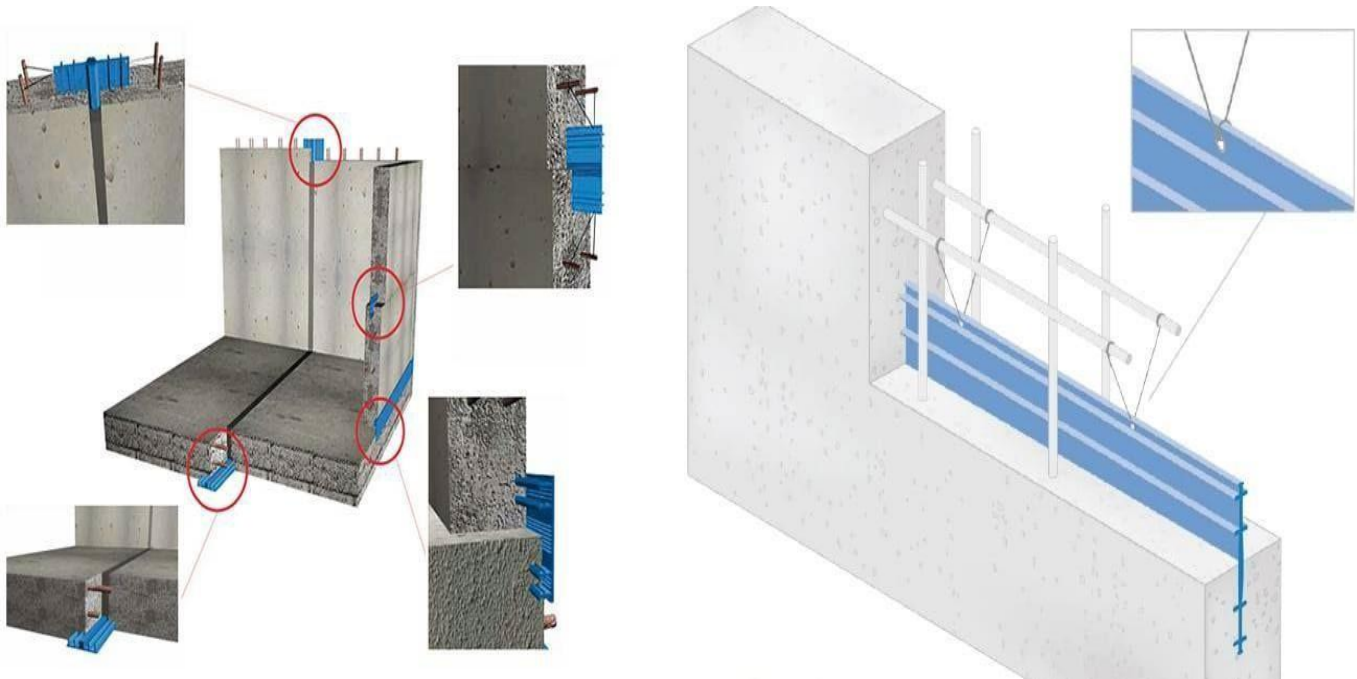
D* = MINIMUM BEND DIAMETER TO
ACI 318M-19

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3.8.3 Installation of water stops

Install approved flexible & self-expanding water stop in construction joints as indicated to form a continuous diaphragm as per drawing from approved materialises. Install in longest lengths practicable. Support and protect exposed water stop during progress of Work. Field-fabricate joints in water stops according to manufacturer's written instructions.

- Prepare the Surface: Ensure the surface where you'll be installing the water stop is clean and free of any debris or obstructions.
- Cutting: Cut the water stop to the required length using a saw or PVC pipe cutter. Make sure to measure accurately before cutting.
- Place the water stop in the desired position, ensuring it fits snugly against the surface where you want to stop the water.
- holding water stop and tie enough to keep striate & alignment as per manufacture requirement.
- Securely tied with wires to the neighbouring reinforcement bar to ensure the water stop do not bend under pressure of concrete pouring.



- AICC Quality team will internal inspection for installation of steel & embedding item to verify acceptance of activity and will submit WIR for CSC inspections and approval

3.9 Concrete placement

Reference Contract Specification Ref. (DD-2023-118_Diriyah Gate II Infrastructure Package 1_Vol 2 – Specification (4 of 9).

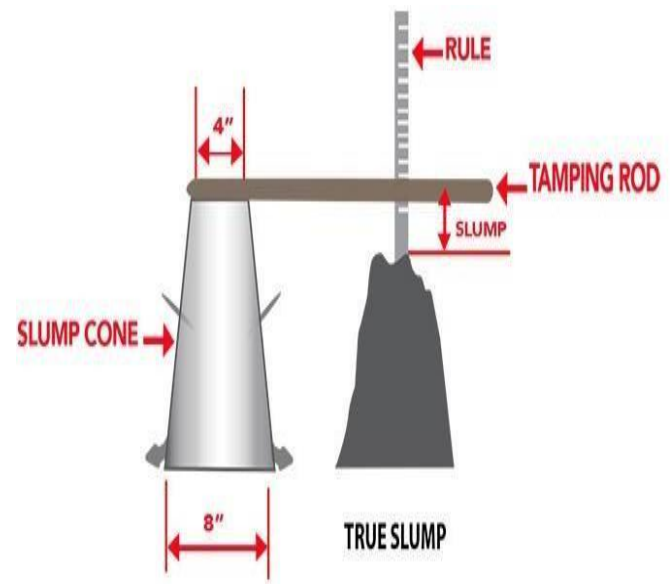
- Concrete delivery shall be Approved Mix design with approved concrete supplier and all site and lab results should be passed as per DG specifications.

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- Concrete height of fall should not exceed 1.0m to avoid segregation. Area of the concrete pump positioning location should be well compacted and concrete pump and concrete mixer should be away at least 2m from the edge of excavation or equivalent to the depth of excavation. Area should be barricaded around pump and mixer location during concrete operation, and we must
- assign one red man & Green man for the safe vehicle movement.

3.9.1 Field Quality Control tests

- AICC shall place the inspection request 24 hours in advance of the time of placing concrete.
- Concrete delivery as per Approved Mix design and from approved concrete supplier.
- Delivered concrete shall be tested by approved 3rd party according to the following requirement:
- Testing Frequency: Obtain one sample for 50 cu. m of each concrete mix or each day's pour.
- Slump: ASTM C 143; one test at point of placement for each delivery, perform additional tests when concrete consistency appears to change.
- Concrete Temperature: ASTM C 1064; one test for each delivery. Maintain concrete temperature below 90 deg F (32 deg C) at time of placement.
- Compression Test Specimens: ASTM C 31/C 31M; cast and laboratory cure one set of six standard cylinder specimens for each sample.
- Cast and field cure one set of six standard cylinder specimens for each sample if needed.



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3.9.2 Foundation And Rafts

Due to Foundation variety Levels and as per top levels of roof, AICC shall execute the concrete process for Rafts and foundations, Roofs through phases as below: -

Phase 1: Pump room & Reservoir

Foundation Stage- 1

Phase 2: Pump room & Reservoir

walls Stage- 1

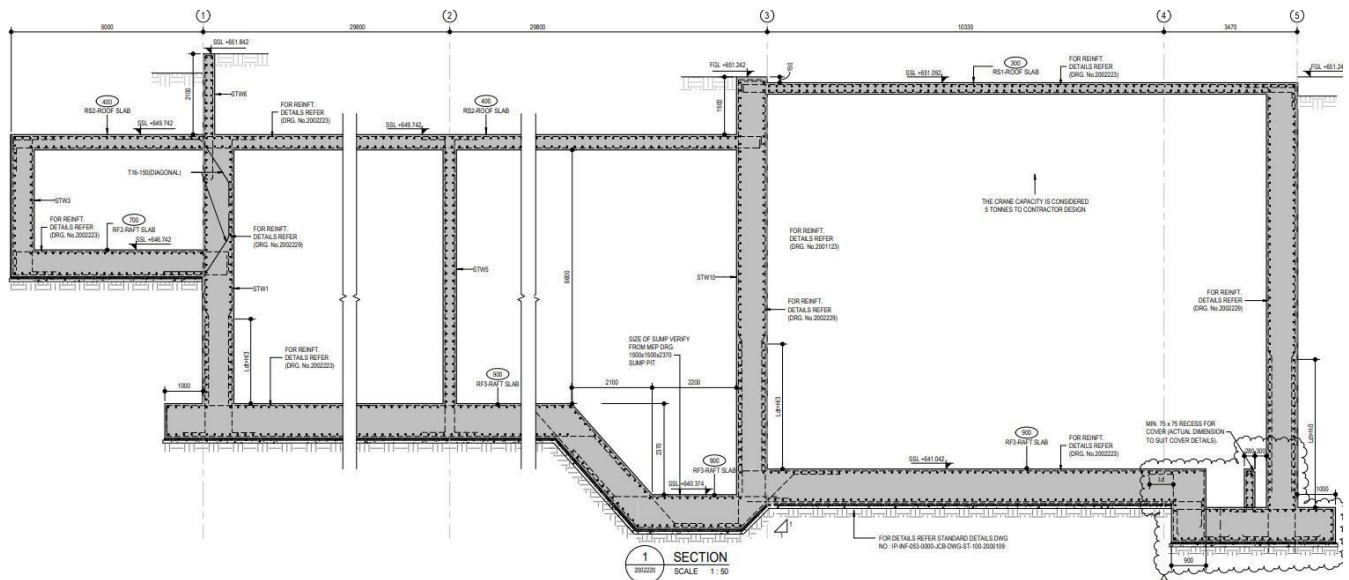
phase 3: inlet chamber & chlorination room raft and walls

Phase 4: Roof slab Reservoir and inlet chamber

Phase 5: Roof pump room and chlorination room.

Phase 6: fuel room, genset room, mcc room, transformer room, operating room & domestic water pump room

- AICC shall allocate the number and type of cranes bucket, Vibrators, pumps, equipment, and layout required to facilitate the rate of supply subject to area of pour.
- Assembly and fixing of reinforcement shall be inspected and approved prior to final pour.
- Concrete shall be cast by direct tipping from the ready-mix truck in the area which is easily accessible, and concrete shall be cast by a mobile concrete pump in the area which is inaccessible.
- The number of pumps required will be determined by the size of pour.
- The concerned section for concreting shall be properly cleaned.
- Do not add water to concrete at site.



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- Deposit concrete continuously in layers of uniform depth not exceeding 600mm in thickness. If a section cannot be placed continuously, provide construction joints as specified. Deposit concrete to avoid segregation.
- Consolidate placed concrete with mechanical vibrating equipment according to ACI 301.
- Vibrator to be used to achieve full compaction, particular attention around reinforcement, spacer, corners of formworks and joints.
- Field Surveyor / Concrete Foremen shall be controlling the concrete level with the levelling machine.
- Concrete placing will be planned in such a manner that cold joints are eliminated during pour.
- Every effort shall be undertaken to avoid emergency stop pours. In the unlikely event of an emergency stop pour, the concrete will be compacted in proper position. If emergency stop ends are occurring horizontally, the concrete will be compacted to achieve a reasonable level. Laitance will be removed from emergency stop end with water jet and surface cleaned prior to continuation of the pour. Approved bonding agents shall also be used on these areas.
- Vibrator to be used to achieve full compaction, particular attention around reinforcement, spacer, corners of formworks and joints.
- The top surface of the concrete shall be brought to a correct level using the mentioned tools and equipment.
- Detailed records shall be kept by AICC of delivered concrete of each mix.
- During the concert work, QC engineer and site engineer be on site during concreting to ensure that the reinforcement and other embedded items are kept in their position.

3.9.3 Concreting of Columns, Walls and Top Slab

A. Preparation:

- Works Inspection Request shall be submitted to consultant, 24 hours prior to concrete.
- The forms will be cleaned, and a form release agent applied prior to erecting formwork.
- A working platform shall be erected at the head of each wall on two sides complete with handrails to facilitate the placement and vibration of the concrete in a safe manner. This work platform shall be of such size as to accommodate three laborers. Access to the platform shall be by a suitably sized ladder secured to the platform. The platform is made such that labor and vibrator person can work safely.
- The formwork shall be checked for verticality with the help of a plumb bob.
- The verticality shall be checked during the concreting and after the finishing of the concreting.
- The formwork shall be checked for concrete cover. Cover will be obtained using approved spacers.
- Concreting shall be continuous between the joints. The concrete bonding agents shall be applied onto the hard surface prior to placing further concrete against it.

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3.9.4 Method of Placing concrete and pump

methodology for placing concrete using a pump involves several steps:

- 1. Preparation:** Ensure that the pump and all associated equipment are in good working condition. Set up the pump at a suitable location, considering factors such as access, distance to the pour site, and obstacles.
- 2. Setting up the pipeline:** Connect the delivery hose to the pump outlet and lay out the pipeline to reach the pour site. Make sure the pipeline is supported properly to prevent sagging or kinking.
- 3. Prime the pump:** Fill the pump hopper with concrete and prime the pump by purging air from the system. This ensures a continuous flow of concrete once pumping begins.

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3.9.5 Commence pumping: Start the pump and gradually increase the rate of concrete delivery to match the pour rate at the construction site. Monitor the flow of concrete to prevent blockages or spills.

3.9.5 Adjustment: Adjust the pump's speed and pressure as needed to maintain a steady flow of concrete and to accommodate changes in elevation or distance.

3.9.6 Safe working arrangement

- Risk assessment: Conduct a thorough risk assessment to identify hazards associated with working at height, such as falling objects, unstable platforms, or electrical hazards.
- Guardrails and barriers: Install guardrails, toe boards, and barriers around the perimeter of elevated work areas to prevent falls.
- Platform stability: Ensure that scaffolding, elevated work platforms, or temporary structures are properly erected, secured, and inspected for stability before use.
- Regular inspections: Conduct regular inspections of elevated work areas, equipment, and safety systems to identify and address any hazards or deficiencies.
- The pouring section shall be cleaned properly. A working platform shall be erected along the full length of the wall on one side complete with handrails to facilitate the placement and vibration of the concrete in an acceptably safe manner. Access to the platform shall be by a suitably sized ladder secured to the platform.
- Area of the concrete pump positioning location should be well compacted.
- pump and concrete mixer should be away at least 2m from the edge of excavation or equivalent to the depth of excavation.
- Handles of concrete buggies shall not extend beyond the wheels on either side of the buggy. Knuckle guards are to be installed on buggy handles to protect the worker's hands. Temporary access shall be protected on either side to prevent falls and shall be inspected by TWC ensuring the loading capacity.
- Concrete shall be placed in horizontal layers not more than 600mm thick except as hereinafter provided. When less than a complete layer is placed in one operation, it shall be terminated in a vertical bulkhead. Each layer shall be placed and compacted before the preceding batch has taken initial set to prevent damage to the green concrete and avoid surfaces of separation between the batches. Each layer shall be compacted to avoid the formation of a construction joint with preceding layer which has taken initial set.
- The verticality shall be checked during the concreting.

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- Concreting shall be continuous between the joints. If we are to place in such circumstances, concrete bonding agents (shall be submitted for approval) shall be applied onto the surface prior to placing further concrete against it.
- Concrete, during and immediately after depositing, shall be thoroughly compacted. The compaction shall be done by mechanical vibration subject.
- Vibrators shall be manipulated to thoroughly work the concrete around the reinforcement and embedded fixtures, and into the corners and angles of the forms.
- Vibration shall be applied only be experienced operators under close supervision, at the point of deposit and in freshly deposited concrete. The vibrators shall be inserted and withdrawn out of the concrete slowly. The vibration shall be of sufficient duration and intensity to thoroughly compact the concrete but shall not be continued to cause segregation. Vibration shall not be continued at any point to the extent that localized areas of grout are formed.
- Vibrator shall be supplemented by such spading as is necessary to ensure smooth surface and dense concrete along form surfaces and in corners and locations impossible to reach with the vibrators.
- Field Surveyor / Concrete Foremen shall be controlling the concrete level with the levelling machine.
- The top surface of the concrete shall be brought to a correct level using the mentioned tools and equipment's. The finished level will be marked on formwork edges and level pads will be formed during the placing of concrete.
- During concreting of all reinforced work, Site engineer/QC engineer shall be in present at site throughout. It shall be their duty to ensure that the reinforcement and other embedded items are kept in position during concrete work.

3.9.7 Cold Weathering Concrete placement

When air temperature has fallen to or is expected to fall below 5 deg C, uniformly heat water and aggregates before mixing to obtain a concrete mixture temperature of not less than 10 deg C and not more than 27 deg C at point of placement.

Do not use frozen materials or materials containing ice or snow. Do not place concrete on frozen subgrade or on subgrade containing frozen materials.

3.9.8 Hot Weathering Concrete placement

Maintain concrete temperature below 90 deg F (32 deg C) at time of placement.

Chilled mixing water of chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water.

Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade moisture uniform without standing water, soft spots, or dry areas.

3.9.9 Control temperature

The effective temperature control program, which may be used.

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We must spray cold water frequently to maintain moisture content inside concrete to avoid evaporation of water.

We must cover freshly poured concrete with wet gunny bags with two layers of polyethylene sheet to make surface of concrete cool and maintain the temperature. Insulation of surfaces to minimize thermal differentials between the interior and the external surfaces.

Cementitious material content control, where the choice of type and number of cementitious materials can have the less heat-generating potential of the concrete.

Construction management, where efforts are made to protect the structure from excessive temperature differentials by knowledge of concrete handling, construction scheduling, and construction procedures.

The temperature control such as restricting placing operations to cool periods at night or during cool weather. On the other extreme, control measures that can include the prudent selection of a low-heat- generating cement system including: The use of pozzolans.

The use of air-entraining and other chemical admixtures to improve both the fresh and hardened properties of the concrete.

3.9.10 Concrete Finishing

The finish of the walls shall be determined by the requirements of the approved drawings. The top of the wall shall be matched with the drawing level.

Laitance to the top of the walls shall be removed by either a water spray or brushing with a stiff brush where the concrete has not yet hardened. On hardened concrete, the laitance will be removed by mechanical chipping.

3.9.11 Concrete Curing

Reference Contract Specification Ref. (DD-2023-118_Diriyah Gate II Infrastructure Package 1_Vol 2 – Specification (4 of 9)

All concrete shall be cured for a period required to obtain the full specified strength, but not less than (7) consecutive days. Unformed surfaces shall be covered with Wet burlap, or other approved fabric mats.

Begin curing immediately after finishing concrete. Cure unformed surfaces, including floors and slabs, concrete floor toppings, and other surfaces, by one or a combination of the following methods:

Moisture Curing: Keep surfaces continuously moist for not less than seven days with one of the following materials:

3.9.12 Water

Absorptive cover, water saturated, and kept continuously wet. Cover concrete surfaces for area which can be covered and edges with 300-mm lap over adjacent

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absorptive covers.

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Cure formed concrete surfaces, supported slabs, and other similar surfaces. If forms remain during curing period, moist cure after loosening forms. If removing forms before end of curing period, continue curing for the remainder of the curing period.

3.10 Removal of forms

Reference Contract Specification Ref. DD-2023-118_Diriyah Gate II Infrastructure Package 1_Vol 2– Specification (4 of 9)

- Formwork, for sides parts of the Work, that does not support weight of concrete may be removed after cumulatively curing at not less than 10 deg C for 24 hours after placing concrete provided concrete is hard enough to not be damaged by form-removal operations and provided curing and protection operations are maintained.
- Leave formwork for slabs minimum 14 days and other vertical structural elements formworks will be removed 24-48 hrs after assurance of concrete should be fully dried and settled.
- At least 70 percent of 28-day design compressive strength.

TABLE 5.03-1

MINIMUM REQUIRED PERCENTAGE OF SPECIFIED 28-DAY STRENGTH (f'c)

Structural Element	Percentage of Specified 28-Day Strength
Columns and wall faces (not yet supporting loads)	50
Mass piers and mass abutments (not yet supporting loads) except pier caps	50
Sidewalk on bridges Sidewalk forms shall, in all cases, be released before the main girder and slab forms are released	70
Box girders	80
T-beam girders, slabs, cross-beams, caps, pier caps not continuously supported, struts, and top slabs on concrete box culverts	80
Trestle slabs, when supported on wood stringers	70
Slabs, when supported on steel stringers or prestressed concrete girders	70
Pier caps continuously supported	60
Arches	90
Rail bases, traffic railings and median barriers*	40

- Start by removing any bracing or supports attached to the formwork.
- Carefully remove the formwork panels or Molds, starting from the top and working downwards. Use appropriate lifting equipment or manpower as needed.
- Ensure that the removal process is gradual and controlled to avoid damaging the concrete structure or causing sudden loads.
- Check for any embedded items, such as bolts or tie rods, and remove them carefully to prevent damage to the concrete.

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3.10.1 Permit to Unload

Prior to start any DE shuttering work, we must invite our TWC and Formworks third party designer to ensure that concrete have achieved its design strength and its safe for shutter removal, Its TWC responsibility to issue the permit to unload once he found everything is as per design and safe. Then only work will be started.

Formwork, for sides parts of the Work, that does not support weight of concrete may be removed after cumulatively curing at not less than 10 deg C for 24 hours after placing concrete provided concrete is hard enough to not be damaged by form-removal operations and provided curing and protection operations are maintained.

Leave formwork for slabs minimum 14 days and other vertical structural elements formworks will be removed 24-48 hrs after assurance of concrete should be fully dried and settled.

Removing form for walls, first we will hold the form with the help of approved lifting tackles connected with H-beam as per approved formworks design. Lifting belt & D-shackles shall be used with 25 tons crane to hold the forms then we will start removing side support and tie rods and other bracing. After removal of all side support, we will erect the forms with approved lifting tackles in presence of lifting competent person & TWC.

Removal of forms for top slab will be commenced manually after completion of 14 days, we will start first to lower/loose the top jack for plywood removal from the concrete base one by one with safe manner. After removal of plywood we will remove H beam and then we will start to dismantle scaffolding manually from top to bottom and all formworks and scaffolding materials will be stored separately in designated place with proper stacking method.

Carefully remove the formwork panels or Molds, starting from the top and working downwards. Use appropriate lifting equipment or manpower as needed.

Ensure that the removal process is gradual and controlled to avoid damaging the concrete structure or causing sudden loads.

Check for any embedded items, such as bolts or tie rods, and remove them carefully to prevent damage to the concrete.

3.10.2 Safety precautions formwork removal

Throughout the de-shuttering process, adhere to appropriate safety measures, such as using personal protective equipment (PPE), securing work areas, and following safe work practices to prevent accidents or injuries.

a. Personal protective equipment (PPE):

Require all personnel involved in de-shuttering to wear appropriate PPE, including hard hats, safety glasses, gloves, and steel-toe boots, to protect against falling objects, impact hazards, and hand

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injuries If working at height or in confined spaces, provide additional PPE such as fall protection equipment and respiratory protection.

b. Stability assessment:

Before removing formwork, assess the stability of the structure and surrounding areas to prevent collapse or instability. Consider factors such as soil conditions, nearby loads, and weather conditions Ensure that the structure has achieved sufficient strength and stability to support its own weight and any imposed loads after formwork removal.

c. Safe work practices:

Use proper lifting techniques and equipment to handle formwork panels and components, preventing strains and musculoskeletal injuries. Avoid overcrowding work areas to minimize the risk of tripping hazards, collisions, and entrapment Prohibit unauthorized personnel from entering the de-shuttering zone to maintain a controlled and safe work environment.

d. Fall prevention:

Implementing fall prevention measures, such as guardrails, safety nets, or personal fall arrest systems, when working at height during de-shuttering activities Ensure that workers are trained in fall protection procedures and equipment usage to mitigate the risk of falls from elevated work areas.

e. Supervision and monitoring:

Assign competent personnel to oversee de-shuttering activities and monitor compliance with safety protocols Conduct regular inspections of the work area and equipment to identify potential hazards and address them promptly.

3.10.3 Joint Filling

Prepare, clean, and install joint filler according to manufacturer's written instructions.

Remove dirt, debris, saw cuttings and sealers from joints; leave contact faces of joint clean and dry.

Apply Approved joint filler materials as per manufacturer recommendation.

3.11 Quality Control and Inspection

- Conduct regular inspections during each construction phase.
- To be Ensure compliance with project specifications and relevant standards.

3.12 Chipping & Surface Preparation

- After DE shuttering if in case of any honeycomb or any defects found in concrete surface, then we will repair it as per approved method of statement for concrete repairs, in addition of that we will use cleared face shield with safety goggles for the workers during chipping and for the surface preparation of concrete we will use good condition buffing machine with safety guard and we will provide face mask, safety earplug (Hearing Protective device) the working crews for the safe works.

3.13 Documentation

- Maintain detailed records of construction activities, inspections, and quality control measures.

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3.14 Final Inspection and Handover

- Conducting a thorough final inspection to verify that all components meet the required standards.
- Obtain approval from the relevant authorities before final acceptance.

3.15 Lifting plan

Crane Specifications 130 Ton & 50 Ton

Crane Specifications for 130T crane:

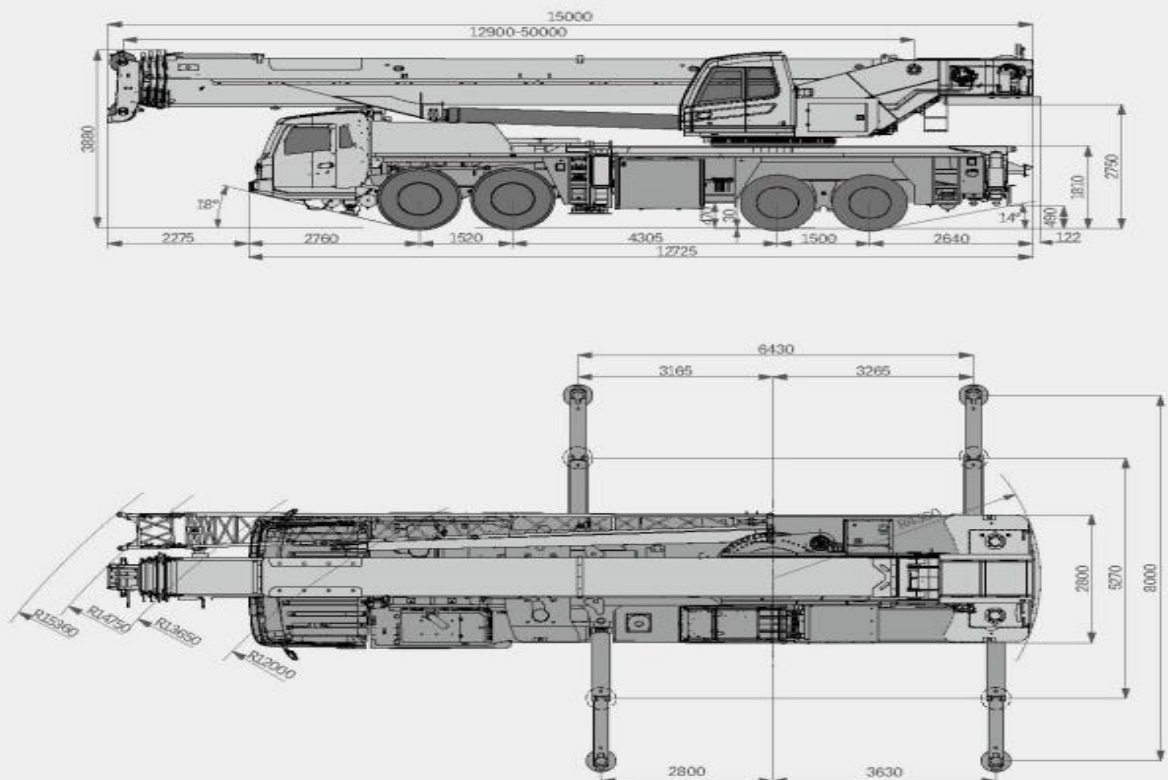
- **Crane Capacity:** 130 tons
- **Crane Model:** SANY STC1300T5
- **Boom Length:** 36.1 meters
- **Radius:** 20 meters

Objective: For lifting, placing and lowering steel and shutter panels using 130t & 50t cranes

130T crane overall dimensions

MODEL STC1300T5 SERIES

Overall Dimensions 整机尺寸图



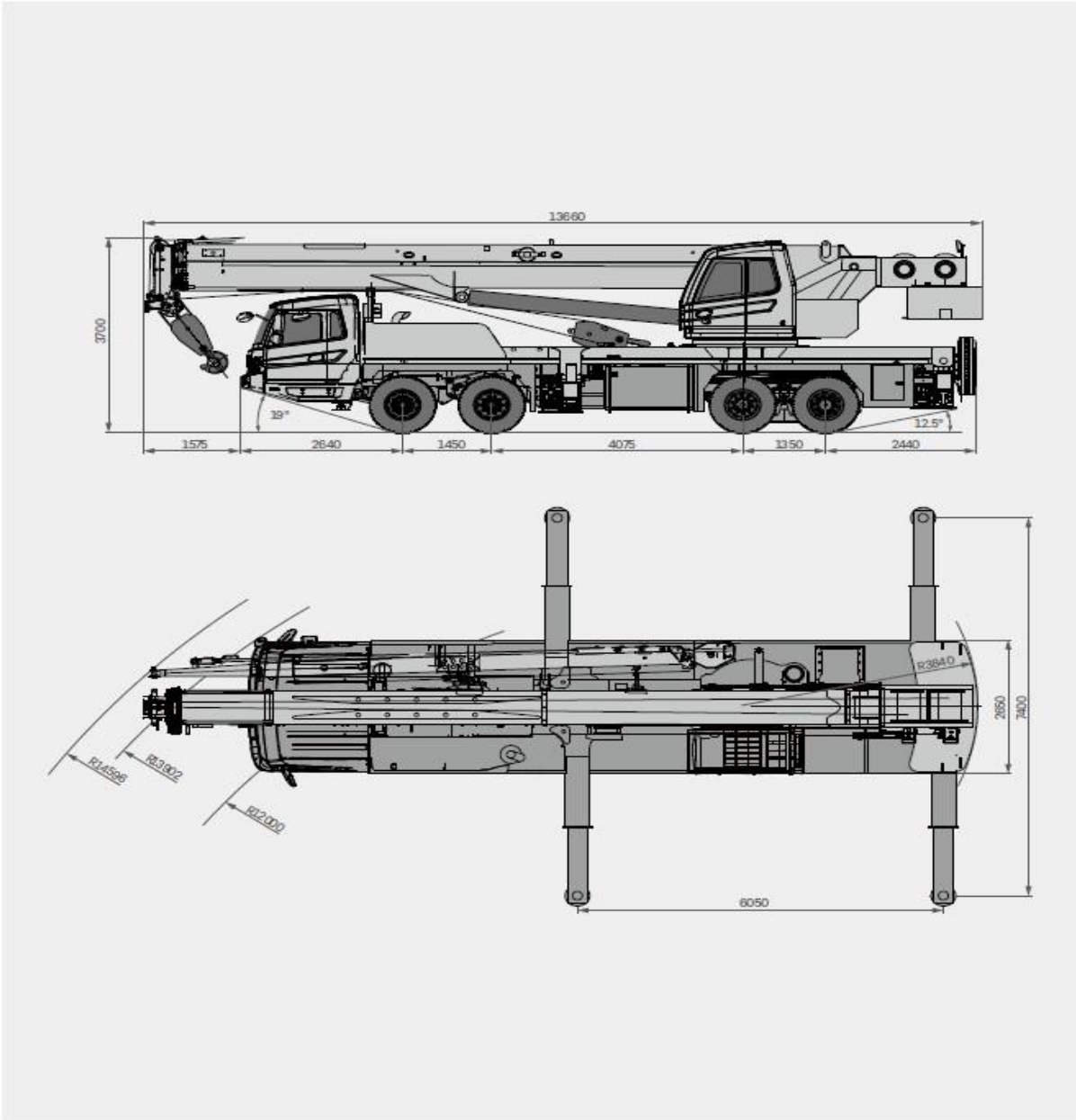
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Crane Specifications for 50T crane:

- Crane Capacity: 50Tons
- Crane model: SANY STC500
- Boom Length: 31.7
- Radius: 26 meters

Model STC500C5 | SERIES

Overall Dimensions 整机尺寸图



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Load Details

- Load Description: Steel bundles and shutter panels and accessories
- Load Weight: 0.065 T to 2.0 T
- Load Dimensions:



Note:

Valid Third-party inspection certificates will be provided for all the lifting tools & tackles.

As per DC we will follow the Crane Quarterly inspection.

Lifting supervisor will carry out Pre inspection of lifting tools and tackles.

Lifting Procedure

Site Preparation

- Ground Conditions: Ensure the ground is stable and compacted to support the crane's weight and load.
- Access Routes: Clear and secure all access routes for the crane's movement.
- Work Area: Establish a clear work area around the lifting site. Ensure no unauthorized personnel are within the lifting zone.
- Obstacles: Identify and remove any obstacles that could interfere with the lifting operation

Pre-Lift Inspection:

- Inspect the crane, rigging equipment, and lifting accessories for any defects or damage.
- Ensure all personnel wear appropriate PPE.
- Ensure all personnel are aware of the sequence of work and associated hazards
- Review emergency escalation procedures and ensure full understanding and compliance of all workers

Load Rigging:

- Attach slings to the load and secure with shackles.
- Ensure the load is balanced and secured.

Crane Setup:

- Position the crane on a stable, level surface.
- Extend the boom and set the radius.
- Engage the outriggers and level the crane.
- Avail adequate clearance between excavation edges to the crane.
- Protect the lifting area and ensure competent supervision throughout the lifting activity

Lift Execution:

- Lift the load slowly and steadily to prevent swinging.

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- Signal person to be assigned and located (opposite side) of the trench to ensure adequate communication with crane operator following international standards.
- Maintain a lifting area free of workers and ensure a safe distance maintained from the load during the lift.
- Crane operator to follow the signal person directions and stop immediately in case of any abnormality in the crane
- Following lead of signal person to position the load at the desired location and lower it slowly.

Post-Lift Procedure:

- Secure the load in its final position.
- Dismantle the rigging and inspect for any damage.
- Conduct debriefs with the lifting team to review the operation and identify any areas for improvement.

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Load chart 130T crane

MODEL STC1300T5 SERIES

Load Chart-Telescopic Boom 主臂性能表

Unit: kg



T



Fully extended
完全伸展



360°



20.7t

EN

Radius (m) 幅度	12.9	17.5	22.2	26.8	31.3	36.1	40.8	45.5	50	17.5	Radius (m) 幅度
3	130000	69000								40000	3
3.5	80000	68000	52000							40000	3.5
4	76000	66000	52000							40000	4
4.5	71000	63500	52000	40000						40000	4.5
5	66000	60000	50000	40000						40000	5
5.5	62000	55500	48000	38000	37000					40000	5.5
6	58000	51500	46500	37000	36000					40000	6
6.5	53000	49000	45500	36000	35000	30000				40000	6.5
7	48000	46500	44500	35000	34000	29000				40000	7
7.5	44500	43500	42500	34500	33000	28000				39000	7.5
8	41500	40500	40500	33000	32500	27000	22500			38500	8
9	35500	35800	35500	31000	30000	26000	21500	20000		37000	9
10		31000	31000	29000	28000	25000	21000	19000	14300	32500	10
11		26300	25800	26500	26000	24000	20500	18000	14000	28500	11
12		22400	21900	23100	23900	22700	20000	16500	13500	24500	12
14			16300	17400	18200	18800	17200	15000	13000		14
16			12500	13600	14300	14800	15200	13500	12000		16
18			9700	10800	11500	12000	12400	12500	11000		18
20				8600	9300	9800	10200	10600	10200		20
22				6900	7600	8100	8500	8800	9100		22
24					6300	6800	7200	7500	7700		24
26					5100	5700	6000	6300	6600		26
28						4700	5100	5400	5600		28
30						3900	4300	4600	4800		30
32							3600	3900	4100		32
34							3000	3300	3500		34
36								2700	3000		36
38								2300	2500		38
40									2100		40
42									1700		42
44											44
46											46
Telescoping status 伸缩模式											
Cylinder I 一号缸	0%	50%	100%	100%	100%	100%	100%	100%	100%	0%	Cylinder I 一号缸
Cylinder II 二号缸	0%	0%	0%	16.5%	33%	50%	67%	84%	100%	16.5%	Cylinder II 二号缸
Rope rate 钢丝绳倍率	12	10	7	6	5	4	3	3	3	6	Rope rate 钢丝绳倍率

Load Chart 50T Crane

Model STC500C5 | SERIES

Load Chart-Telescopic Boom 主臂性能表



EN

Unit: kg

Radius (m) 幅度	11.3	15.4	19.4	25.6	31.7	37.9	44	17.4	23.6	29.7	35.8	21.5	27.6	33.8	39.9	Radius (m) 幅度
2.5	50000															2.5
3	43000	35000														3
3.5	36000	34500	28000					23000								3.5
4	35000	34000	28000					23000	22000			23000				4
4.5	34000	33000	28000					23000	21500			23000				4.5
5	32500	31500	27200	23000				23000	21000			22500	19000			5
5.5	29700	28500	26000	22000	19000			23000	20000	16000		22000	18000			5.5
6	26500	26500	25000	21500	18000			23000	19800	15000		21000	17500	13000		6
7	22000	22000	22200	20500	16500	13000		22200	18500	14000	11000	20000	17000	12500		7
8	18600	19500	20000	18000	15000	12000		20000	17900	13000	10000	18500	16000	12000		8
9	14900	16500	17000	16000	14000	11500	8400	17000	16300	12500	9500	15700	15500	11300	8400	9
10		14400	14500	14800	13000	10800	8200	15600	15000	10800	9000	14000	14000	10800	8000	10
12		10300	10300	11300	11000	10000	7900	11900	12700	9400	7700	10800	10500	9600	7700	12
14			7500	8400	9000	8500	6800	9200	9600	8150	7000	8200	8300	8200	7000	14
16			5500	6300	7200	7200	5800		7600	7300	6200	6600	6600	6500	6200	16
18				4900	5700	5800	5200		6200	6400	5600	5200	5300	5400	5400	18
20				3900	4500	4700	4700			5300	5100		4400	4400	4500	20
22					3550	3600	4100			4350	4500		3500	3600	3700	22
24					2900	3000	3450			3700	3950			3000	3000	24
26					2300	2450	2800				3300			2500	2650	26
28						2000	2300				2850			2000	2150	28
30						1650	1900				2400				1800	30
32						1300	1500								1450	32
34							1200								1250	34
36							950									36
38																38
40																40
Telescoping status (%) 伸缩模式																
Cylinder I 油缸 I	0	50	100	100	100	100	100	0	0	0	0	50	50	50	50	Cylinder I 油缸 I
Cylinder II 油缸 II	0	0	0	25	50	75	100	25	50	75	100	25	50	75	100	Cylinder II 油缸 II
Rope rate 钢丝绳倍率	11	10	8	6	5	4	3	6	6	4	3	6	5	4	3	Rope rate 钢丝绳倍率

Remark:

Wind speed ≤32km/h, safe work.

Wind speed 33-47km/h, load shall be reduced to 40% of rated capacity.

Wind speed ≥48km/h, boom is not allowed to luff up.

备注:

风速 ≤32km/h, 安全风速。

风速 33-47km/h, 允许按额定 40% 作业。

风速 ≥48km/h, 禁止起臂。

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Safety Measures

- **Weather Conditions:** Monitor weather conditions and halt operations if winds exceed safe limits or if there is poor visibility.
- **Communication:** Use clear and standardized hand signals or radios for communication between the crane operator and signal person.
- **Exclusion Zone:** Establish an exclusion zone around the lifting area and ensure it is free of unauthorized personnel.

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4 Resources and logistics

4.1 Materials & tools

1	BOTTOM JACK	43	BITUMEN PAINT
2	TOP JACK	44	DOWEL BAR
3	UNIVERSAL CLAMP	45	PVC PIPE 25MM DIA
4	VERTICAL STANDARD	46	AIR COMPRESSOR
5	LEDGER	47	WATER PUMP
6	WOODEN BEAM	48	GRINDING MACHINE
7	ALUMINIUM BEAM	49	GRINDING STEEL CUTTING DISC
8	SCAFFOLDING PIPE	50	GRINDING DIAMOND DISC
9	PIPE CLAMP	51	PLYWOOD CUTTING MACHINE
10	LADDER	52	PICK AXE
11	ALUMINIUM PORTABLE STEP LADDER	53	HANDWOOD SHAW
12	SPIGOT CONNECTOR	54	HAND SHOWEL
13	DOUBLE BASE PLATE	55	MASKING TAPE
14	DOUBLE COUPLER	56	KNIPEX CUTTER
15	RIVERT PIN	57	BINDING WIRE
16	SOLDIER	58	CLEANING BRUSH
17	WOOD PLANK	59	SILICON
18	PUSH PULL JACK	60	DRILLING MACHINE
19	PLYWOOD BROWN	61	DRILL BIT
20	WOOD BATTERN	62	HILTI MACHINE
21	CONCRETE NAIL	63	HAND GLOVES
22	WOOD NAIL	64	TIE ROD
23	HAMMER WITH NAIL REMOVER	65	PVC PIPE
24	HAMMER	66	LINE DORI
25	SPRIT LEVEL	67	PVC BUCKET
26	CROWBAR	68	MASTER EMACO 301
27	GENERATOR	69	TAURPAULIN SHEET
28	VIBRATOR	70	NUT & PLATE FOR TIE ROD
29	VIBRATOR NEEDLE	71	WATER BARRIER
30	TOWER LIGHT	72	CONCRETE RACKER
31	FOCUS LIGHT	73	GUNNY BAG
32	ELECTRICAL CABLE	74	POLYETHYLENE SHEET
33	HAND ROLLER	75	POLYETHYLENE SHEET
34	MASTERKURE 106J	76	MASON TROWEL
35	SIKA LATEX (BONDING AGENT)	77	MASON TROWEL (SQUARE TYPE)
36	MEASUREMENT TAPE	78	SCRAPER
37	WATER STOPPER	79	GUMBOOT
38	WATER STOPPER WELDING KNIEF	80	CHAMFER
39	INJECTABLE HOSE	81	CHAMFER NAIL
40	SWELLING BAR	82	ROLLER BRUSH
41	FILLIR BOARD 20MM	83	STEEL BRUSH
42	FILLIR BOARD 10MM	84	

4.2 Logistics

- The logistics team shall facilitate the day-to-day logistics activities to increase productivity and performance.
- The logistics team shall update the logistics plan if any change is required based on site conditions.

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- and new areas handed over by DGCL.
- A separate logistics layout for the work location will be updated and submitted along with
- pedestrian/traffic/storage/signages/etc.

Logistics Site access for the Pump station area will be as following:



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4.3 Plant and Equipment

Items	Specific Requirements
Testing Equipment for Concrete Slump & Temperatures.	For checking concrete sample at site (workability, Mix, Temp etc.)
Concrete cylinders	Taking sample for test
Concrete vibrators	For Concrete compaction
Concrete Mixer & Pump	For concrete Shifting & Placing
Level instruments	For level checking
Trowel	For Finishing of concrete
Wheelbarrow	For taking sample
Cut saw	For Formworks and support
Hammers & crowbar	For Formworks and support
Water Tanker	For Curing
Scaffolding	For Formworks and support
Generator	For small electrical tools
Air Compressor	For Cleaning
Roller & Plat Compactor	For Bed Preparation
Aluminum Bar & other tools required for finishing of concrete	For concrete finishing and leveling

4.4 Manpower

Site Civil Engineer	1
QC Civil Engineer	1
Surveyor	1
Safety officer	1
Supervisor	1
Foreman	1
Workers	1
Driver/operators	1

4.5 Material and PPE Resources

PPE		
1	Safety vest	Yes
2	Safety shoes	Yes
3	Gum boots	
4	Safety gloves	Yes
5	Safety goggles	Yes
6	Safety helmet	Yes
7	Safety ear plugs	Yes
8	Disposable Coverall	
9	Safety harness	
10	Welfare facility (Rest Shelter, Water & Toilet)	

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5. Occupational Health & Safety Requirements

5.1 General Safety Requirements

5.1.1 Lockout-Tagout (LOTO)

- A suitable isolation process, including LOTO, shall be implemented prior to performing activities on live systems which could cause injury from unexpected start-up, energization, or releasing stored energy. Isolations can only be performed by formally authorized persons under their scope of work.
- Contractors shall provide such authorization form to GE for their concerned staff to ensure competency of all personnel required to work under the AICC LOTO Program shall receive training to ensure they understand their expectations concerning LOTO.
- All energy sources shall be disconnected and LOTO in place prior to any servicing or maintenance activity.
- All locks used for LOTO shall be unique to LOTO and unlike other locks used on site. All LOTO locks shall have only one key. Each location shall have either a business or site-specific written LOTO program. The following requirements shall be included for ensuring safe work on equipment or systems that may be energized:
 - Schedule of required isolations developed and selection of LOTO devices required, considering positive isolation as the preferred option as far as reasonably practicable and selecting energy isolation points to be as close as possible to the workplace.
 - Identification of personnel assigned to LOTO roles on site, with clearly defined responsibilities for each role.
 - Where equipment or system responsibility is transferred between construction and commissioning teams or other entities such as customers, a written procedure shall be developed to define interfaces and controls required to always ensure that energy control is under the responsibility of one person/entity.
 - Planning and preparation of the LOTO equipment, ensuring all energy-isolating devices and systems involved in isolation schemes are uniquely identified and clearly labeled at site. When isolation activities concern hazardous substances, a contingency plan (or rescue plan) shall be prepared and include any additional measures beyond the site standard emergency arrangements for loss-of-containment incidents.
 - Installation of isolation(s), securing them with appropriate LOTO equipment. Safety locks used by an authorized person for isolation purposes shall be operated by a unique key. No spare keys must be present at the site.

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- Stored energy removal/control through draining, venting, purging, flushing, earthing, blocking, or any other suitable method. This may include the use of a properly engineered chock or scotch should a risk for movement due to gravity remain.
- Verification, testing, and monitoring effectiveness of the isolation. Prior to starting work, the absence of energy must be verified using suitable equipment or a process adapted to the machine and the kind of energy to be checked. For electrical isolation testing, the authorized person shall check that the test device including all leads, probes, and connectors is suitably rated for the voltages and currents which may be present on the system under test. For low voltages, two-poles voltage detectors complying with IEC 61243-3 rated as a minimum as 600V CAT IV according to NF EN IEC 61010-1 shall be used. The use of multi meters is forbidden for such checks. The device used for proving dead should itself be proved immediately before and after testing. If there is no means for effective testing, the system shall be considered as potentially live, and this shall be reflected in the risk assessment and method statement.
- Any person who may be affected by the isolation of a system shall be notified that LOTO process is going to be utilized and the reason why.
- Unauthorized removal of a LOTO device is considered by AICC as a high-risk deviation and will therefore be treated accordingly, if appropriate the site disciplinary process will be applied.
- All Utilities' main connection like electricity and water will be disconnected by the DGDA, AICC will work after removed main line from the Electrical & Water meter by (DGDA).

5.2 Training & Competency

5.2.1 Training and Briefings

- The site supervisor will ensure that all operatives attend the mandatory site-specific Induction training course before mobilization on-site/commencement of operations.
- Also, the following mandatory training courses will be undertaken.

5.2.2 Training

INHOUSE TRAINING	Team	THIRD PARTY TRAINING	team
HSE Induction Training	All	Equipment operators	Backhoe, JCB, Boom loader
Heat stress	All		

5.2.3 Briefings

- General Weekly Toolbox Meeting
- STARRT Card
- Housekeeping

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- Unsafe Behavior
- Emergency Response Procedure
- Toolbox meeting

5.2.4 COSHH Requirements for Concrete

Health warning: Caution for everybody handling cementitious material, concrete and screed.

- Fresh concrete or screed can cause serious burns to skin and eyes as well as skin disease and dermatitis. Do not swallow.
- Keep out of reach of children and animals. Immediately wash off any fresh cementitious material from the skin and thoroughly washout any affected eye. If swallowed do not induce vomit but seek medical advice. Make sure everyone handling or meeting fresh concrete or screed receives this warning! Concrete is highly alkaline and can cause burns. Prolonged skin contact can cause severe irritation.
- Wear protective clothing (such as goggles, gloves, impervious boots, trousers and long-sleeved clothing). Immediately remove any clothing or item saturated with cementitious material and wash thoroughly before reuse.
- Seek medical attention if symptoms persist or in the case of doubt.
- We strongly recommend that you wear non-porous clothing with long sleeves plus gloves, boots and knee pads.
- If concrete splashes onto your skin or into your eyes, wash without delay.
- Remember, concrete is heavy. Do not risk hurting yourself by attempting to move more than you can comfortably wheel in a barrow.

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COSH Assessment

Material / Company Name : Saudi Ready Mix AICC

Chemical Name / Ingredient : Cement

Classification as Per MSDS

								
Corrosive	Explosive	Harmful to Environment	Flammable	Compressed Gas	Irritant	Harmful to health	Oxidizing Flammable	Toxic
☒	☐	☒	☐	☐	☒	☒	☐	☒

Measurements and Attitude

						
Dust mask	Gloves	Face Shield	Overalls	Safety Boot	Goggles	Respirator

Hazard Type

Gas	Vapour	Mist	Dust	Liquid	Solid	Fibre	Biological
☐	☐	☐	☒	☒	☒	☐	☐

Route of Exposure

Inhalation	Skin	Eyes	Ingestion	Injection	Other
☒	☒	☒	☒	☐	☐

Control Measures	Do not inhale. Avoid contact with the skin, eyes and clothing. Avoid exposure - obtain special instructions before use. Handle in accordance with good building materials hygiene and safety practice. Wearing of closed work clothing is recommended. When using, do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.	
First Aid	Eyes: Immediately flush eyes thoroughly with water. Continue flushing eye for at least 15 minutes, including under lids, to remove all particles. Call physician immediately. Skin: Wash skin with cool water and pH-neutral soap or a mild detergent intended for use on skin. Seek medical treatment in all cases of prolonged exposure to wet cement, cement mixtures, liquids from fresh cement products, or prolonged wet skin exposure to dry cement. Inhalation of Airborne Dust: Remove to fresh air. Seek medical help if coughing and other symptoms do not subside. ("Inhalation" of gross amounts of Portland cement requires immediate medical attention.) Ingestion: Do not induce vomiting. If conscious, have the victim drink plenty of water and call a physician immediately.	
Fire	Information about fire - and explosion protection: No special measures required.	
Environment	Environmental precautions: Inform respective authorities in case of seepage into water course or sewage system. Change the Ph of soil so should not use on bare ground	

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	Information concerning hazards for human and environment: The product has to be labelled in accordance with applicable regulations.	
Storage	Keep Portland cement <i>dry</i> until used. Normal temperatures and pressures do not affect the material. Promptly remove dusty clothing or clothing which is wet with cement fluids and launder before reuse. Wash thoroughly after exposure to dust or wet cement mixtures or fluids.	
Disposal	Dispose of waste material according to local, state and federal regulations. (Since Portland cement is stable, uncontaminated material may be saved for future use.) Dispose of bags in an approved landfill or incinerator.	
Training	All the Operatives including Warehouse workers should attend MSDS/COSHH training conducted by HSSE Department prior to use/handle the product.	
Emergency Action	Call _____ In Case of Accidents. (In all cases, please seek immediate medical advice)	

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MATERIAL SAFETY DATA SHEET (MSDS) OF OUR CEMENT PRODUCTS

Section 1 - IDENTIFICATION

Product names for which this MSDS is to be used for:

1. OPC – Ordinary Portland Cement
2. SRC – Sulfate Resistance Cement
3. White Cement
4. Finishing Cement

OPC, SRC & White Cement consists of finely ground Portland cement clinker mixed with a small amount of calcium sulfate dihydrate (Natural Gypsum) while Finishing Cement also contains Natural Limestone in addition to above. All above stated products are Hydraulic in nature and are classified under the category of Portland Cement.

Manufacturer:

Saudi White Cement Company, P.O.Box No. 17775, Riyadh, Kingdom of Saudi Arabia

Chemical family:

The major compounds in our entire Product are Calcium Silicate and other calcium compounds containing iron and aluminum. Major compounds are:

(CaO)₃(SiO₂) Tricalcium silicate
 (CaO)₂(SiO₂) Dicalcium silicate
 (CaO)₃(Al₂O₃) Tricalcium aluminate
 (CaO)₄(Al₂O₃Fe₂O₃) Tetra calcium aluminoferrite
 CaSO₄ * 2H₂O Calcium sulfate dihydrate, that is, Natural Gypsum
 CaCO₃ Only in Finishing Cement

Section 2- HAZARDOUS COMPONENTS

Since Portland cement is a powder, thus it contains dust which may be hazardous, under certain circumstances.

Information about the hazardous dust is:

It contains approximately 10mg total dust/m³
 It contains approximately 50million particles/ft³
 It contains approximately 5mg respirable dust/m³
 It contains approximately 0.10mg respirable quartz dust/m³

Trace Elements:

Portland cement is made from materials mined from the earth and is processed using energy provided by fuels. Trace amounts of naturally occurring potentially harmful chemicals might be detected during chemical analysis. For example, our all products contain up to 0.75% insoluble residue, some of which may be free crystalline silica. Other trace constituents may include calcium oxide (also known as free lime or quick lime), free magnesium oxide, potassium and sodium sulfate compounds, chromium compounds, and nickel compounds.

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Material / Company Name : Saudi ready Mix - AICC

Chemical Name / Ingredient : **AR Crete D-8**

Classification as Per MSDS

								
Corrosive	Explosive	Harmful to Environment	Flammable	Compressed Gas	Irritant	Harmful to health	Oxidizing Flammable	Toxic
☐	☐	☒	☐	☐	☒	☒	☐	☒

Measurements and Attitude

						
Dust mask	Gloves	Face Shield	Overalls	Safety Boot	Goggles	Respirator

Hazard Type

Gas	Vapour	Mist	Dust	Liquid	Solid	Fibre	Biological
☐	☐	☐	☐	☒	☐	☐	☐

Route of Exposure

Inhalation	Skin	Eyes	Ingestion	Injection	Other
☒	☒	☒	☒	☐	☐

Control Measures	Do not inhale. Avoid contact with the skin, eyes and clothing. Avoid exposure - obtain special instructions before use. Handle in accordance with good building materials hygiene and safety practice. Wearing of closed work clothing is recommended. When using, do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.	
First Aid	After inhalation: Take affected persons into fresh air and keep quiet. Seek immediate medical advice. After skin contact: Immediately wash with water and soap and rinse thoroughly. Seek immediate medical advice. After eye contact: Rinse opened eye for several minutes under running water. Seek immediate medical advice.	
Fire	Suitable extinguishing agents: CO2, powder, or water spray. Fight larger fires with water spray. Protective equipment: Wear self-contained respiratory protective device. Additional information: Collect contaminated firefighting water separately. It must not enter the sewage system	
Environment	Measures for environmental protection: Inform respective authorities in case of seepage into water course or sewage system. Dilute with plenty of water. Collect contaminated firefighting water separately. It must not enter the sewage system	
Storage	STORAGE Information about storage in one common storage facility: No special measures required. Further information about storage conditions: Protect from frost. Store in cool, dry conditions in well-sealed receptacles.	

	AECOM	Contract No.		DD-2023-118		Jacobs	
		Project		Diriyah Gate II Infrastructure Package 1			
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		Doc. No.		IP-INF-054-0000-AIC-MST-CE-000032	Rev: 02		

Disposal	Observe national and local legal requirements. Residues should be disposed of in the same manner as the substance/product. Contaminated packaging should be emptied as far as possible; Must not be disposed together with household garbage. Do not allow product to reach sewage system.	
Training	All the Operatives including Warehouse workers should attend MSDS/COSHH training conducted by HSSE Department prior to use/handle the product.	
Emergency Action	Call _____ In Case of Accidents. (In all cases, please seek immediate medical advice)	



Safety Data Sheet according to 1907/2006/EC, Article 31

1 IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY

PRODUCT DETAILS

Trade name: ARCrete D-8
Manufacturer/supplier:
Arkaz AlSharq Building Material
Dammam 2nd Industrial area, Road 135
P.O. Box 2572, Khobar 31952
Saudi Arabia
Tel: +966 3 812 6543
Fax: +966 3 812 6541 www.arkaz.com
Further information obtainable from: info@arkaz.com
Information in case of emergency:
Civil Defense: 997-Fire, 999-Police, 998-Ambulance DEWA: 991-Electricity & Water
JAFZA: 881622-Fire, 8816846-Security DPA, 8833111-Emergency
+44 (0)1235 239 670 / +44 (0)1235 239 671 (Arabic - Dedicated Number)

2 HAZARDS IDENTIFICATION

Information concerning particular hazards for human and environment:

The product does not have to be labelled due to the calculation procedure of the "General Classification guideline for preparations of the EU" in the latest valid version.

3 COMPOSITION/INFORMATION ON INGREDIENTS

CHEMICAL CHARACTERIZATION

Description: For further information, please refer to the product technical data sheet.
Dangerous components: Not applicable.

4 FIRST-AID MEASURES

After inhalation:

Take affected persons into fresh air and keep quiet. Seek immediate medical advice.

After skin contact:

Immediately wash with water and soap and rinse thoroughly. Seek immediate medical advice.

After eye contact:

Rinse opened eye for several minutes under running water.
Seek immediate medical advice.

Safety Data Sheet according to 1907/2006/EC, Article 31

Trade name: ARCrete D-8
(Contd. of page 1)

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After swallowing:

Wash out mouth with water.

Do not induce vomiting; call for medical help immediately.

Never give anything by mouth to an unconscious person.

5 FIRE-FIGHTING MEASURES

Suitable extinguishing agents: CO₂, powder or water spray. Fight larger fires with water spray. Protective equipment:

Wear self-contained respiratory protective device.

Additional information:

Collect contaminated firefighting water separately. It must not enter the sewage system.

6 ACCIDENTAL RELEASE MEASURES

Person-related safety precautions: Wear protective equipment. Keep unprotected persons away.

Measures for environmental protection:

Inform respective authorities in case of seepage into water course or sewage system. Dilute with plenty of water.

Measures for cleaning/collecting:

Absorb with liquid-binding material (sand, diatomite, acid binders, and universal binders).

Send for recovery or disposal in suitable receptacles.

Additional information: No dangerous substances are released.

7 HANDLING AND STORAGE

HANDLING

Information for safe handling:

Open and handle receptacle with care. Ensure good interior ventilation. Store in original containers.

Information about fire - and explosion protection: No special measures required.

STORAGE

Information about storage in one common storage facility: No special measures required.

Further information about storage conditions:

Protect from frost.

Store in cool, dry conditions in well-sealed receptacles.

Safety Data Sheet according to 1907/2006/EC, Article 31

Trade name: ARCRETE D-8

8 EXPOSURE CONTROLS/PERSONAL PROTECTION

Additional information about design of technical facilities: No further data; see item 7.

Ingredients with limit values that require monitoring at the workplace:

The product does not contain any relevant quantities of materials with critical values that have to be monitored at the workplace.

Additional information: Based on the lists valid at the date of MSDS creation.

PERSONAL PROTECTIVE EQUIPMENT

General protective and hygienic measures: Avoid contact with the eyes and skin.

Respiratory protection: In case of brief exposure or low pollution use respiratory filter device.

Protection of hands: Protective gloves. Material of gloves: PVC gloves.

Eye protection:

Safety glasses. Face protection.

Body protection: Protective work clothing.

9 PHYSICAL AND CHEMICAL PROPERTIES

GENERAL INFORMATION

Form: Liquid.

Color: Brown.

Odor: Characteristic.

Change in conditions: -

Melting point/Melting range:

Undetermined.

Boiling point/Boiling range:

> 100°C

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		Project Diriyah Gate II Infrastructure Package 1					
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		Doc. No. IP-INF-054-0000-AIC-MST-CE-000032		Rev: 02	Date: 14-01-2025		

Flash point:

> 100°C

Self-igniting:

Product is not self-igniting.

Danger of explosion:

Product does not present an explosion hazard.

Density at 20°C:

~1,15 g/cm³

Solubility in/Miscibility with:-

Fully miscible.

Water:

pH-value at 20°C:

~6.5

Additional information:

These are typical values and do not constitute a specification.

10 STABILITY AND REACTIVITY

Thermal decomposition / conditions to be avoided: No decomposition if used according to specifications.

Materials to be avoided:

Dangerous reactions: No dangerous reactions known.

Safety Data Sheet according to 1907/2006/EC, Article 31

Trade name: ARCRETE D-8

Dangerous decomposition products:

Carbon monoxide and carbon dioxide.

Sulphur oxides (SO_x).

11 TOXICOLOGICAL INFORMATION

ACUTE TOXICITY

Primary irritant effect:-

On the skin: May be irritating to the skin on prolonged contact. On the eye: May be irritating to the eyes.

Additional toxicological information:

The product is not subject to classification according to the calculation method of the General EU Classification Guidelines for Preparations as issued in the latest version.

When used and handled according to specifications, the product does not have any harmful effects to our experience and the information provided to us.

12 ECOLOGICAL INFORMATION

ADDITIONAL ECOLOGICAL INFORMATION

General notes:

Water hazard class 1 (German Regulation) (Self-assessment): slightly hazardous for water.

Do not allow undiluted product or large quantities of it to reach ground water, water course or sewage system.

13 DISPOSAL CONSIDERATIONS

PRODUCT

European waste catalogue:

07 00 00 WASTES FROM ORGANIC CHEMICAL PROCESSES

07 01 00 wastes from the manufacture, formulation, supply and use (MFSU) of basic organic chemicals 07 01 99 wastes not otherwise specified

UNCLEANED PACKAGING

Recommendation: Disposal must be made according to official regulations.

14 TRANSPORT INFORMATION

LAND TRANSPORT ADR/RID (cross-border)

ADR/RID class: -

-

Inland shipping ADN:

ADN/R Class: -

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Safety Data Sheet according to 1907/2006/EC, Article 31

Trade name: ARCRETE D-8

MARITIME TRANSPORT IMDG

IMDG Class: -

Marine pollutant: No.

AIR TRANSPORT ICAO-TI AND IATA-DGR

ICAO/IATA Class: -

15 REGULATORY INFORMATION

Labelling according to EU guidelines: Observe the general safety regulations when handling chemicals.

National regulations:

Water hazard class: WGK 1 (VwVwS Anhang 4 vom 19.05.1999): slightly hazardous for water.

Other regulations, limitations, and prohibitive regulations:

All components EINECS listed or exempt.

16 OTHER INFORMATION

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

Department issuing MSDS:

Arkaz AlSharq Building Materials Technical Department

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		Doc. No.	IP-INF-054-0000-AIC-MST-CE-000032	Rev: 02	Date: 14-01-2025		

Material / Company Name : Saudi ready Mix - AICC

Chemical Name / Ingredient : FLUIDUM PC175

Classification as Per MSDS

								
Corrosive	Explosive	Harmful to Environment	Flammable	Compressed Gas	Irritant	Harmful to health	Oxidizing Flammable	Toxic
☐	☐	☒	☐	☐	☒	☒	☐	☒

Measurements and Attitude

						
Dust mask	Gloves	Face Shield	Overalls	Safety Boot	Goggles	Respirator

Hazard Type

Gas	Vapour	Mist	Dust	Liquid	Solid	Fibre	Biological
☐	☐	☐	☐	☒	☐	☐	☐

Route of Exposure

Inhalation	Skin	Eyes	Ingestion	Injection	Other
☒	☒	☒	☒	☐	☐

Control Measures	Do not inhale. Avoid contact with the skin, eyes and clothing. Avoid exposure - obtain special instructions before use. Handle in accordance with good building materials hygiene and safety practice. Wearing of closed work clothing is recommended. When using, do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.	
First Aid	After inhalation: Take affected persons into fresh air and keep quiet. Seek immediate medical advice. After skin contact: Immediately wash with water and soap and rinse thoroughly. Seek immediate medical advice. After eye contact: Rinse opened eye for several minutes under running water. After swallowing: Wash out mouth with water. Do not induce vomiting; call for medical help immediately. Never give anything by mouth to an unconscious person.	
Fire	Information about fire - and explosion protection: No special measures required.	
Environment	Environmental precautions: Inform respective authorities in case of seepage into water course or sewage system. Dilute with plenty of water. The product has to be labelled in accordance with applicable regulations.	
Storage	Information about storage in one common storage facility: No special measures required. Further information about storage conditions: Protect from frost. Store in cool, dry conditions in well-sealed receptacles.	

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Disposal	Observe national and local legal requirements. Residues should be disposed of in the same manner as the substance/product. Contaminated packaging should be emptied as far as possible; Must not be disposed together with household garbage. Do not allow product to reach sewage system.	
Training	All the Operatives including Warehouse workers should attend MSDS/COSHH training conducted by HSSE Department prior to use/handle the product.	
Emergency Action	Call _____ In Case of Accidents. (In all cases, please seek immediate medical advice)	



Material Safety Data Sheet

1- Identification of the substance/mixture and of the company/undertaking

Application of the substance/the preparation: Concrete Admixtures

Details of the supplier of the safety data sheet: Manufacturer/supplier:

Arkaz AlSharq Building Materials Damman 2nd Industrial Area, Road 135

P. O. Box 2572, Khobar 31952

Kingdom of Saudi Arabia

Tel: +966 138200089

Fax: +966 138126541

Email: info@arkaz.com

www.arkaz.com

Further information obtainable from: info@arkaz.com

Emergency telephone number:

+44 (0)1235 239 670 / +44 (0)1235 239 671 (Arabic - Dedicated Number)

998-Fire, 999-Police, 997-Ambulance

8122161-Al Mana Clinic

2 Hazards identification

Classification of the substance or mixture:

Information concerning hazards for human and environment:

The product has to be labelled in accordance with applicable regulations.

Label elements:

Labelling according to EU guidelines:

Observe the general safety regulations when handling chemicals.

The product has been classified and marked in accordance with EU Directives / Ordinance on Hazardous Materials.

Code letter and hazard designation of product:

Hazard-determining components of labelling:

a mixture of -5-chloro-2-methyl-4-isothiazolin-3-one [EC No 247-500-7] and 2-methyl-2 H -isothiazol-3-one [EC No 220-239-6] (3:1)

Risk phrases:

43 May cause sensitization by skin contact.

Safety data sheet According to 1907/2006/EC, Article 31 Version number 1.1 Printing Date: 1.1.2019 Revision: 1.1.2020

Trade name: FLUIDUM PC175

Safety phrases:

2 Keep out of the reach of children.

23 Do not breathe gas/fumes/vapor/spray.

24 Avoid contact with skin.

37 Wear suitable gloves.

46 If swallowed, seek medical advice immediately and show this container or label.

56 Dispose of this material and its container to hazardous or special waste collection point.

60 This material and its container must be disposed of as hazardous waste.

Other hazards:

Results of PBT and vPvB assessment: PBT: Not applicable.

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vPvB: Not applicable.

Chemical characterization: Mixtures: Description:

Mixture of substances listed below with non-hazardous additions.

For further information, please refer to the product technical data sheet. **Dangerous components:**

55965-84-9 a mixture of: 5-chloro-2-methyl-4- 0.0-1.0%
isothiazolin-3-one [EC No 247-500-
7] and 2-methyl-2 H - isothiazol-3-
one [EC No 220-239-6] (3:1)
T R23/24/25
C R34
Xi R43
N R50/53

4 First aid measures

Description of first aid measures:

After inhalation:

Take affected persons into fresh air and keep quiet. Seek immediate medical advice.

After skin contact:

Immediately wash with water and soap and rinse thoroughly. Seek immediate medical advice.

After eye contact: Rinse opened eye for several minutes under running water.

After swallowing:

Wash out mouth with water.

Do not induce vomiting; call for medical help immediately. Never give anything by mouth to an unconscious person.

Safety data sheet According to 1907/2006/EC, Article 31

Trade name: FLUIDUM PC175

Information for doctor:

Most important symptoms and effects, both acute and delayed: No further relevant information available.

Indication of any immediate medical attention and special treatment needed:

No further relevant information available.

5 Firefighting measures fighting.

Extinguishing media:

Suitable extinguishing agents: CO₂, powder, or water spray. Fight larger fires with water spray.

Special hazards arising from the substance or mixture: No further relevant information available.

Advice for firefighters:

Protective equipment: Wear self-contained respiratory protective device.

Additional information:

Collect contaminated firefighting water separately. It must not enter the sewage system.

6 Accidental release measures

Personal precautions, protective equipment, and emergency procedures:

Wear protective equipment. Keep unprotected persons away.

Environmental precautions:

Inform respective authorities in case of seepage into water course or sewage system. Dilute with plenty of water.

Methods and material for containment and cleaning up:

Send for recovery or disposal in suitable receptacles.

Dispose contaminated material as waste according to item 13.

Reference to other sections

See Section 7 for information on safe handling.

See Section 8 for information on personal protection equipment. See Section 13 for disposal information.

7 Handling and storage

HANDLING

Precautions for safe handling:

Open and handle receptacle with care. Prevent formation of aerosols.

Ensure good interior ventilation. Store in original containers.

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Material / Company Name : Saudi Ready-mix - AICC
 Chemical Name / Ingredient : FLUIDUM R-123

Classification as Per MSDS

								
Corrosive	Explosive	Harmful to Environment	Flammable	Compressed Gas	Irritant	Harmful to health	Oxidizing Flammable	Toxic
☒	☐	☐	☐	☐	☒	☒	☐	☒

Measurements and Attitude

						
Dust mask	Gloves	Face Shield	Overalls	Safety Boot	Goggles	Respirator

Hazard Type

Gas	Vapour	Mist	Dust	Liquid	Solid	Fibre	Biological
☐	☐	☐	☐	☒	☐	☐	☐

Route of Exposure

Inhalation	Skin	Eyes	Ingestion	Injection	Other
☒	☒	☒	☒	☐	☐

Control Measures	Do not inhale. Avoid contact with the skin, eyes and clothing. Avoid exposure - obtain special instructions before use. Handle in accordance with good building materials hygiene and safety practice. Wearing of closed work clothing is recommended. When using, do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.	
First Aid	After inhalation: Take affected persons into fresh air and keep quiet. Seek immediate medical advice. After skin contact: Immediately wash with water and soap and rinse thoroughly. Seek immediate medical advice. After eye contact: Rinse opened eye for several minutes under running water. After swallowing: Wash out mouth with water. Do not induce vomiting; call for medical help immediately. Never give anything by mouth to an unconscious person.	
Fire	Information about fire - and explosion protection: No special measures required.	
Environment	Environmental precautions: Inform respective authorities in case of seepage into water course or sewage system. Dilute with plenty of water. Information concerning hazards for human and environment: The product has to be labelled in accordance with applicable regulations.	
Storage	Conditions for safe storage, including any incompatibilities: STORAGE. Information about storage in one common storage facility:	

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	Ensure good interior ventilation. Store in original containers. Further information about storage conditions: Protect from frost	
Disposal	Observe national and local legal requirements. Residues should be disposed of in the same manner as the substance/product. Contaminated packaging should be emptied as far as possible; Must not be disposed together with household garbage. Do not allow product to reach sewage system.	
Training	All the Operatives including Warehouse workers should attend MSDS/COSHH training conducted by HSSE Department prior to use/handle the product.	
Emergency Action	Call _____ In Case of Accidents. (In all cases, please seek immediate medical advice)	



Material Safety Data Sheet

Revision: 1.7.2019

1 Identification of the substance/mixture and of the company/undertaking

Product identifier:

Trade name: *Fluidum R-123*

Application of the substance/the preparation: Concrete Admixtures **Details of the supplier of the safety data sheet:**

Manufacturer/supplier:

Arkaz AlSharq Building Materials Dammam 2nd Industrial Area, Road 135

P. O. Box 2572, Khobar 31952

Kingdom of Saudi Arabia

Tel: +966 138086970

Fax: +966 138126541

Email: info@arkaz.com

www.arkaz.com

Further information obtainable from: info@arkaz.com

Emergency telephone number:

+44 (0)1235 239 670 / +44 (0)1235 239 671 (Arabic - Dedicated Number)

998-Fire, 999-Police, 997-Ambulance

8122161-Al Mana Clinic

2 Hazards identification

Classification of the substance or mixture:

Information concerning particular hazards for human and environment:

The product has to be labelled in accordance with applicable regulations.

Label elements:

Labelling according to EU guidelines:

Observe the general safety regulations when handling chemicals.

The product has been classified and marked in accordance with EU Directives / Ordinance on Hazardous Materials.

Code letter and hazard designation of product:

Hazard-determining components of labelling:

a mixture of -5-chloro-2-methyl-4-isothiazolin-3-one [EC No 247-500-7] and 2-methyl-2 H -isothiazol-3-one [EC No 220-239-6] (3:1)

Risk phrases:

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43 May cause sensitization by skin contact.

DRPage 2/7 Safety data sheet According to 1907/2006/EC, Article 31 Version number 1.1 Printing Date: 1.7.2019

Revision: 1.7.2019

Trade name: FLUIDUM R-123

Safety phrases:

2 Keep out of the reach of children.

23 Do not breathe gas/fumes/vapor/spray.

24 Avoid contact with skin.

37 Wear suitable gloves.

46 If swallowed, seek medical advice immediately and show this container or label.

56 Dispose of this material and its container to hazardous or special waste collection point.

60 This material and its container must be disposed of as hazardous waste.

Other hazards:

Results of PBT and vPvB assessment: PBT: Not applicable.

vPvB: Not applicable.

3 Composition/information on ingredients Chemical characterization: Mixtures: Description:

Mixture of substances listed below with non-hazardous additions.

For further information, please refer to the product technical data sheet. **Dangerous components:**

55965-84-9	a mixture of: 5-chloro-2-methyl-4-isothiazolin-3-one [EC No 247-500-7] and 2-methyl-2 H - isothiazol-3-one [EC No 220-239-6] (3:1)	0.0-1.0%
	T R23/24/25	
	C R34	
	Xi R43	
	N R50/53	

First aid measures

Description of first aid measures:

After inhalation:

Take affected persons into fresh air and keep quiet. Seek immediate medical advice.

After skin contact:

Immediately wash with water and soap and rinse thoroughly. Seek immediate medical advice.

After eye contact: Rinse opened eye for several minutes under running water.

After swallowing:

Wash out mouth with water.

Do not induce vomiting; call for medical help immediately. Never give anything by mouth to an unconscious person.

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Information for doctor:

(Contd. of page 2)

Most important symptoms and effects, both acute and delayed: No further relevant information available.

Indication of any immediate medical attention and special treatment needed:

No further relevant information available.

5 Firefighting measures

Extinguishing media:

Suitable extinguishing agents: CO2, powder or water spray. Fight larger fires with water spray.

Special hazards arising from the substance or mixture: No further relevant information available.

Advice for firefighters:

Protective equipment: Wear self-contained respiratory protective device.

Additional information:

Collect contaminated firefighting water separately. It must not enter the sewage system.

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6 Accidental release measures

Personal precautions, protective equipment and emergency procedures:

Wear protective equipment. Keep unprotected persons away.

Environmental precautions:

Inform respective authorities in case of seepage into water course or sewage system. Dilute with plenty of water.

Methods and material for containment and cleaning up:

Send for recovery or disposal in suitable receptacles.

Dispose contaminated material as waste according to item 13.

Reference to other sections

See Section 7 for information on safe handling.

See Section 8 for information on personal protection equipment. See Section 13 for disposal information.

7 Handling and storage

HANDLING

Precautions for safe handling:

Open and handle receptacle with care. Prevent formation of aerosols.

Ensure good interior ventilation. Store in original containers.

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Information about fire - and explosion protection: No special measures required. **Conditions for safe storage, including any incompatibilities: STORAGE**

Information about storage in one common storage facility: No special measures required.

Further information about storage conditions:

Protect from frost.

Store in cool, dry conditions in well-sealed receptacles.

Specific end use(s): No further relevant information available.

(Contd. of page 3)

8 Exposure controls/personal protection

Additional information about design of technical facilities: No further data; see item 7.

Control parameters:

Ingredients with limit values that require monitoring at the workplace:

The product does not contain any relevant quantities of materials with critical values that have to be monitored at the workplace.

Additional information: Based on the lists valid at the date of MSDS creation.

Exposure controls:

PERSONAL PROTECTIVE EQUIPMENT

General protective and hygienic measures: Avoid contact with the eyes and skin. **Respiratory protection:** In case of brief exposure or low pollution use respiratory filter device. **Protection of hands:** Protective gloves.

Material of gloves: PVC gloves.

Body protection: Protective work clothing.

9 Physical and chemical properties

Information on basic physical and chemical properties:

GENERAL INFORMATION Appearance:

Form: Viscous Liquid.

Color: Brown.

Odor: Characteristic. **Odor threshold:** Not determined.

pH-value at 20°C: ~5.5

Change in conditions: -

Melting point/Melting range: Undetermined.

Boiling point/Boiling range: > 100°C

Flash point: > 100°C

Flammability (solid, gaseous): Not applicable.

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Ignition temperature:

Decomposition temperature: Not determined.

Self-igniting: Product is not self-igniting.

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Danger of explosion: Product does not present an explosion hazard.

EXPLOSION LIMITS

Lower: Not determined. Upper: Not determined. VOC Content: Not determined.

Vapor pressure: Not determined.

Density at 20°C: ~1.09 g/cm³

Relative density: Not determined. Vapor density: Not determined. Evaporation rate: Not determined. Solubility in/Miscibility with: -
Water: Fully miscible.

Segregation coefficient (n-octanol/water): Not determined.

VISCOSITY

Dynamic: Not determined.

Kinematic: Not determined.

Other information: No further relevant information available.

10 Stability and reactivity

Reactivity:

Chemical stability:

Thermal decomposition / conditions to be avoided: No decomposition if used according to specifications.

Conditions to avoid: No further relevant information available. Incompatible materials: No further relevant information available.

Hazardous decomposition products: Carbon monoxide and carbon dioxide.

11 Toxicological information Information on toxicological effects:

ACUTE TOXICITY

Primary irritant effect: -

On the skin: May be irritating to the skin on prolonged contact.

On the eye: May be irritating to the eyes.

Sensitization: Sensitization possible through skin contact.

Safety data sheet

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12 Ecological information

Toxicity:

AQUATIC TOXICITY No further relevant information available. Persistence and degradability: No further relevant information available. BEHAVIOUR IN ENVIRONMENTAL SYSTEMS Bioaccumulative potential: No further relevant information available.

Mobility in soil: No further relevant information available. ADDITIONAL ECOLOGICAL INFORMATION General notes:

Do not allow undiluted product or large quantities of it to reach ground water, water course or sewage system. Water hazard class 1 (German Regulation) (Self-assessment): slightly hazardous for water.

Results of PBT and vPvB assessment:

PBT: Not applicable.

vPvB: Not applicable.

Other adverse effects: No further relevant information available.

13 Disposal considerations Waste treatment methods: Recommendation:

Must not be disposed together with household garbage. Do not allow product to reach sewage system.

European waste catalogue:

07 00 00

WASTES FROM ORGANIC CHEMICAL PROCESSES

07 01 00

wastes from the manufacture, formulation, supply and use (MFSU) of basic organic chemicals

07 01 99

wastes not otherwise specified

UNCLEANED PACKAGING

Recommendation: Disposal must be made according to official regulations.

14 Transport information

UN-Number

ADR, ADN, IMDG, IATA Not applicable.

UN proper shipping name

ADR, ADN, IMDG, IATA Not applicable.

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Transport hazard class(es)

ADN

ADN/R Class: Not applicable.

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Packing group

ADR, IMDG, IATA Not applicable.

Environmental hazards:

Marine pollutant: No.

Special precautions for user: Not applicable.

Transport in bulk according to Annex II of

MARPOL73/78 and the IBC Code: Not applicable.

15 Regulatory information

National regulations:

Waterhazard class: WGK 1 (VwVwS Anhang 4 vom 19.05.1999): slightly hazardous for water.

Other regulations, limitations and prohibitive regulations: All components EINECS listed or exempt.

Chemical safety assessment: A Chemical Safety Assessment has not been carried out.

16 Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

Relevant phrases:

R23/24/25 Toxic by inhalation, in contact with skin and if swallowed. R34 Causes burns.

R43 May cause sensitization by skin contact.

R50/53 Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Department issuing MSDS:

Arkaz AlSharq Building Materials

DR

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6. Emergency Contact details

1. Emergency Telephone Numbers		
Service	Location/Authority	Contact Number
Police	Emergency	999/911
Civil Defense	Emergency	998 /911
Ambulance	Emergency	997 /911
Local Hospital	Interhealth Hospital	8001110666
Local Medical Centre	Markaz Sahi Alraed	0114805889
Utilities		
Title	Name	Contact Number
Electricity	Riyadh City Elec Dept	933 / 0118073332
Water	National Water Company	939 /920001744
Gas		+966114664999
CSC/PMC & DGCL - Project Management Team		
Title	Name	Contact Number
DGCL Emergency Management Area Leader	Abdulrahman Aalarjani	050 137 7000
DGCL Project Management Lead	Clive Walsh	054 068 5918
DGCL OHS Lead	Metaeb Alharbi	059 041 8666
PMC Project Management Lead	Rory Poole	055 014 4173
PMC OHS Lead	Seljan George	054 237 5720
CSC Project Management Lead	Theo Stergiopoulos	050 734 9294
CSC OHS Lead	Reneesh Kumar	055 932 5339
Project Team		
Title	Name	Contact Number
Project Director -Incident Commander (IC)	Sherif Abdussattar	0500294285
Project Manager (Chief Warden)	Abdullah Sabry	0505420707
Construction Manager (Deputy Chief Warden)	Alaudeen	0593358665
OHS Manager	Sampath Kumar	050 014 8699
After Hours Contacts		
Title	Name	Contact Number
Security	Sultan Saud	0537212154
Site Manager	Alaudeen	0593358665
OHS Manager	Sampath Kumar	050 014 8699
General Contacts		
Title	Name	Contact Number
Site Engineer	Alamgeer	0597874528
Office Secretary	Matoq	0506273142
Storekeeper	Sameer	0572837605
Safety Officer (Day)	Adnan Hakmi	0533888274
Safety Officer (Day)	Khuram Shehzad	0559062258
Safety Officer (Night)	Wasif Sheikh	050 374 5218
Male Nurse (Night)	TBA	TBA
Male Nurse (Day Shift)	Mohd Shuja	050 159 7914
Security	Ali Haider	0559949593

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7 Emergency Preparedness.

7.1 First Aid & Emergency Response

- As a minimum, AICC will be implementing the organization structure for developing, implementing, and maintaining the Emergency Response Plan (ERP).
- All communication shall be through emergency contact numbers in case of any emergency by using mobile network or internal walkie talkie communications between the sites.

a. FIRST AID

- Call first aid and inform manager / supervisor.
- Give an emergency treatment / evacuate / await ambulance.
- Complete AICC / AL-Diriyah Combined Accident Form As soon as possible Maximum 24hours.
- Review to prevent further occurrence (in line with safety management systems).
- Record to be kept for Management review.
- No matter how minor the injury is, it needs to be immediately reported to the supervisor or first aider and treated. Medical attention will be provided, and the incident must be documented.
- In the event of any injury or illness the person must be given First Aid treatment. Never move an injured or seriously ill person unless necessary to prevent further Injury. In such a case, get help from a person trained in First Aid.
- Patients requiring further professional treatment will be sent to assigned hospital and Emergency cases shall be referred to the nearest hospital.

b. Emergency Response Plan (ERP)

- Accidents/incidents will be dealt with in accordance with the ERP and the relevant accident/incident reports.
- The workforce shall be aware of the ERP and emergency contact numbers. Plans shall be posted in every possible place with the contact numbers.
- All personnel must familiarize themselves with the locations / information of the following as a minimum, but not limited to:
 - Emergency Assembly Point
 - Emergency Contact Personnel
 - First aid Boxes
 - Fire-fighting equipment.
 - Spill Control Kits
 - Exits, Muster points and Rescue equipment such as stretchers.
- Any deficiencies discovered must be immediately reported to the Supervisor or the Project HSE Manager.

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- Spill kits, stretchers, First Aid Boxes, and fire extinguishers must always be fully accessible. Do not stack materials in front of these items or obstruct access to them.
- The Project Manager shall ensure that the workplace meets all local regulatory and project insurance safety and environmental requirements with regards to emergency preparedness.
- This may mean that the project / facilities need to supply its own first aid clinic, ambulance service, rescue equipment and/or rescue squads, wet risers etc., as required.

7.2 Heat Stress Management

HSE Plan shall contain a general procedure for reducing exposure to heat and cold injuries for work conditions in KSA.

- The following factors shall be considered:
- Mean temperatures.
- Wind speeds.
- PPE and clothing that must be worn.
- Health of work force.
- Level of heat and cold injury awareness of the work force.
- Any employees who have symptoms of any healing or cold injury shall seek immediate medical attention from a professional medical service provider.
- AICC is to provide awareness training on symptoms and first aid treatment for heat injuries on a regular basis.
- AICC is to comply - Welfare, especially provision of shade and cold potable water.

a. HEAT STRESS:

- Heat Cramps - cramps are a warning sign that your body has lost too much salt through sweating. The lost effects the muscles that you use most often. If you experience muscle stops, tell someone, and take a break.
- Heat Exhaustion - this is a sign that your system is being stressed. Symptoms include exhaustion, nausea, pale and clammy skin, rapid pulse, and dizziness. Do not ignore these signs in you and your co-workers. Move to cool area, lose, or remove excessive or restrictive clothing. Drink cool liquid.
- Heat Stroke - Can be FATAL. It occurs when your body's cooling system is overwhelmed shut down. Signs include red, hot, and dry skin, no respiration, hallucination seizures vomiting, rapid and pounding heartbeat and loss of consciousness. This is a critical emergency. Medical help is immediately needed. The victim must be cooled rapidly. Douse them with water, place ice packs on their neck groin and armpit areas and fan them vigorously. You must bring their body temperature down. If they begin to shiver, STOP cooling them and cover them lightly.
- AICC Company will adopt color code flag system as follow by the heat index system.

Heat Risk Category		WBGT	Light Work		Moderate Work		Heavy Work	
			Work / Rest	Water Intake (ml/hr)	Work / Rest	Water Intake (ml/hr)	Work / Rest	Water Intake (ml/hr)
Very Low	Unacclimated	25.5-26.5	50/10 min	500	40/20 min	750	30/30 min	750
	Acclimated	25.5-26.5	Continuous work	500	continuous work	750	50/10 min	750
Low	Unacclimated	26.6-29.5	40/20 min	500	30/30 min	750	20/40 min	1000
	Acclimated	26.6-29.5	Continuous work	500	50/10 min	750	40/20 min	1000
Moderate	Unacclimated	29.6-31.0	30/30 min	750	20/40 min	750	10/50 min	1000
	Acclimated	29.6-31.0	Continuous work	750	40/20 min	750	30/30 min	1000
High	Unacclimated	31.1-32.2	20/40 min	750	10/50 min	750	Avoid work	1000
	Acclimated	31.1-32.2	Continuous work	750	30/30 min	750	20/40 min	1000
Extreme	Unacclimated	>32.2	10/50 min	1000	avoid work	1000	Avoid work	1000
	Acclimated	>32.2	50/10 min	1000	20/40 min	1000	10/50 min	1000

Table 2 - Exposure category and possible heat syndrome

8 Environmental & Sustainability Requirements

8.1 Waste Management

Waste Management details have been provided in the Logistics and Security Plan and more detail will be provided in the Construction Environment Management Plan.

- AICC assigned contaminated products collection to an approved contractor, however, all waste generated from construction shall have such wastes managed by a waste management contractor, and NO waste generated from construction shall be discharged at the job side.
- Minimize the amount of waste generation at the site.
- Ensure to maximize the reusing/recycling of the waste in coordination with the client.
- Ensure that the waste management plan is being implemented, and all the personnel understand how the site waste management system works.

8.2 Hazardous Substance Management

- All hazardous waste will be stored in bunded areas and separate skip containers and will be segregated from all other types of waste.

Examples of hazardous waste include:

- Soil contaminated with hazardous pollutants if any (diesel fuel, oil, etc.)
- Cans/containers containing waste paints, solvents, thinners, machinery lubricants, oil, fuel, chemical disinfectant, drain openers, air fresheners, pest control (insect or cockroach killers), etc.
- Waste kitchen oil (oil left after frying, cooking, etc. that can't be used)
- All the above will be stored separately in a container and will be labelled at the site as hazardous

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material.

8.3 Dust Management

To mitigate the dust problem the following measure has been taken.

- Water sprinkler at feed point
- Water Sprinkler at Discharge point
- Pre wetting the surface of the crushed material.
- Avoid crashing in high winds.
- Limit material drop height at feed point.
- Dust screens can be placed in case of excessive dust.

8.4 Sustainability Requirement

Water and fossil fuel and soil/rock are the main consumable and are main sustainability concerns while crushing operation. The following steps will be taken to reduce consumption or natural resources.

- Treated Sewage Effluent or dewater effluent for dust suppression when possible.
- Hight pressure mist/fog system will be used to suppress the dust.
- Crusher and Generator are in good working condition.
- Generator maintenance is done regularly.

8.5 Culture and Heritage Sites management measures

Within the site, if archaeological finds, such as heritage walls, historical water wells, graves, pottery, stone tolls amongst others are identified, a “no-go” archaeological zone and tents must be established, for protection of historical artefacts. After collection of the findings, the zone can be cleared, except for architectural remains.

Key requirements will be followed:

- Any archaeological findings near worksites or access roads are fenced with a 50 m buffer zone to prevent unauthorized access and accidental harm.
- Temporary fences or barriers shall be placed at the works sites to limit the work areas. Hoarding will be provided by Contractor, at each work site, as per DGDA specifications, fixed at 3 m and 5 m heights.
- No persons, machinery, equipment, material, or waste shall enter in “no-go” archaeological zones and tents. In addition, stationary noise, vibration, or dust sources, including generators and compressors.
- Adhere to the Chance Finds Procedure developed in line with the recommendation provided above.
- Contractors should be required to undertake awareness training on the region’s archaeological and cultural importance for key personnel, with attention given to artefacts and sites of interest that have

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been reported in the region.

- If any artefact / site of interest is encountered during construction, work should be stopped immediately at the location where the artefact has been reported.
- To be taken on the artefact is agreed upon between the Project management team and the department.
- Should archaeological / cultural heritage sites / artefacts be encountered during construction and demolition, the following steps should be followed:
- During excavation works, site supervisors, machinery operators and other staff involved in the works will be watching for any potential heritage / archaeological items.
- In the event of discovery of remains / objects of cultural heritage or archaeological importance, the works shall stop, and the E&S Manager shall be informed.

During construction, noise and vibration will be generated from equipment during land clearing, compaction, and construction of the development. Vehicles and machinery such as graders, compactors, concrete pumps, generators, etc. will represent the major noise sources.

The key construction noise and vibration mitigation measures will be followed:

- Pre-construction survey at Project boundary points located near identified SRs.
- Coordinate with public relation offices or their representative office/s and notify the public about the construction activities and duration of activities.
- Limit working hours near SRs where possible i.e., agree on appropriate working times with respect to the SRs etc.
- Construction compound sites / logistic areas should be located at distances of more than 25 m away from the nearest noise / vibration sensitive receptor.
- During the process of selecting methodologies / equipment, less noise and vibration generating equipment/ methods, shall be opted for.
- Construction vehicles and equipment (including noise and vibration reducing devices, silencers, or mufflers) used on site shall be in good working condition and properly maintained, according to manufacturer's recommendations, and used in a manner as to avoid causing excessive noise and vibration emissions.
- Impacting works shall be restricted to daytime hours (between 7:00 and 19:00), during weekdays, if possible.
- When selecting methodologies / equipment, less noise and vibration generating equipment / methods, shall be used, as much as possible, such as using surface mining machines over rock breakers, utilizing vibratory or hydraulic pressed sheet piling methods over driven sheet piling, or using dead rolling

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rather than vibratory rolling for ground compaction.

- Number of equipment working in proximity of heritage items shall be limited, to avoid supplementary vibration (e.g., only one equipment (rock breaker) shall be used during demolition works and using drum cutter attachment instead of rock breaker and a noise attenuation attachment).
- Should noise or vibration standard levels be exceeded for a prolonged period, residents, or other sensitive receptors (within 200 m) to be adversely impacted by the works (even after application of mitigation measures) will be communicated in advance of the works commencing on site.
- Construction contractors should be required to use equipment that is in good working order, is properly maintained according to the equipment's manufacturer's requirements and that meets current best practice noise emission levels.

9. Security Arrangements

AICC will secure the construction site, the entry and exit points shall be controlled through 24/7 security arrangements, Hoarding arrangements will be carried out all around the site, AICC will implement site security management as per DGCL standards.

9.1 Document Records

9.1.1 Inspection Test Plan (ITP)

Inspection and test plan for construction of concrete work along with checklist shall be submitted as separate document for approval.

Quality Control Staff and Notification Period

- Quality Inspectors/Engineers will inspect the work procedure and quality of work at the construction site to assure conformity with the contractual requirements given in the General and Special Specification, and the approved Design and Shop Drawings.
- The QA/QC Manager is responsible for the overall quality Management system of the project.
- Quality of material for concrete work during the construction operation and check with the related method sent, and testing requirements.

10.HSE Risk Assessment

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10.1 HEALTH SAFETY AND ENVIRONMENT (HSE) RISK ASSESSMENT

Project:		Diriyah Gate (DG-II)					Risk Assessment Ref:		17				
Company Name:		Al-Ayuni					Date:		28-07-2024				
Activity Name:		Cast In-Situ Concretes works for potable water Pump Station02					Site Location		PMPS2				
Sr. No	Task	Hazards Identified	Harm	Risk Rating without control measures				Control Measures	Controlled by whom	Residual Risk			
				LXS	L	M	H			LXS	L	M	H
01	Preparationfor the job	Poor Planning Lack of proper supervision Lack of awareness from the legal/ contractual obligations	Waste of manpower/ resources Financial consequences Rework Legal consequences/ fines/ NCRs etc	2x3		6		- Getting prior approvals and NTS before start of works. - Availability of RAMS - Scope of work briefed among the crew. - Supervision and knowledge sharing from the team leader. - Maintaining a two way-communication channel between the work force and management	Site Engineer/Constructi on Manager	1x3	3		
02	Surveying Works	Slip, Trip & Fall from heights. Slip, Trip, and fall from the same level. During fixing survey points. Man and equipment interface. Heat stress. Traffic Risk. Snake and other poisonous insects.	Personal Injury Equipment damage Property damage Serious injuries	2x3		6		- Isolate the working area with barricades. - Ensure all equipment follows the speed limit (20km/hr). - Inspect the working area before allowing the survey team on site. - Not to allow lone working at the site. - Legs of the tripod are to be securely bedded into the ground, and mercury accuracy is to be checked.	Survey Engineer/ HSEO	1x3			

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		Survey equipment falls. Manual handling of survey equipment. Personal Injury due to working near roads. Lack of welfare facilities						- No activity during wind or sandstorms. - Adequate training must be conducted. - An adequate walkie-talkie/ Bravo or satellite device was provided during the activity. - Persons are allowed to wear long boots in bushes to safeguard from any snake bites. - Adequate personal protective equipment must be worn. - Emergency Contact No provided to the survey team. - A drinking water bottle was provided for each personnel member. - A fire extinguisher and first aid box were provided to the survey team. 3rd party training provided to survey team members like: (First aid with CPR and fire warden).				
03	Personnel Access to Site	Uneven surfaces Lack of proper access and egress Exposure to weather conditions Lack of welfare facilities	Personnel injury Slip, trip and fall Heat stress Dehydration Sun burn	5x4			20	- Pre task briefing before start of every activity - Provision of safe access and egress - Provision of welfare facilities - Adherence with PTW system	Site Engineer/Construction Manager	1x4	4	

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		Lack of communication arrangements Lack of planning Unavailability of Proper PPEs	Entrance in a restricted area					• Trained employees • Provision of emergency arrangements • Availability of supervision • Use of proper PPEs for the job				
04	Traffic Circulation on the Job site	Lack of traffic control management Lack of proper parking Lack of traffic signs Unauthorized driving Damaged vehicle	Fumes and Pollution, drips and spills, dust. Vehicle collision Hurt/ injury to persons Traffic fines/ insurance premiums	4x4			1 6	• All drivers and operators must be following the site safety rules and speed limit. • All drivers have Saudi government license Access roads clear from any obstruction • Make grading and compaction • display speed limit and driving guidelines signs • Provide two way road and control the traffic • SUV and inspected vehicles to be used for the job • Monitor weather condition, sprinkle water on roads	Logistics/ supervisor/ HSEO	1x4	4	
05	Precheck list use before concrete	Poor access Poor communication No use of approved checklists	Work delays Client penalties or legal actions Property damage.	3x4			12	• Ensure NTS approved for the above activity. • Ensure safe access provided for the employees to reach to the concrete pouring location.	Site Engineer/Concrete engineer/construction manager/Project manager/ HSEO	1x4	4	

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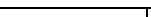
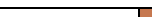
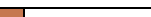
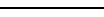
		Not following the MST	Personnel injury.				• Provision of adequate welfare facilities like drinking water, rest shelter, Toilets in place(250mt). • Ensure the safe distance maintained from the edges or site, is free from any soil collapse hazards. • Ensure work area protected or barricaded to avoid fall of persons or material/equipment. • Ensure safe access provided for equipment or transit mixer to reach to pouring location. • Check Boom placer operator & equipment have a valid third-party document. • Ensure the activity not planned during mid-day break (12Am to 3pm from June 15 to sept 15. • Ensure all temporary Electrical connections are properly routed,30MA RCD was installed in power distribution board, No damaged cables being used.				
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06	Working near and in excavation	Cave-ins or Collapses Falls into Excavation Falling Objects Mobile equipment operation at edge Manual Handling Access and Egress	Burial, serious injury, or death Injury or death from falling Injury from tools, equipment, or materials falling into the excavation Struck by or caught between equipment Musculoskeletal injuries Difficulty in exiting excavation in emergency	4x4	1 6	• Use protective barricade edge protection • • Ensure proper sloping of excavation walls • • Inspect excavation sites daily and after any event that could increase hazard (e.g., rain, equipment vibration) • • Install barriers, guardrails, or covers around the excavation • Use fall protection systems when necessary • Ensure proper signage and lighting around excavation • • Keep tools and materials away from the edge of the excavation • • Use toe boards and guardrails • Use spotters and ensure proper communication • Keep workers out of the swing radius of equipment • Equip mobile equipment with warning alarms and lights •	Site Engineer/ operators / HSEO	1x4	4	
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							• Use mechanical lifting aids where possible • • Train workers on proper lifting techniques • • Provide ladders or other safe means of access/egress within 25 feet of workers • • Ensure access and egress routes are clear and unobstructed • • Plan tasks to minimize manual handling				
07	Preparation of FDT and Blinding Concrete	Risks associated with improper handling or equipment failure Risks related to the operation of equipment used for FDT, including mechanical failures or improper use. Risk of falls due to unstable surfaces or edges around the area where concrete is being placed or tested.	Hand injury, personal injuries cause skin burns and irritation severe injuries	3x4		12	• Regularly inspect and maintain, transport, and FDT equipment to ensure proper functioning. • • Implement procedures for use of suitable equipment and PPE. • • Provide appropriate personal protective equipment (PPE), such as gloves, goggles, and long sleeves, to protect against dust. •	Site Engineer/Quality Engnr / HSEO	1X4	4	

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		Risks from inadequate access to the testing or blinding areas. Risks associated with exposure to wet concrete dust generated during the preparation and testing processes Poor site conditions, such as uneven ground or wet areas					- Implement measures to control dust, such as using water sprays or dust suppressants. - Prepare the site to ensure it is stable and suitable for concrete work and testing. - Provide training for all personnel on concrete handling, FDT procedures, and the use of safety equipment. - Ensure constant supervision to verify that safety measures are followed and to address any issues that arise. - Encourage the reporting of near-misses and incidents to improve safety practices.				
08	Concrete Bomplacer placement	Poor Access / Egress, / Non-competent. Ergonomics and Manual Handling. Lifting activity loading and unloading. Slips, Trips, Falls and Abrasions. Unauthorized operation of Boom placer	Property damage, fatality, or injuries during the placement of concrete. poor housekeeping Cuts and laceration.	4x4			- Obtain a work permit prior to starting work. - Ensure assigned competent. - The supervisor or deputy should be on site during working hours. - The area has to be barricaded to avoid unauthorized entry to the moving equipment area	Site Engineer/ HSEO	1x4	4	

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							• Ensure to provide separate pedestrian route. • Use right PPEs for the job and Keep hands away from pinch/crash points. All workers engaged in the task must be provided with the required Hand gloves. • Provide signage and fire extinguisher. • Ensure boom placer is 3rd party inspected. • Operator must have a valid TPI certificate. • Proper communication to be maintained. • Equipment to be placed away from the edge of excavation(min 1.2mt). • Works to be carried under supervision only.				
09	use of boom placer for pump station roof slab/wall concreting	Risk of damage to concrete pump from vehicles & machinery on site Collapse or settlement of pump due to ground conditions/ overturning of pump, or collapse of boom due to insufficient stability Crush or impact by	Personal injury Fall of boom placer or collapse Damage to equipment Serious injuries Crush and finger injuries Fatality	4X5		20	• Ensure sufficient cordoned area is provided to keep other machinery & vehicles away from Concrete Pump. • Ensure ground conditions at set up are adequate for load of pump. Provide sufficient support (timbers, steel sheets) as required and ensure there is adequate space around the pump for	CPO/HSE/Site Engr/Concrete Engr/CM	1X5	5	

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		boom, or struck by placing hose, during set up or during 'off the boom' placement of concrete						- the full deployment of outriggers. - CPO to ensure outriggers are extended to appropriate position. Sole plates are provided for placement below outrigger feet to support the pump. - CPO to operate via remote control unit from a position of safety. - Ensure sufficient set up space and adequate light levels to ensure safe pump set up. Ensure personnel remain clear of area and any operative controlling the position of the placing hose should hold it at arms length. Persons shall be nominated to converse with and signal the CPO for minor boom positioning.				
10	Concrete Pump Operation	Vehicle movements - crushing or trapping of persons the moving pump to work in position Back injury due to moving rubber hose(end hose) Washout of pump	Broken limbs Equipment damage Environmental impacts Eye injury or irritation Personal injuries Entanglement	4X5			20	Keep site personnel out of reversing area. A competent traffic marshall should direct the moving vehicle and look out for anyone in the danger area. Vehicles fitted with reversing sirens and amber flashing lights should have these engaged. High visibility clothing must be work and do not back truck on out of	CPO/HSE/Site Engr/Concrete Engr/CM	1X5	5	

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		Potential blockages at start or restart of pump, use of suspended hose 'off the boom' resulting in hose whipping into						- impact goggles and high visibility clothing,coverall. - Pipe seals to be maintained correctly. - Level of concrete in hopper to be controlled correctly. - Do not kink placing hoses in the ground line. Do not attempt to pump very old concrete or concrete other than a pump mix. - The danger zone is the area around the end hose in which it can strike out. - The diameter of the zone is twice the length of the end hose. - Make sure area is kept clear on restart. Security lanyard must be used on joint between boom and flexible hose. Only use single ended hose off boom. - Keep hopper full and inform mixer driver of risk - Access shall be restricted to "short duration' tasks only. Ensure footwear is clean and free from mud. Unload pipes, one at a time, from the side of the lorry.				
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		Pipeline cleaning with compressed air - pressurised release/ejection of sponge ball/ concrete/water/air Drawn into machinery in motion COSHH. Cement - Irritant						• level, where the CPO is required to position himself remote of • the pump appoint a competent person to monitor deliveries and • hopper level (to ensure air is not pumped into the pipe system) • • CPO to ensure pipeline is depressurised before freeing jointing clips, hirer personnel must not uncouple pipes without the express permission of the CPO. • • A separated and access restricted area shall be provided at the • end of the pipeline that is to be air cleansed. • • Ensure hopper grating is lowered and secured during pumping operations and keep unauthorised personnel away from working area. • • Rubber Gloves to be worn when handling pipes, hoses, grout or washing out •					
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11	Concrete Blinding works	Risk of Trench Collapse- the weight and vibration from the pump truck can destabilize the trench walls. Equipment Malfunction - mechanical failure of the pump truck. Hose blockages or burst hoses. Risk of entrapment or injury from moving parts. Risk of being struck by concrete or equipment. Exposed protruding rebar steels around excavation concrete casting. Spillage of concrete causes environmental pollution. Disruption to local water drainage systems.	Potential for serious injury or death. Potential for equipment damage and project delays. Potential for serious injury or death. Potential for environmental cleanup costs and regulatory fines.	4x4			- Monitor trench conditions regularly during pumping operations. - Limit the size and weight of the pump truck to prevent excessive vibration. - Conduct regular maintenance and inspections of the pump truck and hoses. - Have backup equipment and emergency shutdown procedures in place. - Train operators on proper equipment operation and troubleshooting. - Establish a clear exclusion zone around the pump truck during operation. - Provide personnel with appropriate personal protective equipment (PPE), including safety helmets, high-visibility clothing, and safety footwear. - Conduct regular safety briefings and training for all personnel involved in the casting process. - Provide rebar caps protection all protruding steep exposed around excavated trench.	CM, QC, HSE, Supervisors	1x4	4	
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							• Use spill containment measures, such as berms or barriers, around the trench area. • Have spill response kits readily available on-site. • Provide polythene plastics below from the truck concrete mixer and pump truck concrete bucket to catch spills of concrete. • Ensure proper disposal of any spilled concrete or contaminated materials. • Regularly monitor and inspect the trench, equipment, and personnel safety measures throughout the casting process. • Conduct a post-operation review to identify any lessons learned and update the risk assessment and control measures as necessary.				
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12	First aid care on site	Unavailability of trained medical staff at site Incorrect diagnosis Poor communication	Death Serious medical condition Business rupture Reputation damage	2x2	4	• Deployed one paramedic/Nurse Clinic for providing first aid treatment and one ambulance for handling the emergency case. • Providing emergency contact • Numbers	Site Engineer/ Safety officer	1x2	2	
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							Provision of communication means in case of emergency.				
13	Working on/ near live roads, Road crossing Loan Working Survey vehicle Use of Chalks, warning tapes	Unauthorized use of equipment, untrained manpower Lack of signage and orientation Traffic regulation Improper access No segregated routes for pedestrians Improper supervision Lack of awareness/ training	Car Accident OProperty damage Injuries Run over Cut, puncture and hit injuries Drive over personals Slip, fall and crush	2x3	6		- Proper signage - Traffic directions - Speed limit - Pedestrian route segregation with concrete barriers - One-way system application to avoid reversing if Applicable - Proper training and supervision and use of buddy system. - Reflective signs and jackets utilization.	Site Engineer/ Safety officer	1x3	3	

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14	Walking onsites.	Ground conditions: obstructions (tools, materials, cables, etc....) lack of pedestrian access Weather condiitons	Falls on the same level. Slip, trip and fall. Heat stress, Heat stroke. De-hydration Personal injury	3x2	6	• Keep your hands free (not in pocket) while walking on site • Keep shoelaces tied. • Don't step over obstructions. • walk around. • Ensure that the walking surface does not have oil, grease, or water. • Avoid slippery ground surfaces where possible Use suitable footwear Be aware/look ahead. • Ensure suitable lighting is provided on site. • Avoid unlevel/uneven ground and use an alternate route. • Use designated route/walkway. • Look ahead/be aware Keep hands free (not in pocket) while walking on site • Follow mandatory signs on site where possible. • Provision of PPEs	HSE/CM/Logistics/Site Engr	1x2	2		
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15	Use of chemicals/ paints etc.	Lack of awareness of the properties of flammable liquids Operator error, due to lack of training. Hot work on or close to flammable liquid containers. Exposure to heat from a nearby fire. Damage nozzle pipe, Smoking near fuel tanks and oil tanks. Uncontrolled housekeeping.	Serious skin damage, fires, explosions, burns. property damage Health issues (skin sensitization, respiratory problems)	2x3	6	- Display the Safety data sheet provision (SDS). - Minimize quantities of chemical held to reduce the risk - Ensure the store has suitable signage to warn others of its contents. - Ensure a COSHH assessment is completed - Always ensure that chemical cans. - are returned to the store and stored in dip trays to avoid any spillage or soil contamination. - PTW implementation. - Supervision Firefighting arrangements. - Good stocking practice. - Right PPEs for the job.	Site Engineer/Stores/Environmental/Site Engr/CM	1x3	3	
16	Installing approved flexible & self-expanding water stop in construction joints	Use of Wrong tools and equipment Poor access to the wall or slab. Potential for tool or equipment failure during installation handling and cutting flexible and self-expanding water stop materials, including chemical exposure using adhesives or sealants that may contain chemicals	musculoskeletal injuries delays or injuries chemical burns or physical injuries. accidents or poor installation quality rework severe injuries or fatalities.	4X5	20	- Develop and follow procedures for the safe use of tools and equipment during installation. - Ensure all personnel are trained in the installation of water stops, including safety procedures and proper handling techniques. - Use only using a saw or PVC pipe cutter to cut the rubber stopper. - Persons or workers working must be trained in Work at height.	Site Engineer/Stores/Environmental/Site Engr/CM/Quality Engr	1x5	5	

		stability of lifting equipment or formwork.						• Ensure all formworks will be lifted/erected as per lifting calculation and safe lifting plan with approved lifting tackles as per formworks design. • Establish clear communication protocols among the crane operator, ground crew, and other workers, using radios or hand signals as necessary. • All Lifting Belt/Sling with D-Shackles used for formworks lifting must be approved by third party. • Work will be supervised by Lifting Competent Person & presence of TWC. • Ensure continuous supervision during lifting operations to verify adherence to safety measures. • Avoid manual handling where possible and use mechanical aids. • Use only designers recommended hook hanging points inside Hbeam.				
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18	Working with Hand/power Tools	Damaged/ Improvised Tools Weakness Illness Uneven surfaces High physical level demand Night shifts Wild animals Dynamic site conditions Dust noise Rotating parts of the machinery Combustible sources Electrical sources Defective vibrators and needle pins	Cold stress Heat stress. Back pain Eye strain Fatigue Fall from height. Burn Cut injuries. Bite by the animals Entangled/ engulfed with Struck/hit by the equipment. Crushed by Accident Incident Low sugar Hi blood pressure. Epilepsy Respiratory problems Hearing problems Death Hand arm vibrations	4x4	16	• Site Induction and toolbox talks should be given prior to starting the job. • Ensure competent workers are assigned for the task. • The area must be barricaded and ensure no unauthorized entry to the working area. • Medical pre assessment before assigning the work. • Provision of adequate welfare facility • Training of manual handling hazards, and other medical oriented issues • Incident reporting compliance • Having emergency contact info with everyone • Provision of fire extinguisher and medical first aid kit in every strategic location. • Availability of paramedic staff 24/7 with ambulance • Availability of life saving drugs • Availability of proper PPEs for the tasks • Use buddy system, no lone working • Availability of proper communication channels • Site familiarization and handing-over to the next shift in charge for dynamic risks awareness • Use of Vibrators and needles after certified by TWC. • No modifications to be done in vibrators or Needle pins for concrete works.	Site Engineer/ HSEO/Electrical Engr	1x4	4
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							• use of power float with Deadman switch, air compressor, vibrators, derailing immediate basis prior to handling & shifting, power tools, saw cutter, grinder will be followed and inspected prior use. • Only industrial plug sockets will be used during the entire operations and checked by electrician. • Proper cable routing or protection will be provided from sharp edges. •				
19	Steel reinforcement concrete/tie rods	Fabricated hand tools used. Incorrect manual handling/lifting of material. Sharp & protruding materials/ objects. Limited Access/Egress	Personal injury to workers and visitors. Loss of lives. Asset damage. Equipment and property damage. Delay in project execution. Loss of productive work time. Eye injury	4x5		20	• Keep eye on the task. • Provide proper steel fixing hand tools. Improvised and fabricated hand tools shall not be used. • Supervisor and Foreman shall ensure that workers are provided with appropriate hand tools. • Wear rubberized cute resistant hand gloves. • Provision for barricade and signage. • Refer to engineering drawing. • Ensure the correct manual lifting procedures and body positioning before the actual lift. • Use the buddy system if the load exceeds. • Use hand protection (cotton gloves minimum) when handling materials. • Keep hands away from pinch/crash points. All workers engaged in the task must be	Site Engineer/ HSEO/CM/	1x5	5	

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								provided with the required Hand gloves.				
								- Supervisor/foreman to ensure proper monitoring of their activities for safe operation - Access pathways are to be kept clear from any obstruction. - Conduct housekeeping before, during, and after work to ensure free access/ egress.				
20	SELECTION, USE, CONTROL AND CARE OF PERSONAL PROTECTIVE EQUIPMENT (PPE's) for all activities	Unapproved PPE's Inadequate PPE's Lack of care/ maintenance Loose PPE's exposed to rotating machines/parts Unfitted PPE	Minor health effect Medical treatment / Temporary or partial disability Personal injuries/serious injuries	4x4			16	. Proper selection and adequate supplies of PPE's for the crew - Personnel training / retraining of PPE's b. Only PPE's which bear ANSI or equivalent approvals are to be supplied: - Fall protection (ANSI Z359) - Respirators (ANSI Z88.2) - Eye and face protection (ANSI Z81.1) - Hard hats (ANSI Z89.1) - Foot protection (ANSI Z41) - Hearing protection ((ANSI S3.19-1974) - Working gloves as per (Part 1 General Safety, Chapter 3- Personal Protective Equipment) - Wearer to inspect PPE before each use, maintain them in good condition. - Wearers of PPEs are to be educated on PPE values and limitations: - Use of PPEs is last line of defense hence nonnegotiable, engineering	Project Manager / Safety Supervisor/Officer / Project Engineer / Site Supervisor	1x4	4	

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								and, administrative controls being the primary controls. :				
								- PPE's do not replace safe behavior. - Defective PPEs will not offer protection - Proper use of or poorly maintained PPEs will not offer protection.				
21	Air compressor use	Air hose whipping if not properly secured or if there is a sudden release of pressure. Exposure to high noise levels from the compressor. Prolonged exposure to vibration from the compressor Carrying heavy air hoses or equipment Air hoses and cables on the ground Risks associated with electrical components and connections	Potential for injury if the air hose becomes uncontrollably disconnected Hearing problem HAVs Airborne Dust Risk of respiratory issues Damage to equipment Slips, Trips, and Falls Electrocution	4X4			16	- Establish clear procedures for connecting and disconnecting hoses to prevent hose whip. - Provide safety goggles or face shields to protect against eye injuries from flying debris. - Provide hearing protection due to noise exposure. - Ensure workers wear suitable clothing to protect against abrasions. - Ensure trained personnel operate the air compressor and follow manufacturer guidelines. - Establish clear procedures for connecting and disconnecting hoses to prevent hose whip. - Keep work areas clean to reduce the risk of slips, trips, and falls. - Use drip trays or absorbent materials under air compressors to catch oil or water leaks. - Provide training on safe use of air compressors, including hazards and control measures.	Site Engineer/ HSEO/Concrete engineer	1x4	4	

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								- Supervise to ensure safe practices are followed and respond promptly to any safety concerns. - Regularly inspect and maintain air compressors to ensure they are in good working condition.				
22	Stacking and storing of materials	Unauthorized, untrained workers Material fall from height Improper stacking Collapse	Crush Cut and abrasion Damage to property	2 x 3		6	- Authorized and trained employees - No materials shall be stacked or stored near the edge of excavation (should maintain minimum 1. Mt). - All materials shall be stored below 2 meter height. - Drums should be stored well drained or concrete surfaces - If the material is too heavy, use hydraulic trolley to shift the material. - Do not stored on the flat side. i.e., with flanges horizontal. - No storage of materials to be done in the work place, only required quantity to be taken for the job. - Personnel protective equipment must worn at all time during the activity. (Helmet, shoe, vest, overall, gloves, goggles, mask.)	Safety Supervisor/Officer / Project Engineer / Site Supervisor	1 x 3	3		

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23	Utilized portable generator for power supply	Risk of carbon monoxide (CO) poisoning Fire and explosion risks. Electrical shock Noise exposure Fuel spills and leaks. Mechanical hazards	Risk of inhalation if high concentration buildup. Fire and damage to property. Burns or electric shock injury pose to the workers engaged. Workers hearing loss if continue expose. Contamination- environmental impact Mechanical failures and accidents.	2x4	8	- Operate generators outdoors in well-ventilated areas away from buildings, windows, and air intake vents to prevent CO buildup. - Educate workers on the symptoms of CO poisoning and the importance of recognizing and responding to early signs of exposure. - Never operate generators indoors or in confined spaces, such as basements or garages, where CO can accumulate. - Provision of drip tray for portable generator use at worksite to catch oil leak if it happened. - A portable generator shall not be placed in the trench. - Provision of adequate fire extinguisher of each generator use at worksite. - Keep generators away from flammable materials, combustible debris, and sources of ignition to prevent fires and explosions. - Store fuel for generators in approved containers and away from heat sources, sparks, or open flames. - Refuel generators only when they are turned off and cool to minimize the risk of fuel spills and ignition. - Inspect generators regularly for leaks, loose connections, or damaged components that could pose fire hazards.	Logistic Manager/ Supervisor Site Engineer Supervisor/ Foreman HSE	1x4	4
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							• Ensure generators are properly grounded and connected to electrical systems by qualified personnel to prevent electrical shock. • Use ground fault circuit interrupters (GFCIs) on generator outlets to protect against electrical shock in wet or damp environments. • Inspect generator cables, plugs, and connections for damage before each use and replace any defective components. • Train workers in safe handling and operation of electrical equipment and provide appropriate personal protective equipment (PPE) such as insulated gloves and footwear. • • Provide hearing protection, such as earplugs or earmuffs, for workers exposed to high noise levels from generator operation. • Enclose or isolate noisy generators in soundproof enclosures or barriers to reduce noise exposure. • Rotate workers to minimize prolonged exposure to high noise levels. • Conduct regular noise monitoring and implement engineering controls to reduce noise emissions from generators. • • No storage of fuel at worksite- in case of refueling drip tray					
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							- containment to prevent spills and leaks. - Inspect fuel storage tanks, hoses, and connections for leaks or damage before refueling generators. - Clean up fuel spills immediately using absorbent materials and proper disposal methods to prevent environmental contamination. - Train workers on proper fuel handling procedures and emergency response protocols for fuel spills and leaks. - Ensure generators are operated and maintained by trained and qualified personnel. - Conduct regular inspections and preventive maintenance of generators according to manufacturer's guidelines. - Train workers on proper startup, shutdown, and maintenance procedures for generators to minimize the risk of mechanical hazards.				
24	use of solar tower lights during night operations	Weather Conditions: Adverse weather such as rain, wind, or storms affecting stability and functionality of the solar tower lights. Electrical Safety: Risks associated	Fall of lighting tower or toppling risk of collapse or tipping over. Poor visibility shock hazards and short circuits.	2x4	8	- Ensure adequate lighting coverage and brightness levels for the operational area. - Properly secure solar tower lights to stable surfaces or use ballast systems to prevent tipping. - Ensure the support legs are fully extended and locked on base.	Electrical Engineer/HSEo/O M/CM	1x4	4		

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			Electrocution					• Use insulated tools and equipment. • Implement a lockout/tagout procedure to isolate electrical circuits. • Only qualified electricians should be permitted to perform electrical work. • Use 3 core cables with proper earthing and cable should be protected from shared edges or protruding objects. • All electrical equipment's to be grounded • All cables to be made overhead with proper insulated supporting rings or cable ties. • Ensure that electricians have the necessary certifications and training to handle electrical systems safely. • Identify potential electrical sources, circuits, and any specific risks related to the lighting systems. • Before any maintenance activity, isolate the electrical circuit supplying power to the lights. • Implement a lockout/tagout procedure to ensure that the electrical circuit is de-energized and cannot be accidentally re-energized. • Before starting work, use voltage testing equipment to verify that the electrical circuit is de-energized.				
			Severe burns									
			Hand injury									

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								• Ensure that testing equipment is regularly calibrated and in good working condition. • Provide electricians with appropriate PPE, including insulated gloves, safety glasses, and flame-resistant clothing. • Ensure that PPE is in good condition and worn consistently during electrical work. • Erect barricades and warning signs to prevent unauthorized personnel from entering the work area. • Ensure that all operatives are familiar with emergency procedures related to electrical accidents, such as electrical shocks or fires. • Provide necessary training to ensure quick and effective response to emergencies. • Use insulated tools designed for electrical work to reduce the risk of electric shock. • Regularly inspect and maintain insulated tools to ensure their effectiveness. • Assign a competent supervisor to oversee electrical work and ensure compliance with safety procedures. • Implement a system for continuous monitoring of electrical safety practices throughout the maintenance activity.				
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								• Require proper documentation and permits before commencing electrical work. • Ensure that the electricians have reviewed and understand the electrical system layout and any associated risks.				
26	Use of Vibrator & Needles	Noise Exposure	Hearing damage	3x3	9	• Use screens or barriers to protect others in the vicinity • • Keep the work area clean and free of debris • • Use hearing protection such as earplugs or earmuffs • • Limit exposure time to noisy activities • • Wear respiratory protection such as masks or respirators • • Ensure the machine is properly grounded • • Regularly inspect the machine and cords for damage • void using the machine in wet conditions • • Limit duration of use • Use anti-vibration gloves • Rotate tasks to minimize prolonged exposure • • Wear tight-fitting clothing • Tie back long hair •	Site Engineers / Supervisors/ Foremen, OHS officer/ Supvs / All Personnel	1X3	3			
		Inhalation of Dust	Respiratory issues									
		Electrical Hazards	Electric shock or electrocution									
		Vibration	Hand-arm vibration syndrome (HAVS)									
		Entanglement	Injuries due to clothing or hair getting caught									
		Exposure Fire Hazard	Fire or explosion due to sparks									
		Contact with Moving Parts	Cuts, abrasions, or amputations									
		Slips, Trips, and Falls	Injuries due to tripping over cords or debris									
		Manual Handling	Musculoskeletal injuries									
		Struck by Equipment	Injuries from equipment being swung or dropped									
		Eye Injuries	Injuries from flying debris									

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							• Ensure machine guards are in place and functional • • Provide training on proper use • Keep hands and clothing away from moving parts • • Keep work area tidy and free of clutter • Secure cords to avoid trip hazards • • Use mechanical aids for moving heavy objects • • Ensure proper lifting techniques are used • • Ensure proper handling and securing of equipment • • Keep a safe distance from the operating area • • Use safety goggles or face shields • • Provision of eye wash facility, ambulance availability for eye wash. • • Ensure the work area is clear of other personnel				
27	Use of Grinding machine	Contact with Moving Parts Flying Particles	Cuts, abrasions, or amputations Eye injuries, cuts, and abrasions			20	• Ensure machine guards are in place and functional • • provide training on proper use • • Keep hands and clothing away from moving parts	Site Engineers / Supervisors/ Foremen, OHS	1x5	5	

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		Noise Exposure	Hearing damage	4x5				• Use safety goggles or face shields • Use screens or barriers to protect others in the vicinity • Keep the work area clean and free of debris • Use hearing protection such as earplugs or earmuffs • Limit exposure time to noisy activities • Wear respiratory protection such as masks or respirators • Ensure the machine is properly grounded • Regularly inspect the machine and cords for damage • void using the machine in wet conditions • Limit duration of use • Use anti-vibration gloves • Rotate tasks to minimize prolonged exposure • Wear tight-fitting clothing • Tie back long hair • Keep flammable materials away from the work area	officer/ Supvs / All Personnel				
		Inhalation of Dust	Respiratory issues										
		Electrical Hazards	Electric shock or electrocution										
		Vibration	Hand-arm vibration syndrome (HAVS)										
		Entanglement											
		Exposure Fire Hazard	Injuries due to clothing or hair getting caught										
			Fire or explosion due to sparks										

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							- Have fire extinguishers readily available - Follow hot work permit procedures if necessary				
28	Use of power actuated tools such as Drill machine, Plywood Cutting machine etc.	Damaged cable insulation, Sockets, electrocution, electric shock, short circuit, trip/fall. Tools uses by untrained workers	Hand injuries, cuts, bruises, strains, property damage.	4x4		16	- All portable electrical equipment must be PAT- tested and color-coded. - Make sure the test dates are visible on the equipment. - Ensure all power tools are not used more than 110V. - Ensure all power tools are in good condition and appropriately maintained. 4 PTW to be obtained, before commencing the task - Damaged industrial sockets and power cables must be removed. - Ensure the electrical cable is not damaged and has not been repaired with insulating tape or unsuitable connectors. - Ensure cables from power tools shall be organized so as not to present a tripping hazard.	Site Engineers / Supervisors/ Foremen, OHS officer/ Supvs / All Personnel	1x4	4	
29	Use of hand tools like Hammer, Hand Saw, Crow bar etc.	Damaged hammer head Flying of hammer head Use of wrong tool for the job Sharp handles Unsafe handling of tools Untrained person using the tools	Hand injury Head injury or eye injuries Serious incidents or accidents Property damage	4x4		16	- Use the right tool for the right job - Only trained and skilled workers are allowed to handle the tools. - Use proper PPE & Job specific PPE's like cut resistant gloves and safety goggles. - Job supervision is must.	Site Engineers / Supervisors/ Foremen, OHS officer/ Supvs / All Personnel/Electric al Engr	1x4	4	

							• Ensure tools are properly grounded • Regularly inspect tools and cords for damage • Avoid using tools in wet conditions				
31	Chipping & Surface Preparation	Poor Planning. Lack of proper supervision. Lack of awareness about the work criticality. Unavailability of proper PPE, Lack of TBT about activity. Flying rock or granules	Risk of injury to the working persons working within the work limit Equipment, and property damage. Delay in project execution. Loss of productive work time Eye injury ,hand injury.	4x4		16	• Ensure all site personnel are provided and are using required PPE, including High Visibility Vest, Face Sheild, Face mask, Earplug, Safety Helmet, Safety shoes. • Ensure Eye Flushing equipment is readily available. • Provide TBT about work. • Provide adequate firefighting equipment (Fire Extinguisher) near the work area with proper access. • If Power and Rotating type Buffing Machine is to be used, a control switch must be provided that will automatically shut off power whenever the operator removes his hand. • All electrical DB will be routed and grounded properly. • Dust masks to be used during the chipping & preparation works.	Site Engineer/ HSEO	1x4	4	

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32	Use of electrical equipment	Electrical shock, Burns and fire	Operatives and others risk potentially fatal injuries if they receive a shock from faulty electrical equipment or while working with live electrical sources.	4X4			16	- A pre-use check conducted by users. - All electrical equipment brought to the site must have been electrically tested. - Ensure that workers know how to use the electrical equipment safely. - Ensure there are no trailing cables that can cause people to trip or fall. - Stop using the equipment immediately if it appears faulty. - have it checked by a competent person. - Consider using a residual current device	Site Engineers / Supervisors/ Foremen, OHS officer/ Supvs	1x4		4	
33	Working on top slab or deck	Access to roof/plant room by unauthorized persons Risk of workers falling from the top slab or deck. Inadequate or improperly installed fall protection systems. Exposure to Wet Concrete Poorly trained work force Adverse weather	Fatal or severe injuries. chemical burns and slips. include accidents or delays. Sunstroke	4X5			20	- Take care to secure access points to prevent unauthorised persons accessing roof areas. - Ensure the slab is designed and constructed to handle the expected loads. - Conduct regular inspections of the slab to check for stability and integrity. - Install appropriate guardrails, barriers, and edge protection to prevent falls. - Use harnesses and fall arrest systems where guardrails cannot be installed.	Site Engineers / Supervisors/ Foremen, OHS officer/ Supvs	1X5		5	

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							• Provide training on working at height, concrete handling. • Implement safe procedures for handling and placing concrete, including spill management and proper curing practices. • Provide personal protective equipment (PPE) such as gloves, goggles, and long sleeves to protect against wet concrete. • • Use full body harness while working on deck edges. • • Avoid working during od weather conditions • Follow mid day break as per DGCL guidelines. • Follow work schedule breaks. •				
34	Hot works	Ignition source Availability of combustible materials Combination of incompatible materials at one place Competency issues SSW implementation issues Ergonomic issues Environmental	Fire Explosion Inhalation problems Skin problems Eye problems Stress Fatigue	4x4		16	• Implement a permit-to-work system for hot works, ensuring proper authorization, risk assessment, and controls are in place. • • Clear combustible materials and ensure fire extinguishers are readily available. • Ensure workers are trained in safe hot work practices and emergency procedures. • Regularly inspect and maintain hot work equipment to ensure safety standards are met. • • Ensure clear communication between workers conducting hot	Site Engineer/ HSEO/Concrete engineer/Operations & Maintenance	1x4	4	

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		Extreme weather conditions Sensitizing agents						- works and other personnel on site. • Ensure TPI is available for al the welders/welding machine and team perform hot works are trained to carry the job. • • Inspected and verified equipment usage • • Firefighting provisions & Supervision of job. • • Availability of fire watcher • • Earthing and ELCB provisions for the equipment-welding M/c. • • Proper tool for the job • Provision and usage of proper PPEs for the job. • Specific PPE s like Leather hand gloves, Apron and Fire blanket and face shield etc.				
35	Carpentry works	Lack of proper supervision Flying of hammerhead Inadequate, improvision of tools Inhalation hazards Sharp edges Electrical hazards	Personal injury Inhalation problems Asthma Eye injury Head injury Eye irritation Puncher injuries Cut	2x3		6		• Ensure all works are carried under proper supervision. • • Provision of right tools for the right job • Use of proper PPEs and dust mask. • Avoid the use of damaged and modified tools. • Dust control at the source of dust generation • Use hard hats and safety goggles. •	Site Engineer/ HSEO	1x3	3	

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		Noise Vibrations	Scratches Hearing problems HAVS					• Provide of emergency stop button for tools used to control any critical issues. • • Wear cut-resistant gloves. • • Only trained people are allowed to carry the job. • • Wear earmuffs to protect individuals from being exposed to excessive noise. • • Allow work shift patterns and frequent breaks.				
36	Formworks	Poor Planning/unsafe access/platform Lack of proper supervision/wrong design Inhalation hazards Sharp edges/unsafe handling of materials. Contact with Cement/mortar without proper PPE Electrical hazards Noise Vibrations	Electrocution Hand Injury Personal Injury Fatality/Serious Injuries	5x4		20	• Use platforms that are certified and ensure no overlading of platforms. • • Provide necessary control measures like guardrails an to avoid Workers falling during steel fixing and the erection of formwork. • • All Designs to built and approved by third party to avoid any Collapse of the formwork/falsework. • Secure the Materials to control falling during the striking of formwork. • Like Workers must be provided with safe access to the work. • Guard rails must be put in place as work progresses. • Access ladders must be properly erected, tied and project at least 1	Concrete Manger HSE Officer	1x4	4		

								- metre above the landing platform. - Ladders or an access scaffold must be used for access. - Equipment must be in good order before use. - Wet sweeping where possible using water. Wet materials before disturbing, unless that creates significant muscular skeletal strain, slip or other safety hazards. - Silica dust from scabbling operations. - Manual handling of shutters, reinforcing bars etc. - Cement burns from wet cement. - Arm and back strain for steel fixers. - Unsafe power tools/use of damaged electrical cables				
37	Scabbling and grinding concrete	High levels of noise from scabbling and grinding machines Hand-arm vibration syndrome (HAVS) from operating handheld scabbling or grinding tools. Exposure to concrete dust containing silica	hearing damage. Hand-arm vibration syndrome (HAVS) respiratory issues like silicosis. Electric shock Slips, Trips, and Falls Contact with Chemical Hazards cause skin irritation and allergies	4X4			16	- where possible, or implement noise-reducing measures and provide earmuffs to the crew. - ensure workers take regular breaks to minimize exposure. - Provide respiratory protective equipment (RPE) where necessary. - Provide training on safe lifting techniques and use of mechanical aids.	HSEO/Concrete engineer/CM/For eman	1X4	4	

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		Lifting and moving heavy scabbling or grinding equipment. Risks associated with using electrical equipment in a potentially damp environment. Uneven surfaces or debris on the ground can increase the risk of accidents. Exposure to chemicals from coatings or contaminants on the concrete surface.						- Inspect electrical equipment regularly and maintain in good condition. - Maintain good housekeeping and Keep work areas clean and clear of debris. - Identify and manage any chemicals present on the concrete surface. Provide appropriate personal protective equipment (PPE) and ensure adequate ventilation. - Ensure workers are trained in the safe operation of scabbling and grinding equipment, as well as the hazards associated with concrete work				
38	Shutter & de shuttering works	Poor Planning/unsafe access/platform Lack of proper supervision/wrong design Unsafe access/egress Improper selection of lifting equipment Use of damaged lifting tools & tackles Unsafe Hooking points	Fall of material on people. Collapse of equipment/ ground giveaway. Hit by equipment/ vehicle. Noncompliance may have legal consequences. Personal injury, hand injuries Cut, pinch and other injuries. Fatigue. Back pain other non-musculoskeletal injuries. Heat stress.	4X4			16	- Provide safe access to the work area - Use of Third party inspected Lifting equipment & tools & tackles. - Prior to start of work all Lifting tools & tackles to be re inspected i.e should be free from damages. - Ensure proper supporting is provided while shuttering or deshuttering. - Only trained workers will be engaged in the job. - Permit system: Permit to load/unload and Lifting permit will be followed during the activity. - All works will be supervised by TWC.	Concrete Manger HSE Officer	1X4	4	

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								- Lifting hook will be used as per the approved Dokha/Acrow supplier specifications. - Proper housekeeping will be maintained around the work areas.				
39	WORK AT HEIGHTS (≥1.2 mtr) during the construction process	Fall hazard / drop objects Personnel injury Defective fall protection Personnel not trained Ineffective supervision Sub-standard access ladder/scaffolding. Environment/weather status. (sand storm, raining, strong wind and foggy) Slippery surface. Unsafe access and egress	Workers who assign to work at heights and workers who are working underneath	4x4			16	- Approved full body harness (FBH) with lanyard provided and worn at 1.8 meter and above heights from ground or lower level. - FBH inspected by wearer prior to each use. - FBH color coded monthly by assigned competent person. - FBH lanyard safely secured to safe anchor point. Do not use structures/ members under construction partially supported as anchor point - Lifeline /anchor point to be install prior to start working on work at height. - Qualified/certified scaffolding inspector should check and tag the scaffolding. f. Proper access/ work platform / scaffold provided and shall comply with GI 8.001 - Scaffolding Safety Requirements. - Proper barricades, signage and control below and around overhead work to be maintained - Ensure that all workers are wearing the basic PPE. (hardhat, gloves, safety glass, anti-slipped safety shoes) - Install sling to hand and power tools to avoid from falling	Project Manager / Safety Supervisor/Officer / Project Engineer / Site Supervisor / Permit Receiver Qualified Scaffolding inspector	1X4	4	

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							• Make sure that materials will be stored in the middle to avoid from falling. • Suspend working at heights during bad weather condition and wait until the weather is safe to continue the activity. • As per DGCL Fall protection systems will be applied as follows: • Above 1.2 meters - standard railing systems with fitted toe board around working platforms • Above 2.0 meters - collective guardrail systems around working platforms • Fall arrest systems - selected based on the type of work, fall potential, injury severity, fall and clearance distances • Safety nets - only when guardrail systems are not reasonably practicable) • • Provision of proper access and egress with approve infrastructural engineer design.				
40	Handling and storage of hessian cloths	Dust Fibers Micro organisms Smoking or wrong storage Manual handling issues	Inhalation problems Biological infections Back pain Stress Fire Strain	4x4		16	• Storing of hessians after use by cleaning properly • Making sure that the hessians are dry before storing. • Store the hessian cloth in steel containers and provide fire extinguisher near to storage to avoid any fire. • Assign no smoking signages around the storage areas. • Training and awareness to the employees for lifting/ handling	Site Engineer/ HSEO/Concrete engineer/Foreman	1x4	4	

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		Encountering with insects etc. Damp/wet conditions.	Engulfment Byte by insects					• Use of proper PPEs while curing, handling and storage • • Disposing of the contaminated hessian • according to the environmental protocols				
41	Formwork with scaffold for Reinforcement Concrete(wall & top slab)	Structural Integrity: Collapse or failure of formwork due to design flaws, improper installation, or overloading. Manual Handling: Injuries from lifting heavy formwork components, handling sharp tools (e.g., tie wires), and repetitive strain injuries. Environmental Factors: Adverse weather conditions affecting stability and safety.	Potential injuries, structural damage, delays in project timeline, Musculoskeletal Disorders (MSDs) are injuries of to the body's musculoskeletal system. and environmental impact. Hand injuries or serious injuries Fall cause serious and fatality Electrocution or electrical hazards Tripp of cable and falls Slip trip fall	5X5			25	• Ensure all structures are buit and approved as per design by third party. • Prior installation of any structures need to access the surrounding which will have impact. • Provide and enforce the use of PPE such as helmets, gloves, safety boots, harnesses, and lanyards. • Enough manpower to be engaged in the task to avoid over work or perosns carrying over weight. • Follow Mid day breaks 12 to 3 pm rest schedule. • Provide training on safe handling of formwork and scaffold erection, use, and dismantling. • Regularly inspect formwork, scaffold components, and anchorage points for defects or damage. • Develop contingency plans for emergencies such as formwork or scaffold collapse, and ensure all workers understand emergency procedures. • Allow only trained team to allow on scaffold ,with valid third party certificates.		1X5	5	

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		Collapse: Scaffolding collapse due to improper assembly, overloading, or inadequate maintenance. Falls: Workers falling from heights during scaffold erection, use, or dismantling. Slips and Trips: Due to uneven surfaces, debris, or improper access points. Electrical Hazards: Proximity to power lines or electrical equipment. Fall of materials						- Ensure the structures are provided with Guardrails, fall arrest systems, training - Ensure proper stacking of materials at work place no storage of materials or excessive material storage is allowed on the deck or roof. - Follow the instructions of the structural design engineer. - Ensure Weather monitoring, TWL and rest schedules are in place. - Ensure standard scaffolding system approved by Engineer and inspected by Scaffolding competent person & inspector.. - Only erect scaffolding under supervision of competent person. - provide access & egresses for working crews at interval of 25 ft. - Provide proper working platform for the workers to work safely at height with safety harness.					
42	WORKING OUTDOOR IN THE HOT SUN/ EXTREME WEATHER CONDITION	Lack of proper planning / heat stress management Serious heat illness – heat exhaustion / stroke Lack of welfare facilities Lack of emergency preparedness Lack of administered	Major Health Effect, Temporary or partial disability or worst fatality	4x4			16	- Proper planning and look ahead Supervisor, prior to working outdoors in the hot sun/peak summer, he should plan for: - Workers' heat stress shelters / cool welfare facility / water stations - Mid day break 12 to 3pm) June 15 to sept 15 must be followed. - Electrolyte water distribution and Electrolyte station to be made ready for workers.	Project Manager / Safety Supervisor/Officer / Project Engineer / Site Supervisor / Permit Receiver	1X4		4	

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							• Work-rest schedule(minutes): safe for all continuous self-paced light work. • Rehydration Schedule: Heavy work >1.2Liters /hour • Work-rest schedule(minutes): Continuous paced work. • 45 mins work 15 mins rest • TWL: <115 • HIGH RISK ZONE • Flag: Red  • Rehydration Schedule: All work – >1.2 Lts/hour • Work-rest schedule(minutes): • Light Work - 45 mins work 15 mins rest • Work-rest schedule(minutes): Heavy Work - 20 mins work 40 mins rest.				
43	Hot works	Ignition source Availability of combustible materials Combination of incompatible materials at one place Competency issues SSW implementation issues Ergonomic issues Environmental	Fire Explosion Inhalation problems Skin problems Eye problems Stress Fatigue	4x4		16	• Implement a permit-to-work system for hot works, ensuring proper authorization, risk assessment, and controls are in place. • Clear combustible materials and ensure fire extinguishers are readily available. • Ensure workers are trained in safe hot work practices and emergency procedures. • Regularly inspect and maintain hot work equipment to ensure safety standards are met. • Ensure clear communication between workers conducting hot works and other personnel on site.	Site Engineer/ HSEO/Concrete engineer/Operations & Maintenance	1x4	4	

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		Extreme weather conditions						- Ensure TPI is available for all the welders/welding machine and team perform hot works are trained to carry the job. - Inspected and verified equipment usage - Firefighting provisions & Supervision of job. - Availability of fire watcher - Earthing and ELCB provisions for the equipment-welding M/c. - Proper tool for the job - Provision and usage of proper PPEs for the job. - Specific PPEs like Leather hand gloves, Apron and Fire blanket and face shield etc.					
		Sensitizing agents											
44	Environmental Issues	Dust	Respiratory issues.	2x3		6	- Spray water - Use the right tool and PPEs for the right job	Construction Engineer / HSEO	1x3	3			
		Oil Spillage	Respiratory problems	2x3		6	- Provide MSDS of the chemicals in use at site. - Train workers about safe handling of chemicals. - Chemical spillage drills to be conducted. - Use the right procedure for chemical spillage control as per manufacturer's instructions. - Only trained and experienced workers will do this job. - Use of spill kits. - Right PPEs for the job	Construction team/HSE officer	1x3	3			
			Skin sensitization										
			Soil and water contamination										
			Fines, SORs / NCRs										
			Quality defects										
			Slip, trip and fall										

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		Doc. Title	Method Statement					
			Cast In-Situ Concretes works for potable water Pump Station02					
		Doc. No.	IP-INF-054-0000-AIC-MST- CE-000032		Rev: 00	Date: 09-11-2024		

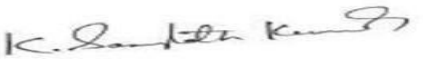
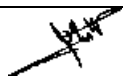
		Noise	Hearing loss	2x3	6	- Wear appropriate ear protection. - All tools and equipment to be in a good condition. - Use the tools and equipment according to manufacturer's instructions.	Construction Engineer / HSEO	1x3	3	
			Tinnitus							
			Psychological issues							
			Mental health issues							
		Transportation - Use of Natural Resources (fuel)	Depletion of Natural Resources	5x4	20	- Local Procurement of Project Materials - Car Sharing Schemes among Project Teams - Awareness on Eco-driving - Consideration of Eco-driven Vehicles	Procurement Manager	1x4	4	
		Air/ Greenhouse Gases Emission from Project Driving	Air Pollution				QC Manager			
			Contribution to Climate Change				HSE officer			
			Air Pollution (Greenhouse Gases) etc.				Environmental Engineer / Manage			
			Global Warming							
			Contribution to Climate Change							
			Air Pollution (Greenhouse Gases) etc.							
			Global Warming							

Risk Assessment Team

Sl.No	Name	Designation	Signature
1	Mohmed Biome	Senior Civil Engineer	
2	Sampath Kumar	HSE Manager	

Likelihood (L)		Severity (S)	Risk Level	Risk Matrix				
Rare – 1		Insignificant – 1	Extreme Risk 15 - 25	5	5	15	20	25

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Possible – 2		Minor – 2	High Risk 8 - 12		4	4	1 2	1 6	20
Likely – 3		Moderate – 3	Moderate Risk 4 - 6		3	3	9	1 2	15
Often – 4		Major – 4	Low Risk 1 - 3		2	2	6	8	10
Frequent/Almost certain - 5		Catastrophic – 5			1	1	3	4	5
						1	1	3	4
					Consequence (C)				
Assessed by:	Sampath Kumar		Approved By:	Abdullah Sabry					
Designation:	HSE manager		Designation:	Construction Director					
Signature:			Signature:						
Date:	07/01/2025		Date:	02/01/2025					